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An Efficient Deep Learning Approach with Modified CNN for Multi-Class Brain Tumor Classification

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Rabi' al-Awwal 1448 AH

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

{ يَا أَيُّهَا الَّذِينَ آمَنُوا إِذَا قِيلَ لَكُمْ تَفَسَّحُوا فِي
الْمَجَالِسِ فَافْسَحُوا يَفْسَحِ اللَّهُ لَكُمْ وَإِذَا قِيلَ انشُرُوا
فَانشُرُوا يَرْفَعِ اللَّهُ الَّذِينَ آمَنُوا مِنْكُمْ وَالَّذِينَ أُوتُوا
الْعِلْمَ دَرَجَاتٍ وَاللَّهُ بِمَا تَعْمَلُونَ خَبِيرٌ }

صَدَقَ اللَّهُ الْعَظِيمُ

سورة المجادلة

الآية (11)

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Dedication

To:

The man who lit up my life with his unconditional love.....

(My father)

**The woman who stood by my side along the way and never let me
down, being both a mother and a friend to me.....**

(My mother)

**To my beloved sister, Dr. Tayseer Shamran Atheab, for her
invaluable support, guidance, and inspiration.**

With deepest gratitude and love.

Those who have filled my life with their kindness.....

(My brothers and sisters)

My brothers' children

(Youssef, Hussein, Zine El Abidine, Taha, Ali)

With my love.....

Mahmoud 

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In the name of Allah, the beneficent, the merciful

First of all, praise be to Allah who offered me patience; power and faith in a way that words cannot express.

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Mahmoud 

Abstract

Accurate multi-class classification of brain tumors from Magnetic Resonance Imaging (MRI) is central to timely clinical decision-making, yet most high-performing deep learning models remain too large for deployment in resource-constrained clinical settings. This thesis proposes an ultra-lightweight Modified CNN built around three purpose-designed modules: the Micro Adaptive Feature Extractor (MAFE), which reverses the sequential ordering of the Convolutional Block Attention Module (CBAM) by refining spatial attention before channel attention; the Micro Multi-Scale Processor (MMSP), which fuses three parallel receptive-field pathways to capture multi-resolution tumor morphology; and the Context-Aware Intelligent Pooling (CAIP) module, which replaces static pooling with input-adaptive, per-sample aggregation. Trained and evaluated on the Mendeley Brain Tumor MRI dataset (12,064 images, four classes), the model achieves a test accuracy of 97.35% and a best validation accuracy of 99.22% (mean $99.12\% \pm 0.22\%$ across three runs), with only 659,228 parameters, representing a 47-fold reduction in parameter count compared with the comparable U-Net-based approach, and a parameter-efficiency score of 147.67. Ablation analysis confirms each module's contribution, with MMSP identified as the primary driver of performance gains. Cross-dataset evaluation demonstrates generalization beyond the training distribution, achieving 98.38% on Kaggle, 90.60% on Br35H in zero-shot evaluation and 97.62% after fine-tuning, and 89.36% on Figshare initially and 99.35% after orientation correction. With a model size of 7.88 MB, a memory footprint of 1.8 MB, and an inference latency of 12 ms, the architecture is suitable for mobile and edge deployment, subject to further clinical validation. These findings suggest that targeted, domain-specific architectural design rather than indiscriminate parameter scaling offers a more effective path toward accurate, efficient brain tumor classification for resource-limited healthcare settings.

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List Of Abbreviations

Abbreviations	Full Terms
AC	Accuracy
AUC	Area Under the Receiver Operating Characteristic Curve
BN	Batch Normalisation
BRATS	Brain Tumor Segmentation Challenge
CAIP	Context-Aware Intelligent Pooling
CBAM	Convolutional Block Attention Module
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
DenseNet	Densely Connected Convolutional Networks
DL	Deep Learning
DWConv	Depthwise Separable Convolution
EfficientNet	Efficient Compound-Scaled Neural Network
F1	F1-Score (Harmonic Mean of Precision and Recall)
FCN	Fully Convolutional Network
FLAIR	Fluid-Attenuated Inversion Recovery
FN	False Negative
FP	False Positive
FPN	Feature Pyramid Network
GAP	Global Average Pooling

GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
MAFE	Micro Adaptive Feature Extractor
ML	Machine Learning
MMSP	Micro Multi-Scale Processor
MobileNet	Mobile Neural Network (Lightweight CNN)
MRI	Magnetic Resonance Imaging
PR	Precision
PyTorch	Python-based Deep Learning Framework
RAM	Random Access Memory
RC	Recall (Sensitivity)
ReLU	Rectified Linear Unit
ResNet	Residual Network
ROC	Receiver Operating Characteristic
SC	Specificity
SE	Squeeze-and-Excitation
SOTA	State of the Art
TN	True Negative
TP	True Positive
U-Net	U-shaped Convolutional Network for Segmentation
VGG	Visual Geometry Group Network
VRAM	Video Random Access Memory
WHO	World Health Organisation

Chapter

One

Introduction

Chapter 1

Introduction

1.1 General Introduction

Brain tumor classification based on Magnetic Resonance Imaging (MRI) data has received considerable attention at the research frontier of deep learning and clinical neuro-oncology. Intracranial neoplasms have substantial pathological effects that impact patient prognosis, life expectancy, and quality of life. Consequently, early, accurate, and reliable diagnosis plays a crucial role in treatment selection (Athwal et al., 2021). MRI has been broadly adopted as the gold standard for brain tumor diagnosis due to its high soft-tissue contrast and non-ionizing nature, enabling detailed detection and characterization of underlying anatomical abnormalities (Işın et al., 2016; Ismail et al., 2022). Deep learning techniques have made significant advances in automated medical image analysis, progressing from early convolutional architectures (Simonyan, 2014; LeCun et al., 2015) through residual and dense connectivity structures (He et al., 2016; Huang et al., 2017) to attention-augmented and multi-scale designs (Hu et al., 2018; Woo et al., 2018; Lin et al., 2017). Despite this progress, model complexity, size, and inconsistent cross-dataset generalization remain open challenges that limit real-world clinical deployment, particularly in resource-constrained healthcare settings (Khan and Auvée, 2024; Sivanantham et al., 2021). A detailed clinical rationale for automated classification is presented in Section 1.3, while the technical background underpinning the proposed architecture including brain tumor taxonomy,

epidemiology, and imaging fundamentals is treated in full in Chapter Two. Section 1.3 below reviews the relevant deep learning literature and identifies the specific research gaps that motivate the architecture proposed in this thesis.

1.2 Thesis Motivation

Brain tumors remain a major global health burden and one of the leading causes of cancer-related death, particularly among children and young adults. While MRI is the diagnostic gold standard due to its high soft-tissue contrast, manual interpretation is subject to notable inter- and intra-observer variability, motivating automated, objective, and reproducible classification systems especially where specialist availability is limited. A detailed discussion of brain tumor taxonomy, epidemiology, and diagnostic imaging is provided in Section 2.2. This clinical need is compounded by a technical gap no existing deep learning architecture simultaneously achieves high accuracy and the computational efficiency required for mobile deployment. Established architectures such as U-Net, ResNet50, and VGG16 exceed the constraints of on-board mobile hardware or resource-limited clinical settings, making lightweight, low-cost diagnostic tools increasingly essential for equitable healthcare access. Motivated by these considerations, this thesis proposes a co-designed ultra-lightweight CNN combining a multi-scale feature extractor (MMSP), an adaptive spatial-channel attention unit (MAFE), and a content-aware pooling mechanism (CAIP) that achieves high four-class brain tumor MRI classification accuracy while remaining suitable for real-time inference on commodity hardware. This directly challenges the prevailing assumption that high diagnostic accuracy requires architectures with millions of parameters, and is validated through ablation experiments, statistical reproducibility testing, and cross-dataset generalization analysis.

1.3 Literature Review

This section presents a critical review of prior deep learning approaches to brain tumor MRI classification, tracing the technical progression from foundational CNN architectures through attention mechanisms, multi-scale processing, and lightweight design strategies, before positioning the present study relative to the state of the art through a comparative analysis and gap identification.

(Simonyan et al., 2014) established that CNNs could learn hierarchical discriminative features directly from raw pixel data, eliminating manual feature engineering. (He et al., 2016) introduced ResNet, resolving the vanishing-gradient problem via skip connections, while (Huang et al., 2017) proposed DenseNet for improved feature reuse through dense connectivity. (Ronneberger et al., 2015) developed U-Net for biomedical image segmentation (>25M parameters), later adapted by (Ilani et al., 2025) into a hybrid brain tumor classifier achieving 98.56% accuracy, though at a computationally prohibitive parameter cost. Attention mechanisms subsequently emerged to enable selective focus on diagnostically relevant regions. Hu et al. (2018) proposed Squeeze-and-Excitation (SE) networks for channel-wise recalibration, later extended to medical imaging by (Roy et al., 2019). (Woo et al., 2018) introduced CBAM, combining channel and spatial attention in a sequential, channel-first pipeline. (Vaswani et al., 2017) introduced self-attention via Transformers, subsequently adapted for brain tumor classification by (Tummala et al., 2022) using Vision Transformers, though at a substantial computational cost due to the quadratic scaling of self-attention.

(Lin et al., 2017) formulated Feature Pyramid Networks (FPN) for multi-scale representation via a full encoder-decoder pathway. (Howard et al., 2017)

introduced MobileNet using depthwise separable convolutions, advanced by Sandler et al. (2018) with MobileNetV2, and by (Tan and Le, 2019) with EfficientNet through compound scaling. Even EfficientNet's most compact variant (EfficientNet-B0, ~5.3M parameters in its standard 1000-class configuration; ~4.01M when adapted to the four-class task used in this study, as reported in **Table 4.11**) substantially exceeds the sub-700K parameter budget targeted by the present framework.

Recent brain tumor classification studies (2019–2025) report high accuracy but at considerable parameter cost: Khan et al. (2025) achieved 99.11% validation accuracy (test accuracy not reported) using a hybrid CE-EEN-B0-ResGANet architecture (~15M parameters) — notably comparable to, though slightly below, the present work's 99.22% validation accuracy; Zahoor et al. (2024) achieved 98.22% with Res-BRNet (~8.2M parameters); Ilani et al. (2025) achieved 98.56% using a U-Net-based hybrid (~31M parameters); Khan and Auvée (2024) achieved 98.09% with a resource-efficient custom CNN (~1.1M parameters), the closest existing combination of accuracy and efficiency to the present work. Table 1.1 summarizes these representative methods alongside their reported architectures, parameter counts, accuracy, strengths, and main limitations.

Table 1.1 - Comparative Overview of Representative Deep Learning Methods for Brain Tumor Classification

References	Architectures	~Params	Accuracy	Dataset	Strengths	Main Limitations
Deepak & Ameer (2019)	GoogLeNet + SVM	~6.8M	~97.1%	Figshare	Transfer learning + classical classifier	Two-stage pipeline overhead

Çinar & Yildirim (2020)	Modified ResNet50 + 8 custom layers	~25M	97.2%	Figshare	Hybrid CNN with fine-tuned depth	High parameter count
Tummala et al. (2022)	Ensemble ViT (B/16, B/32, L/16, L/32)	>300M	98.7%	Figshare	Ensemble attention outperforms individual models	Extremely high computational cost
Zahoor et al. (2024)	Res-BRNet	~8.2M	98.22%	Kaggle	Residual + boundary	Heavy model
Nazir et al. (2024)	Custom CNN + XAI (SHAP, LIME, Grad-CAM)	—	98.67%	Kaggle	Explainable AI enhances clinical trust	Binary classification only
Khan & Auvée (2024)	Custom CNN	~1.1M	98.09%	Kaggle	Resource-efficient	Resource-efficient, but with more than 1 million parameters
Ilani et al. (2025)	U-Net based hybrid classification model	>25M	98.56%	Kaggle, Figshare	High feature extraction capability	Computationally intensive
Khan et al. (2025)	EfficientNet hybrid	>10M	99.11%	Kaggle	Very high accuracy	High parameter count

Table 1.2 further contrasts the design principles of the three proposed modules (MAFE, MMSP, CAIP) against their closest prior-art counterparts (SE, CBAM, FPN) across five key architectural dimensions.

Table 1.2 - Comparison of Proposed Modules vs. Prior-Art Attention and Pooling Architectures

Property	Prior Art (SE / CBAM / FPN)	Proposed Modules
Attention Fusion	Sequential channel-first (CBAM) or channel-only (SE)	Spatial-first sequential attention (MAFE)
Attention Parameters	Tens to hundreds of thousands	~8K lightweight conv (MAFE)
Multi-scale Processing	Full encoder-decoder (FPN)	DWConv 3rd path + learnable fusion (MMSP)
Pooling Strategy	Fixed max/average/static blend	Sigmoid-gated, sample-adaptive (CAIP)
Design Target	General-purpose vision	Clinical-grade efficiency for edge devices

Three specific gaps in the existing literature motivate the present study:

- **Gap 1 — Joint accuracy-efficiency optimization:** While lightweight CNN architectures have been explored, achieving both >99% classification accuracy and a sub-10 MB model size remains challenging. For example, Khan and Auvée (2024) reported approximately 1.1M parameters with 98.09% accuracy, indicating room for further improvement in the joint optimization of accuracy and model efficiency.
- **Gap 2 — Attention ordering within a lightweight budget:** No prior lightweight attention design for brain tumor MRI reverses CBAM's channel-first ordering to apply spatial attention before channel attention.

- **Gap 3 — Content-adaptive pooling in lightweight architectures:** No prior lightweight architecture integrates a learned, sample-adaptive pooling mechanism within a sub 1M parameter budget.

These gaps are addressed respectively through the MAFE, MMSP, and CAIP modules detailed in Chapter Three.

1.4 Thesis Problem Statement

Classification of brain tumors from MRI images remains one of the most challenging diagnostic problems in clinical radiology. Although the four main classes glioma, meningioma, pituitary tumor, and normal brain exhibit distinct morphological characteristics, accurate classification is complicated by subtle inter-class similarities and intra-class variation. While deep learning has advanced this task considerably, a fundamental trade-off persists: lightweight CNNs reduce computational cost at the expense of accuracy, whereas high-performing models remain too computationally expensive for real-world deployment in resource-constrained clinical settings.

This trade-off stems from four interrelated shortcomings in existing approaches. First, leading architectures contain tens or hundreds of millions of parameters, producing large model files and latencies impractical for mobile or edge-computing deployment. Second, standard convolutional filters process the input uniformly rather than emphasizing the tissue boundaries and lesion regions radiologists naturally attend to. Third, most models rely on single-scale features, limiting their ability to capture the multiscale nature of tumor size, shape, and texture. Fourth, conventional pooling operates uniformly regardless of image content, indiscriminately discarding diagnostically salient boundary information.

These limitations cannot be resolved by simply scaling general-purpose architectures; they require a principled co-design of components tailored specifically to brain tumor MRI classification. This is the problem addressed by this thesis: how can a lightweight deep learning architecture achieve high classification accuracy while minimizing parameters, computational complexity, and inference latency for multiclass brain tumor MRI classification?

1.5 Thesis Aims

This thesis addresses the following research aims:

1. To design, implement, and empirically evaluate an ultra-lightweight modified CNN architecture for high-performance multi-class brain tumor classification under parameter constraints suitable for deployment in resource-constrained clinical environments.
2. To design a spatial-then-channel sequential attention module (MAFE) that reverses CBAM's channel-first ordering and investigates its effectiveness for boundary-aware channel recalibration.
3. To design a three-pathway multi-resolution fusion module (MMSP) employing depthwise separable convolutions for efficient multi-scale feature representation.
4. To implement a sigmoid-gated, sample-adaptive pooling module (CAIP) that learns blending weights from a global feature summary.
5. To train the proposed model on the Mendeley dataset and evaluate its cross-dataset generalization on Kaggle, Br35H, and Figshare.

6. To conduct an ablation study to quantify the individual and combined contributions of the proposed modules.
7. To assess model interpretability using the Grad-CAM technique.

1.6 Thesis Scope

This research is scoped to four-class brain tumor classification using T1-weighted MRI images. The study addresses the classification task rather than tumor segmentation, grading, or treatment response prediction. Experimental validation is confined to four publicly available MRI repositories. Multimodal imaging fusion, federated learning, prospective clinical trials, and regulatory approval are identified as future work directions.

1.7 Thesis Contributions

This thesis presents architectural and methodological contributions to improve the accuracy efficiency trade-off in lightweight brain tumor classification. The main contributions are summarized as follows:

1. We propose an ultra-lightweight modified CNN that achieves 99.22% accuracy Validation and Test Accuracy 97.35%, using only 659,228 parameters approximately 2% of U-Net's parameter count (Ronneberger et al., 2015), a 47-fold reduction while attaining a parameter efficiency score of 150.51. We also introduce MAFE, a sequential spatial-then-channel attention unit that reverses CBAM's channel-first ordering (Woo et al., 2018) by refining spatial saliency before channel-wise recalibration, motivated by the hypothesis that spatial region selection improves the reliability of subsequent channel attention for tumor boundary discrimination.

2. MMSP, a three-pathway multi-resolution fusion combining depthwise separable convolutions (Howard et al., 2017), achieving multi-scale representation within a single forward-pass block rather than requiring FPN's complete encoder-decoder pathway (Lin et al., 2017).
3. CAIP, a sigmoid-gated, globally-adaptive pooling mechanism that learns per-sample blend coefficients between max and average pooling from a global feature summary (GAP), instead of using static 50/50 weights, allowing the network to adjust its pooling behavior across different tumour classes and imaging conditions at the whole-feature-map level.

1.8 Thesis Organization

This thesis is organized into five chapters, structured as follows:

1. **Chapter Two** presents a critical review of the related literature and background, including brain tumor taxonomy and clinical context, CNN architectures, attention mechanisms (SE, CBAM, transformers), multi-scale feature extraction, lightweight architecture design, regularization techniques, the datasets employed, and the mathematical formulations underlying the proposed modules and evaluation metrics.
2. **Chapter Three** presents the methodological framework for the proposed Modified CNN, structured across four phases: dataset acquisition and preprocessing; architecture design comprising the three proposed modules (MAFE, MMSP, and CAIP); training and optimization; and the evaluation and validation protocol.
3. **Chapter Four** presents the experimental setup and results: the hardware/software environment, dataset characteristics, detailed architectural design, training configuration, and the evaluation of results

across classification performance, statistical reproducibility, cross-dataset generalization, the ablation study, comparison with existing methods, clinical deployment efficiency, robustness, and Grad-CAM explainability, together with a supplementary benchmark against six standard architectures.

4. **Chapter Five** summarizes the key findings, discusses the study's limitations, and outlines directions for future work.

Chapter
Two
Background

Chapter 2

Background

2.1 Overview

This chapter provides the foundational clinical and technical background necessary to understand the architectural design choices detailed in Chapter Three. It covers brain tumor taxonomy, epidemiology, and diagnostic imaging; the theoretical evolution of convolutional neural network architectures; attention mechanisms; multi-scale feature extraction; lightweight architecture design principles; regularization and optimization techniques; and the datasets employed throughout this study. The critical comparative positioning of the proposed architecture relative to existing methods, and the specific research gaps it addresses, were presented in Section 1.2.

2.2 Brain Tumor Taxonomy, Epidemiology, and Diagnostic Imaging

Brain tumors constitute one of the most challenging diagnostic problems in contemporary neuro-oncology, with a worldwide incidence that continues to rise (Athwal et al., 2021). Primary brain and CNS tumors account for over 300,000 new cases annually, with disproportionately high mortality relative to other oncological diagnoses a consequence of their anatomical location, resistance to systemic therapy, and the critical neurological functions at risk (Sung et al., 2021). In children and young adults under 40, brain tumors represent the second leading cause of cancer-related death, underscoring their public health significance beyond raw incidence figures (Ostrom et al., 2022).

The four-class taxonomy adopted in this study corresponds to the dominant clinical intracranial neoplasm subtypes (Athwal et al., 2021; Sivanantham et al., 2021). Gliomas (~45% of primary brain tumors) span a histological spectrum from WHO Grade I pilocytic astrocytoma's with excellent prognosis to WHO Grade IV glioblastoma multiforme, the most aggressive primary brain cancer, with a median survival of approximately 14–16 months despite aggressive multimodal therapy including surgical resection, radiotherapy, and chemotherapy (Athwal et al., 2021; Louis et al., 2021). This grading dichotomy makes precise classification directly consequential for treatment selection. Meningiomas (~36% of cases) are predominantly well-differentiated dural-based tumors with generally favorable prognosis following surgical resection, though atypical and anaplastic variants exhibit substantially higher recurrence rates (Ostrom et al., 2022). Pituitary adenomas are clinically significant for the endocrine dysfunction and visual field deficits they produce via optic chiasm compression, with treatment selection critically dependent on accurate subtype identification (Melmed, 2020). Normal brain tissue provides the essential negative control category, whose reliable discrimination from pathological tissue is a prerequisite for any clinically deployable classification system (Ismail et al., 2022).

Figure 2.1 (Ilani et al., 2025) shows the most frequently occurring anatomical sites of origin of these tumor types: gliomas (oligodendrogliomas, astrocytomas, brainstem gliomas) are distributed throughout both cerebral hemispheres and the brainstem; meningiomas originate from the meningeal surface; pituitary adenomas are strictly confined to the sellar region. This variety in originating location, together with the morphological variation within individual classes, directly motivates the multi-class classification approach adopted in this thesis.

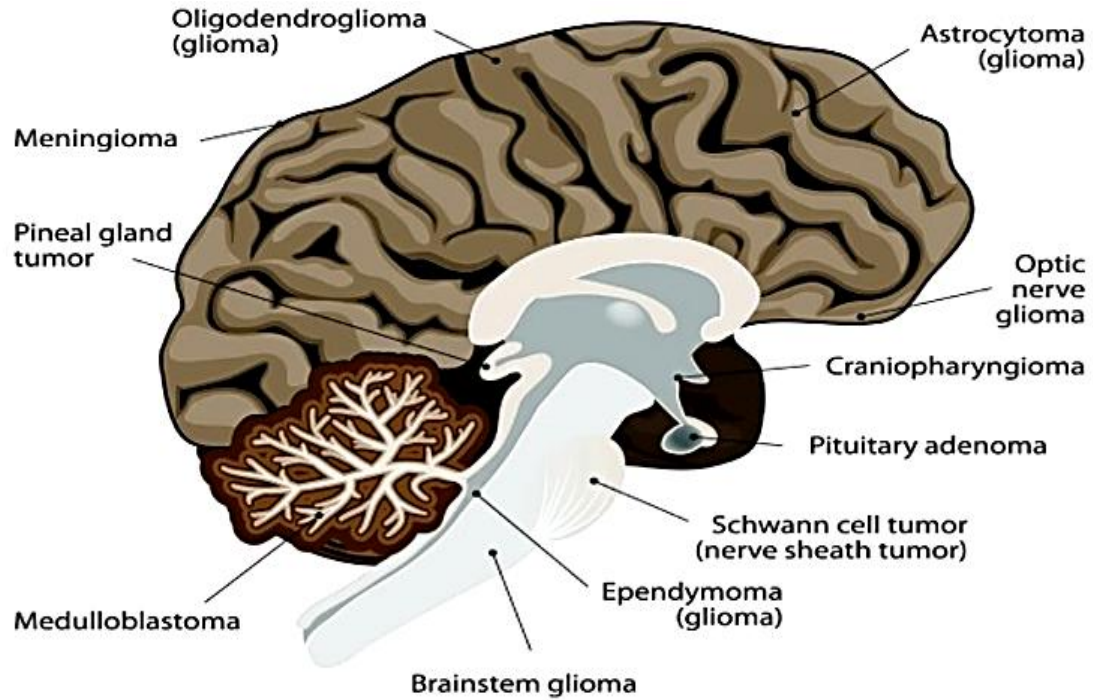


Figure 2.1 - Common anatomical locations of glioma, meningioma, and pituitary adenoma within the brain (Ilani et al., 2025)

MRI has remained the modality of choice for evaluating brain tumors owing to its superb soft-tissue contrast resolution, its non-ionizing nature, and its capacity to image in all three planes (Işın et al., 2016; Ismail et al., 2022). **Figure 2.2** (Ilani et al., 2025) illustrates the categories of medical imaging modalities used in clinical practice for brain tumor diagnosis Computed Tomography (CT), Magnetic Resonance Imaging (MRI), Magnetic Resonance Spectroscopy (MRS), Positron Emission Tomography (PET), Single-Photon Emission Computed Tomography (SPECT), and lumbar puncture analysis — with MRI predominant among them. This study focuses specifically on MRI, using contrast-enhanced T1-weighted (CE-T1) and T2-weighted (CE-T2) sequences, both of which are represented in the dataset used for this project.

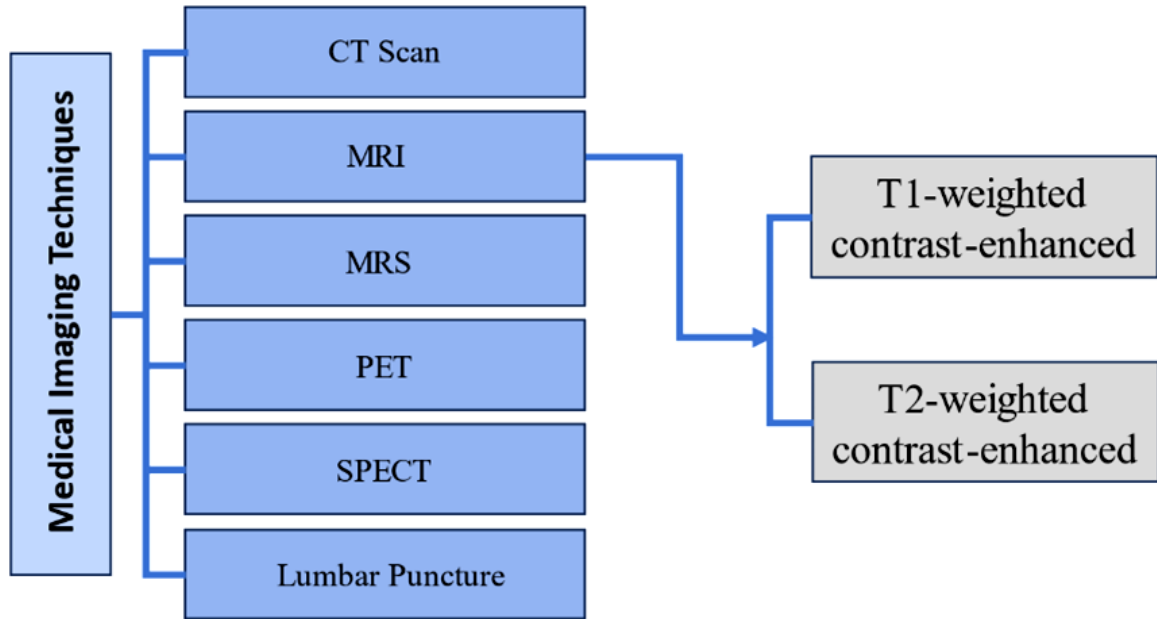


Figure 2.2 - Categories of medical imaging modalities for brain tumor diagnosis in clinic (Ilani et al., 2025).

Despite MRI's diagnostic strengths, interpretation of its complex multi-parametric signal signatures still inherently requires specialist expertise and is subject to well-documented systematic limitations. Clinical studies have consistently reported substantial inter-observer variability among radiologists interpreting brain MRI scans (Çinar and Yildirim, 2020; Ismail et al., 2022), particularly in cases where definitive separation between pathology types is difficult for example, distinguishing high-grade glioma from brain metastasis, or dural-based meningioma from glioma with dural involvement. Even within a single radiologist's interpretations of identical, non-sequential cases, intra-observer error rates as high as 5–15% have been documented on complex neuro-oncology diagnoses, stemming from inherent human perceptual limits rather than lack of training (Çinar and Yildirim, 2020). Misdiagnosis, whether human or machine-derived, may result in inappropriate treatment selection, delayed therapy, and unnecessary exposure to damaging chemotherapy (Athwal et al.,

2021; Khan et al., 2025). This risk is compounded by a global shortage of neuroradiological subspecialty expertise, with the majority of MRI scanners in low- and middle-income countries lacking access to expert human interpretation (Mollura et al., 2012). Automated deep learning-based classification systems, with their inherent objectivity and reproducibility, have significant potential to mitigate these risks and extend advanced AI-driven diagnosis to patients worldwide (Deepak and Ameer, 2019; Shen et al., 2017).

2.3 Foundations of Deep Learning for Image Analysis

The evolution of image-analysis pipelines can be traced through three related waves of architectural design. The first wave, convolutional neural networks, demonstrated that discriminative features could be learned automatically and hierarchically from raw pixel data, overcoming the need for hand-engineered features. The next wave, residual and dense connectivity structures, addressed the vanishing-gradient problem and improved feature reuse in deeper networks. This progression then extended into segmentation-oriented architectures, which prioritize the preservation of fine spatial detail critical to tasks such as tumor boundary delineation, typically at the cost of substantial model size.

2.3.1 Early CNN Architectures

Simonyan et al. (2014) demonstrated that CNNs can automatically learn discriminative, hierarchical features from raw pixel data, eliminating the need for manual feature engineering. LeCun et al. (2015) provided a foundational account of deep learning's broader significance. Litjens et al. (2017) and Shen et al. (2017), in independent survey works, concluded that deep learning models consistently outperform classical machine learning approaches on medical imaging datasets. Albawi et al. (2017) presented the theoretical basis and

operational mechanics of CNNs. Deepak and Ameer (2019) demonstrated strong brain tumor classification performance using transfer learning on pretrained CNN backbones.

2.3.2 Residual and Dense Networks

(He et al., 2016) introduced residual learning via skip connections in ResNet, directly addressing the vanishing-gradient problem in deep networks. (Xie et al., 2017) extended this with aggregated residual transformations. (Huang et al., 2017) proposed DenseNet, employing dense connectivity to improve feature reuse across layers. (Çinar and Yildirim, 2020) demonstrated that hybrid CNN designs can improve sensitivity to small brain tumors, a capability that inherently demands multi-scale representational capacity.

2.3.3 Segmentation-Oriented Architectures

(Long et al., 2015) developed the first fully convolutional network (FCN)-based architecture for image segmentation. (Ronneberger et al., 2015) subsequently introduced U-Net, which retains fine spatial detail through skip connections between encoder and decoder pathways. U-Net remains a foundational reference architecture for medical image segmentation and classification tasks alike, though its substantial parameter count (~31 million) motivates the pursuit of more lightweight alternatives, as pursued in this thesis.

2.4 Attention Mechanisms

Attention mechanisms allow a model to concentrate representational capacity on the most informative features rather than treating all spatial and channel information uniformly. This is particularly valuable in medical imaging, where attending preferentially to lesions and clinically relevant tissue regions aligns

more closely with radiological practice than uniform whole-image processing. This section describes the principal branches of attention mechanism design that inform the architecture proposed in this thesis, from channel-wise recalibration through combined spatial-channel mechanisms to self-attention in Transformers, and their interpretation via gradient-based visualization.

2.4.1 Channel Attention - Squeeze-and-Excitation Networks

(Hu et al., 2018) proposed Squeeze-and-Excitation (SE) networks, which rescale channel-wise feature responses using global average pooling-based descriptors. (Roy et al., 2019) subsequently adapted this mechanism for medical imaging through spatial and channel squeeze-and-excitation blocks within fully convolutional networks. SE-type mechanisms, however, rescale channel relationships without explicitly capturing spatial correlations.

2.4.2 Spatial and Combined Attention – CBAM

(Woo et al., 2018) developed the Convolutional Block Attention Module (CBAM), combining channel and spatial attention in a sequential, channel-first pipeline, in which spatial attention is computed only after channel attention has already been applied. This sequential dependency between the two attention branches has motivated alternative orderings explored in subsequent lightweight architectural designs (see Section 3.4.1). (Gu et al., 2019) demonstrated that context-encoder networks employing attention-like mechanisms can achieve superior performance on 2D medical image segmentation tasks.

2.4.3 Self-Attention and Transformer Architectures

(Vaswani et al., 2017) introduced self-attention for modeling long-range dependencies in the original Transformer architecture. (Tummala et al., 2022) subsequently adapted Vision Transformers for brain tumor classification, though the quadratic scaling of self-attention with input size results in substantially higher computational cost relative to CNN-based counterparts.

2.4.4 Grad-CAM for Visual Explainability

(Selvaraju et al., 2017) introduced Gradient-weighted Class Activation Mapping (Grad-CAM), which produces class-discriminative localization maps by computing a weighted sum of forward activation maps using gradients specific to a target class. Grad-CAM has become a cornerstone technique for assessing the diagnostic plausibility of deep learning models in medical imaging, and is employed in this thesis (Section 4.15) to verify that the proposed model attends to clinically relevant tumor regions.

2.5 Multi-Scale Feature Extraction

(Lin et al., 2017) formulated Feature Pyramid Networks (FPN), employing a top-down pathway within a full encoder-decoder structure to produce multi-scale feature representations. (Fanai and Abbasimehr, 2023) demonstrated that multi-scale processing improves classification accuracy on datasets exhibiting substantial morphological diversity. These findings motivate the multi-scale design pursued in this thesis (Section 3.4.2), which seeks to achieve comparable representational benefits without requiring a full encoder-decoder architecture.

2.6 Lightweight Architecture Design

(Howard et al., 2017) introduced MobileNet, employing depthwise separable convolutions to substantially reduce parameter count relative to standard convolutions. (Sandler et al., 2018) advanced this design with MobileNetV2's inverted residual blocks. (Tan and Le, 2019) proposed EfficientNet, achieving strong accuracy-efficiency trade-offs through compound scaling of network depth, width, and resolution. (Liu et al., 2017b) demonstrated network slimming as a post-training model compression strategy. These lightweight design principles collectively inform the parameter-efficient architecture proposed in this thesis, though, as discussed in Section 1.2, even the most compact contemporary variants substantially exceed the sub-700K parameter budget targeted here.

2.7 Regularization and Optimisation Techniques

(Ioffe and Szegedy, 2015) introduced batch normalization, accelerating training convergence while providing implicit regularization through the reduction of internal covariate shift. (Srivastava et al., 2014) established dropout as a simple and effective technique for preventing co-adaptation of feature detectors. (Gal and Ghahramani, 2016) provided a Bayesian interpretation of dropout, motivating its use for uncertainty quantification in deep learning models. (Kingma and Ba (2015) proposed the Adam optimizer, offering robust convergence properties with minimal hyperparameter sensitivity. (Loshchilov and Hutter, 2017) introduced cosine annealing with warm restarts, improving final convergence quality by smoothly decaying the learning rate according to a cosine schedule. These techniques collectively underpin the training methodology detailed in Section 3.5.

2.8 Datasets for Brain Tumor Classification

This study employs the Mendeley Brain Tumor MRI dataset (T1-weighted) (Cheng et al., 2015), comprising 12,064 images across four unbalanced classes, as the primary dataset. Three additional independent datasets Kaggle (Nickparvar, 2021), Br35H (Nazir et al., 2024), and Figshare (Cheng, 2017) are employed for cross-dataset generalization evaluation, simulating genuine domain shift consistent with recommended practice in the medical imaging literature (Litjens et al., 2017; Shen et al., 2017). **Figure 2.3** (Ilani et al., 2025) presents representative T1- and T2-weighted MRI images for meningioma, glioma, and pituitary adenoma, illustrating both inter-class and intra-class morphological diversity. Visual inspection confirms substantial morphological differences between tumor classes: glioma typically presents as an infiltrative lesion with ill-defined boundaries and variable signal intensity as it invades adjacent brain parenchyma; meningioma arises from the meningeal layers and typically appears as a well-defined extra-axial mass with uniform contrast enhancement; and pituitary adenoma presents as a localized sellar lesion with distinctive T1/T2 signal characteristics, often extending into the suprasellar cistern. Considerable intra-class variation is also present — for instance, gliomas range in histological grade from low- to high-grade, with correspondingly divergent imaging appearances. This inter- and intra-class diversity renders traditional handcrafted-feature or shallow-learning approaches inadequate, motivating the multi-scale, multi-path feature extraction strategy (MMSP) proposed in this thesis (Section 3.4.2).

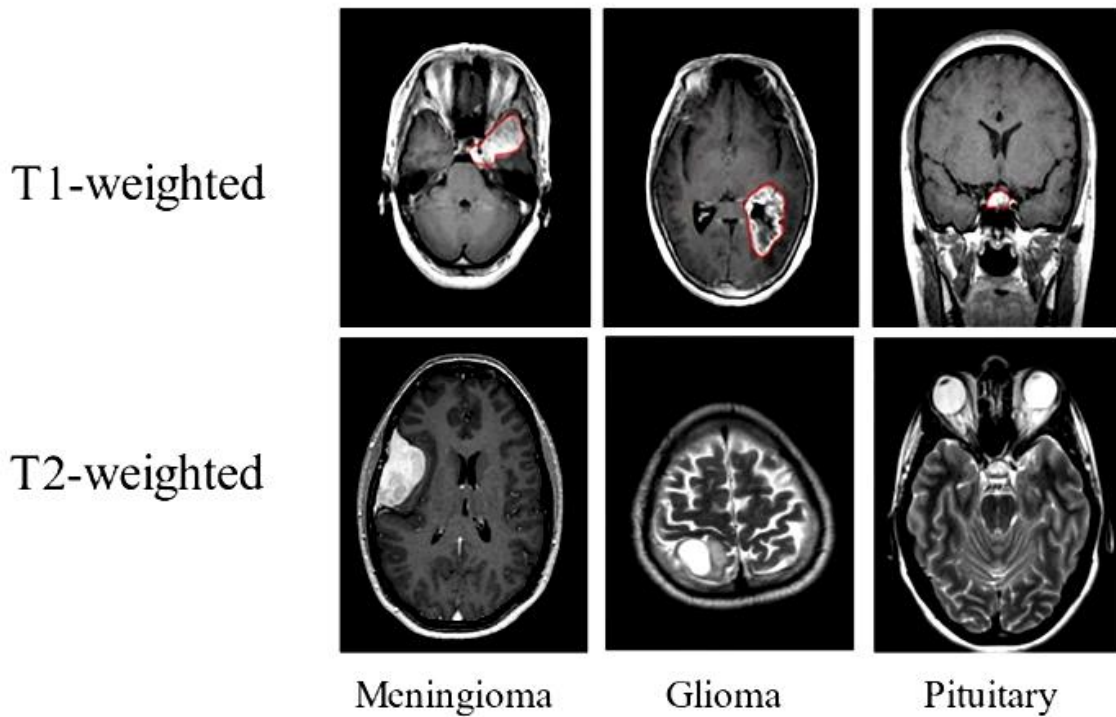


Figure 2.3 - Representative T1- and T2-weighted MRI images of meningioma, glioma, and pituitary adenoma (Ilani et al., 2025)

Figure 2.4 (Ilani et al., 2025) illustrates a representative class-distribution imbalance reported in a related published study (Ilani et al., 2025). It should be noted that this distribution differs from the actual class balance observed in the Mendeley dataset used in the present study (see Table 4.2, Chapter Four), where glioma is in fact the largest class (25.02%) rather than the smallest. The figure is retained here solely to illustrate the general phenomenon of class imbalance commonly encountered in brain tumor MRI datasets, which motivates the class-balanced augmentation strategy described in Section 3.3.3.

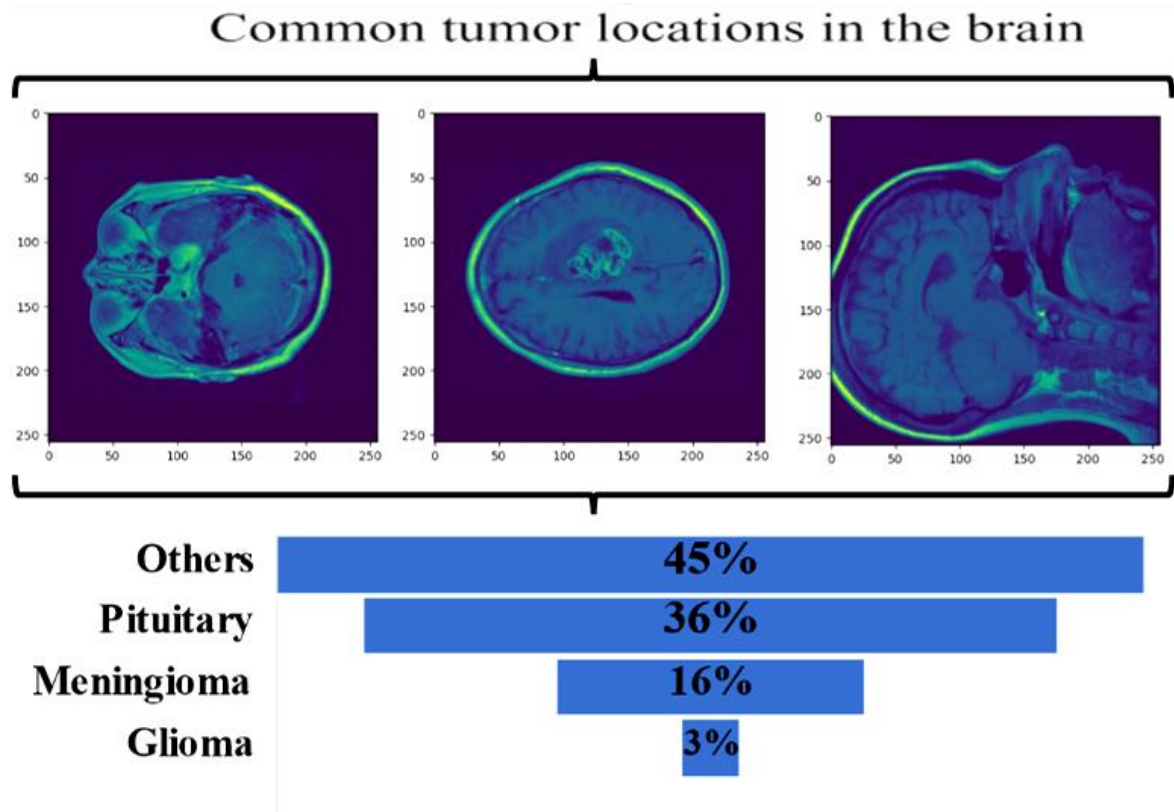


Figure 2.4 - Distribution of brain tumor classes in the study by (Ilani et al., 2025)

2.9 Mathematical Formulations

This section consolidates the complete set of mathematical formulations referenced throughout this thesis: the formal definitions of the three proposed architectural modules (MAFE, MMSP, and CAIP), detailed in Chapter Three, and the standard performance evaluation metrics used to assess the proposed model in Chapter Four. Presenting these formulations together provides a single, unified mathematical reference, while the design rationale, architectural justification, and experimental application of each formulation are discussed in their respective chapters.

2.9.1 Proposed Module Formulations

The following formulations define the three modules proposed in this thesis MAFE, MMSP, and CAIP whose architectural design, motivation, and distinction from prior work are discussed in detail in Chapter Three.

2.9.1.1 Micro Adaptive Feature Extractor (MAFE)

- Input: $\mathbf{x} \in \mathbb{R}^{(B \times C \times H \times W)}$ (2.1)

where the variables denote: $\mathbf{x}$ is the input feature map, $\mathbf{b}$ is the batch size, $\mathbf{c}$ is the number of channels, $\mathbf{h}$ and $\mathbf{w}$ are the height and width of the feature map respectively, and $\mathbb{R}$ denotes the set of real numbers to which the input values belong.

- Spatial Attention:

$$\mathbf{W}_s = \sigma(\text{Conv}_{1 \times 1}(\mathbf{x})) \quad (2.2)$$

- Spatially-Refined Feature:

$$\mathbf{x}' = \mathbf{x} \otimes \mathbf{W}_s \quad (2.3)$$

- Channel Attention:

$$\mathbf{W}_c = \sigma\left(\text{Conv}_{1 \times 1}\left(\text{ReLU}\left(\text{Conv}_{1 \times 1}(\text{GAP}(\mathbf{x}'))\right)\right)\right) \quad (2.4)$$

- Output: $\mathbf{y} = \mathbf{x} + \alpha \cdot (\mathbf{x}' \otimes \mathbf{W}_c)$ (2.5)

where σ denotes sigmoid activation, GAP is Global Average Pooling, $\otimes$ is element-wise multiplication, and α is a learnable scalar initialized to 0.3 (Loshchilov and Hutter, 2017). Total parameter overhead: $\sim 4,956$ per instance (586 at Stage 1 + 4,370 at Stage 3, per Table 4.5). The reduction ratio r is configured per-stage to balance representational capacity against parameter budget; the rationale for this per-stage configuration is discussed in Section 3.2.2.

2.9.1.2 Micro Multi-Scale Processor (MMSP)

MMSP implements three parallel processing pathways (Lin et al., 2017; Howard et al., 2017) emulating multi-level magnification in diagnostic analysis:

Rapid Context Path (Scale 1):

$$S_1 = \text{ReLU}(\text{Conv}_{1 \times 1}(x)) \quad (2.6)$$

Local Pattern Path (Scale 2):

$$S_2 = \text{ReLU}(\text{Conv}_{3 \times 3}(x)) \quad (2.7)$$

Efficient Multi-Scale Path (Scale 3):

$$S_3 = \text{ReLU}(\text{DWConv}_{3 \times 3}(x)) \quad (2.8)$$

Feature Fusion:

$$y = \text{ReLU}\left(x + \beta \cdot \text{Conv}_{1 \times 1}(\text{Concat}(S_1, S_2, S_3))\right) \quad (2.9)$$

2.9.1.3 Context-Aware Intelligent Pooling (CAIP)

CAIP derives its blending weights from a global summary of the feature map rather than a per-pixel context map, so its adaptivity operates at the level of the whole feature map for a given sample, rather than varying across spatial locations within a single feature map:

- Context Analysis:

$$[w_1, w_2] = \text{Softmax} \left(\text{Conv}_{1 \times 1} \left(\text{ReLU} \left(\text{Conv}_{1 \times 1} (\text{GAP}(x)) \right) \right) \right) \quad (2.10)$$

- Dual Pooling:

$$y_{max} = \text{MaxPool}_{2 \times 2}(x) \quad (2.11)$$

$$y_{avg} = \text{AvgPool}_{2 \times 2}(x) \quad (2.12)$$

- Adaptive Combination:

$$\text{Output } y = w_1 \cdot y_{max} + w_2 \cdot y_{avg} \quad (2.13)$$

where w_1 and w_2 are input-dependent, softmax-normalized weights derived from a global summary of the feature map (**Eq. 2.10**), constrained to sum to 1.

2.9.2 Evaluation Metric Formulations

When evaluating the current classification model, often rely on a set of performance metrics to describe the model performance rather than a single measure, which provide us with varied viewpoints. The classification metrics

can be constructed on basis of the four major values of classification namely - True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), and researchers look at a combination of these values to obtain a more sensible perspective over this model's characteristics including efficiency, reliability and strength with emphasis in handling unbalanced dataset.

Accuracy measures the overall correctness of the model as a percentage, calculated by dividing the number of correctly classified samples by the total number of samples. It is a useful overall indicator of performance; however, it can be misleading in imbalanced datasets.

These standard classification metrics (Fawcett, 2006) are computed as follows:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \times 100 \quad (2.14)$$

Precision evaluates how many of the predicted positive samples are actually correct, focusing on the quality of positive predictions and reducing false positives.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (2.15)$$

Recall tests the ability of the model to pick out all true positive examples as much as possible that is to say, minimizing false negatives.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (2.16)$$

F1-score is the harmonic mean of Precision and Recall, and it provides a balanced measure among them, especially when there is an uneven class distribution.

$$F1 = \frac{2PR}{P+R} \quad (2.17)$$

Efficiency To capture the tradeoff between model performance and architectural complexity, this thesis adopts a parameter-efficiency metric commonly used in the lightweight-architecture literature (e.g., Khan and Auvée, 2024; Sandler et al., 2018), defined as.

$$Efficiency = \frac{Accuracy\%}{Parameters(M)} \quad (2.18)$$

Higher values indicate a more favorable accuracy-to-complexity ratio. This formulation is applied here to enable direct comparison across architectures of vastly different scale (from ~659K to ~138M parameters), a scale disparity not typically addressed within a single comparative table in prior brain tumor classification studies.

AUC (One-vs-Rest) (Fawcett, 2006) measures the model's ability to distinguish between different classes. It is calculated for each class against all other classes and then averaged, where higher values indicate better separability between classes.

$$AUC_{OvR} = \frac{1}{K} \sum_{k=1}^K AUC_k \quad (2.19)$$

This evaluation is also taking computational and clinical efficiency into account to achieve the requirement of practical applications, such as parameter

efficiency, inference latency, model size and memory efficiency. Experiments were all executed in and coded by (PyTorch 2.1.0).

2.10 Summary

This chapter established the clinical and theoretical foundations underpinning the methodological framework presented in Chapter Three: brain tumor taxonomy, epidemiology, and diagnostic imaging (Section 2.2); the historical progression of convolutional architectures (2.3); the principal families of attention mechanisms (2.4); multi-scale feature extraction strategies (2.5); lightweight architecture design principles (2.6); standard regularization and optimization techniques (2.7); the primary and external datasets employed throughout this study (2.8); and the complete set of mathematical formulations for the proposed modules and evaluation metrics referenced throughout the thesis (2.9). The critical comparative positioning of the proposed architecture against existing brain tumor classification methods, together with the specific research gaps that motivate its design, were presented in Section 1.2 and are addressed directly through the methodology detailed in Chapter Three.

Chapter

Three

Methodology

Chapter 3

Methodology

3.1 Overview

This chapter presents the complete methodological framework for the proposed deep learning approach to multi-class brain tumor classification. Four sequential phase's structure the research: (1) dataset preparation and preprocessing; (2) modified CNN architecture design; (3) training and optimization; and (4) evaluation and validation. The theoretical justification for each of the three architectural modules is presented in detail, building on the mathematical formulations introduced in Section 2.9.

3.2 Proposed Framework Architecture

The proposed framework integrates MAFE, MMSP, and CAIP within four sequential convolutional stages totaling 659,228 parameters. Each module addresses one specific limitation: MAFE addresses generic feature extraction; MMSP addresses single-scale representation; and CAIP addresses static pooling through a context-adaptive mechanism proposed in this thesis (Section 3.4.3). The key architectural innovation is the co-design of all three modules within a strict parametric budget targeted at brain tumor MRI classification requirements, see **Figure 3.1** presents the complete block diagram of the proposed ultra-lightweight CNN architecture.

Ultra-Lightweight CNN Architecture

Brain Tumor MRI Classification - MAFE + MMSP + CAIP

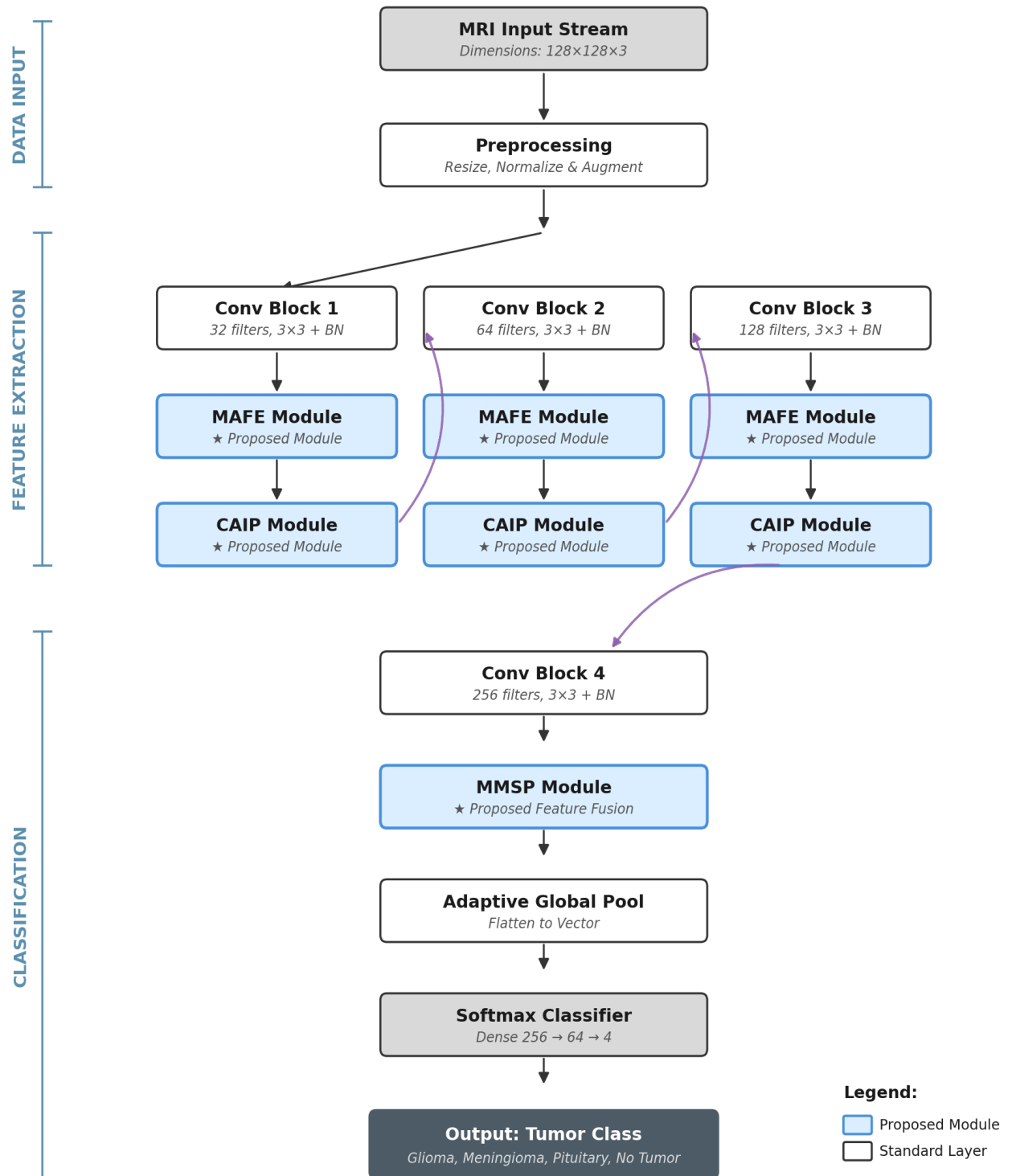


Figure 3.1 - Block diagram of the proposed ultra-lightweight CNN architecture

3.2.1 Phase 1: Dataset Preparation and Preprocessing

Start off with the Dataset Preparation step since all of this experimental results in later chapters are dependent on a well-prepared empirical base, So, it is best to deal with that first. We will walk through the three sequential sub-steps of Data Preparation, as shown in **Figure 3.2**; (1) Get the main MRI dataset and validate for corrupted or duplicated image files to ensure no false positives will be seen from experimentation, (2) Apply spatial and intensity normalization which should level the image size, intensity characteristics of each image, and (3) Develop a bespoke data augmentative pipeline which allows to increase variety in training set but no modification over test sets.

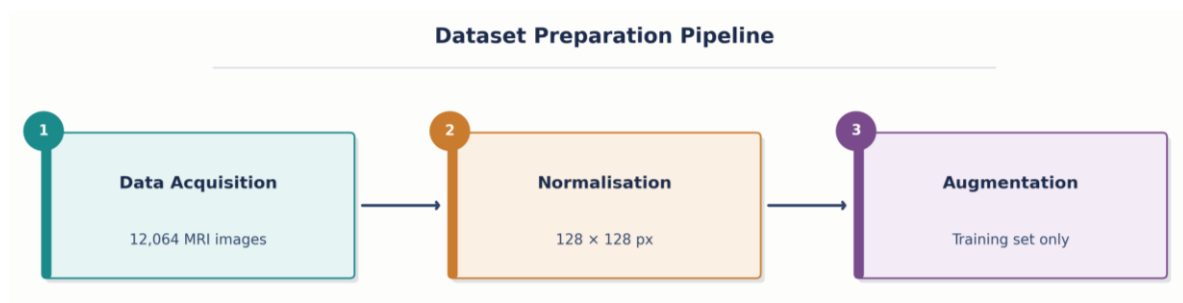


Figure 3.2 - Dataset Preparation Pipeline

3.2.1.1 Data Acquisition and Integrity Verification

The primary dataset that was utilized is the Mendeley Brain Tumor MRI Repository v4; it contains 12,064 pre-processed T1-weighted brain MRI images across four diagnostic classes. Following the 80/20 stratified train-test split described in Section 4.2.2, the train-and-validation subset (9,650 images, detailed in Table 4.2) is composed of 3,018 glioma, 2,183 meningioma, 2,504 pituitary, and 1,945 no-tumor images, exhibiting a moderate class imbalance (maximum-to-minimum ratio of 1.55:1).

Dataset integrity check has been carried out prior to every experiment using MD5 cryptographic hashing of the files - presented in the bottom half of **Figure 3.3**, which indicates 0 duplicated image (0.00% duplication rate) and 100.00% valid pixels in all 9,650 images of the train-and-validation set.

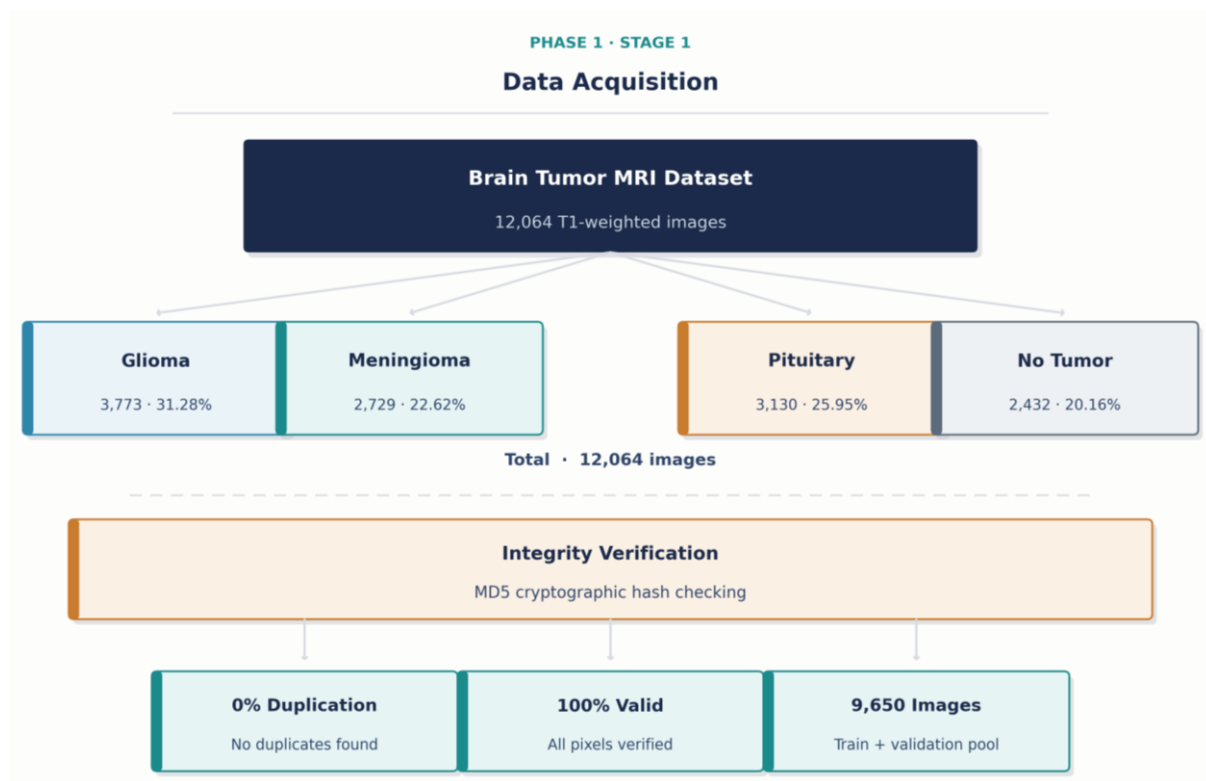


Figure 3.3 - Data Acquisition

3.2.1.2 Spatial and Intensity Normalization

Thus, the two-stage normalization process applied to all images. Panel A depicts the spatial normalization step, in which all images originally of variable size were resampled to a uniform 128×128 pixels using bilinear interpolation. Panel B depicts the intensity normalization step, which employed ImageNet pre-training statistics, see **Figure 3.4** illustrates all that applied channel-wise using the formula $x_{\text{norm}} = (x - \mu) / \sigma$, with per-channel parameters: Red ($\mu=0.485$,

$\sigma=0.229$); Green ($\mu=0.456$, $\sigma=0.224$); Blue ($\mu=0.406$, $\sigma=0.225$). This yields zero-mean, unit-variance normalized inputs across the dataset.

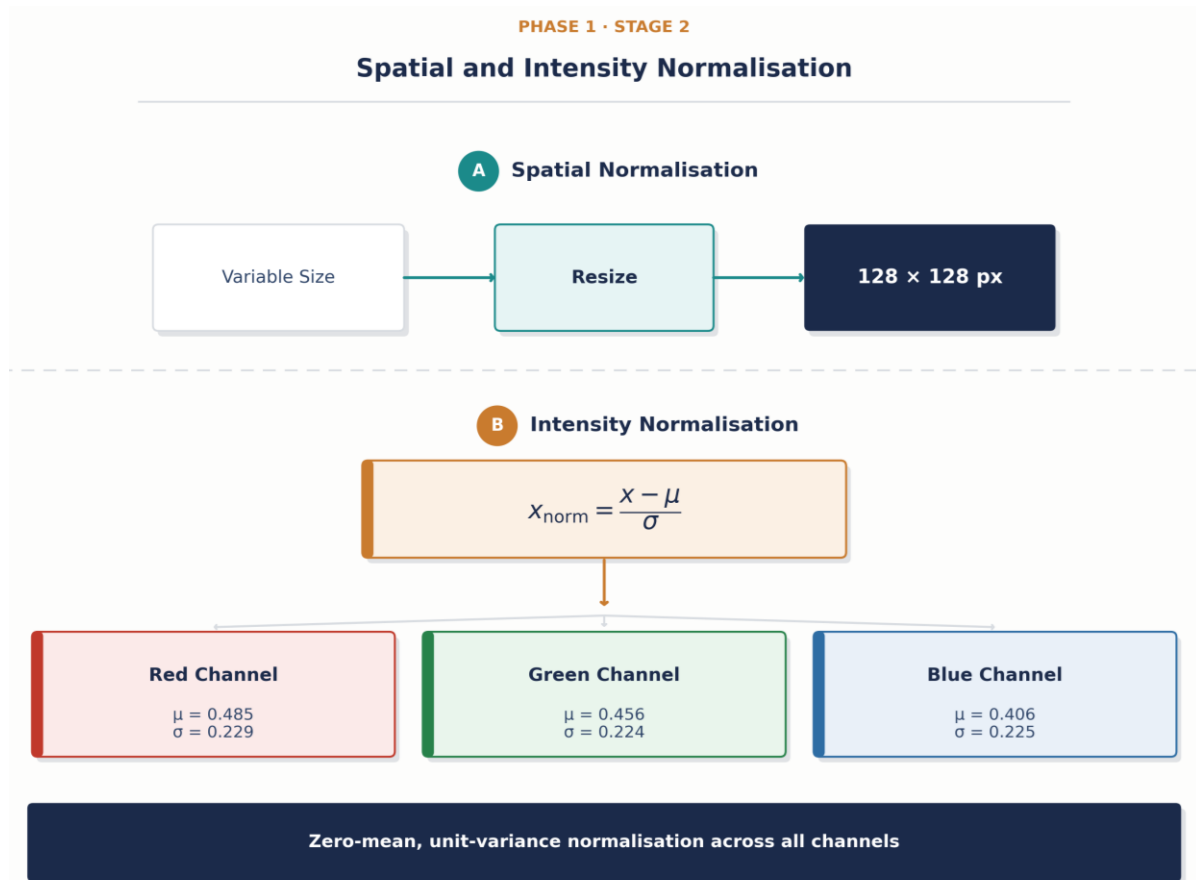


Figure 3.4 - Spatial and Intensity Normalization

3.2.1.3 Data Augmentation Strategy

Figure 3.5 shows the augmentation strategy in which the upper part of the figure explains the dataset split, where augmentation was applied only to the training set (9,650 images), while the validation and test sets were left unchanged. The lower part of **Figure 3.5** lists the augmentation operations used in training, including random horizontal flip ($p=0.40$), vertical flip ($p=0.30$), rotation up to $\pm 20^\circ$, colour jittering for brightness and contrast (0.30) and saturation (0.20), Gaussian blur with a 3×3 kernel, and elastic deformation

($p=0.20$). These transformations were chosen because they can reasonably reflect the natural variations that occur when acquiring MRI images and do not break the overall anatomy of the organs.

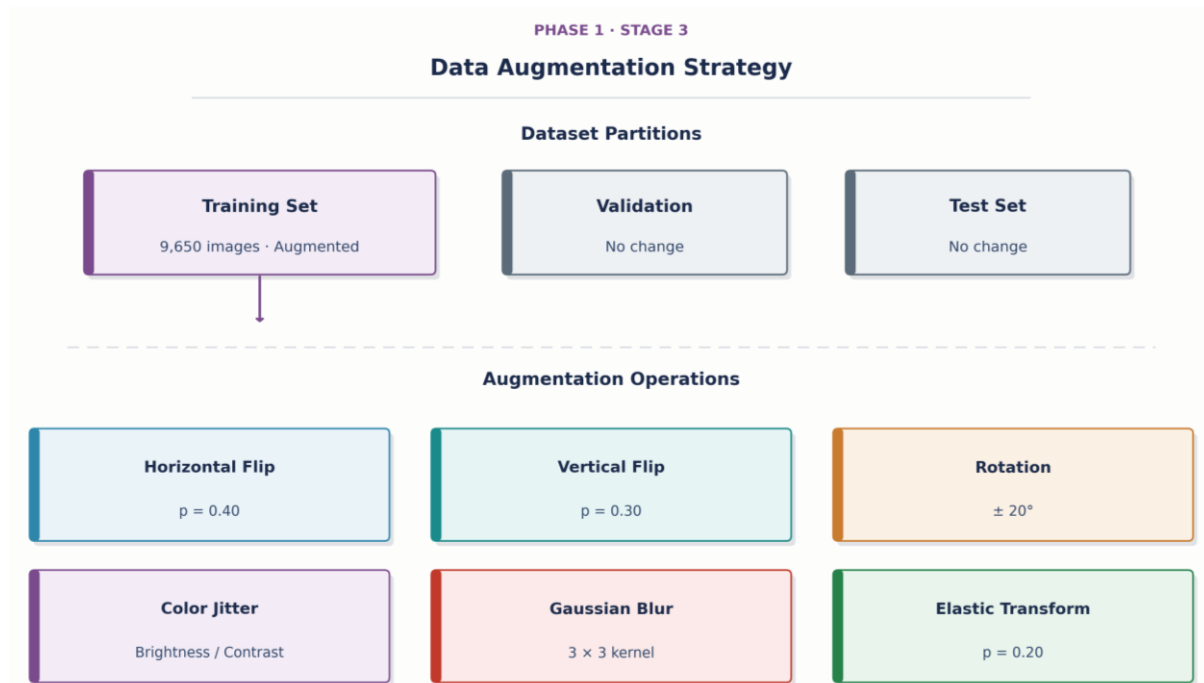


Figure 3.5 - Data Augmentation Strategy (Applied to Training set only)

3.2.1.4 Complete Phase 1

The three stages of Phase 1 are consolidated into a single unified diagram in **Figure 3.6**, providing a complete overview of the dataset preparation pipeline. The top panel (**Stage 1**) reports the class composition alongside the MD5 verification result. The middle panel (**Stage 2**) presents the spatial (128×128 px) and intensity normalization parameters for each color channel, shown side by side. The bottom panel (**Stage 3**) shows the six augmentation operations applied exclusively to the training set, with the validation and test sets explicitly labelled as unmodified. A sequential legend line at the bottom (Acquisition →

Normalization → Augmentation) confirms the strict, non-overlapping order of execution across the pipeline.



Figure 3.6 - Phase 1 Complete Dataset Preparation

Figure 3.6 consolidate all three dataset preparation stages into a single unified pipeline overview, Stage 1 confirms the acquisition of 12,064 brain tumor MRI images across four classes: Glioma (3,773), Meningioma (2,729), Pituitary (3,130), and No Tumor (2,432), with MD5 hashing verifying zero duplicate files, Stage 2 describes the normalization parameters applied uniformly to all

images: spatial resizing to 128×128 pixels, and per-channel intensity normalization with mean (μ) and standard deviation (σ) values recorded for the Red, Green, and Blue channels respectively, Stage 3 applies six data augmentation operations Horizontal Flip ($p=0.40$), Vertical Flip ($p=0.30$), Rotation ($\pm 20^\circ$), Color Jitter, Gaussian Blur (3×3), and Elastic Transform ($p=0.20$) exclusively to the 9,650 training-set images, while the validation and test sets remain unmodified, the sequential legend at the bottom (Acquisition $\rightarrow$ Normalization $\rightarrow$ Augmentation) reinforces the strict, non-overlapping order of execution throughout the pipeline.

3.2.2 Phase 2: Modified CNN Architecture Design

Micro Adaptive Feature Extractor (MAFE) - this thesis reverses CBAM's channel-first sequential ordering, applying spatial attention before channel attention, motivated by the hypothesis that spatial region selection improves the reliability of subsequent channel recalibration for tumour boundary discrimination; Micro Multi-Scale Processor (MMSP) - mimicking multi-magnification diagnostic analysis, a total of three receptive field widths are generated and a depth-wise separable architecture (to cope with the strict parameter budget) are employed at each stage to generate different feature scales; Context-Aware Intelligent Pooling (CAIP) - learning a weighted combination of max pooling and average pooling to retain peak activations, as well as preserving distribution context, in the place of generic fixed pooling operations; these three modules are then stacked in an architectural design and composition intended to provide maximum diagnostic representational power, while remaining computationally lightweight.

3.2.2.1 Micro Adaptive Feature Extractor (MAFE)

MAFE employs a sequential spatial-then-channel attention mechanism, proposed in this thesis. The core architectural distinction from CBAM (Woo et al., 2018) lies in the order of operations: CBAM applies channel attention first, then computes spatial attention on the channel-refined output. In contrast, MAFE applies spatial attention (**Equation 2.2**) to the input x to obtain a spatially-refined feature map x' (**Equation 2.3**); channel attention (**Equation 2.4**) is then computed from this spatially-refined representation rather than from the original input. This ordering spatial refinement preceding channel recalibration differs from CBAM's channel-then-spatial pipeline (Woo et al., 2018) in the order of operations, though both remain sequential rather than parallel formulations. The design choice to refine spatial saliency before channel recalibration is motivated by the intuition that channel-wise importance for tumour discrimination is more reliably estimated once irrelevant spatial regions have already been suppressed. The spatially-refined feature map is then re-weighted by the channel attention map in a final fused gating step (**Equation 2.5**). This reversed sequential topology is the central architectural distinction of MAFE relative to CBAM's channel-then-spatial design and to SE-Networks' (Hu et al., 2018) channel-only design.

The choice of r is scaled with the channel width c to balance representational capacity against computational cost. At Stage 1 ($c=32$), a smaller $r=4$ preserves sufficient representational detail in the intermediate dimension (8 units), since the features here are still low-level and channels are limited. At Stage 3 ($c=128$), a larger $r=8$ keeps the parameter overhead controlled while still yielding an adequate intermediate dimension (16 units), as the features at this deeper stage are more abstract and less sensitive to fine-grained channel-wise detail. This

adaptive scaling of r with c is consistent with the design principle proposed by Hu et al. (2018) in SE-Networks.

3.2.2.2 Micro Multi-Scale Processor (MMSP)

MMSP implements three parallel processing pathways emulating multi-level magnification in diagnostic analysis. The depthwise separable convolution in Scale 3 reduces parameter count by a factor equal to the number of input channels versus standard convolution, enabling multi-scale processing within the parametric budget. The three parallel pathways of MMSP simultaneously preserve broad structural context, localized texture patterns, and computationally efficient multi-scale features within a unified architecture. The feature fusion step (**Equation 2.9**) integrates the complementary representations learned across all three scales, producing a richer and more discriminative combined feature map than any single-scale pathway could achieve independently. This design, grounded in depthwise separable convolutions and multi-magnification principles, directly addresses the challenge of brain tumor heterogeneity by enabling the network to analyse MRI scans at multiple levels of granularity while remaining within practical parametric constraints.

3.2.2.3 Context-Aware Intelligent Pooling (CAIP)

CAIP learns globally optimal, input-dependent blending coefficients between maximum and average pooling. A lightweight decision network (**Equation 2.10**) derives two softmax-normalized weights from a global summary of the input feature map (via Global Average Pooling), rather than from a per-pixel spatial context map. These weights (w_1 , w_2) then blend the standard max-pooled and average-pooled outputs (**Equations 2.11–2.12**) on a per-sample,

per-stage basis. Unlike a fixed 50/50 blend or a static pooling ratio (Lee et al., 2016), CAIP allows the network to learn, for each input image and each stage, how much to favour boundary-preserving max pooling versus noise-suppressing average pooling, based on the overall content of that feature map. By learning these weights rather than relying on a fixed 50/50 pooling ratio, CAIP performs dynamic, per-sample balancing between the peak-activation sensitivity of max pooling and the noise-robustness of average pooling. This formulation allows the model to adjust its overall pooling behavior across different tumor classes and imaging conditions, improving representational robustness without a significant increase in the number of learnable parameters. As noted above, this adaptivity operates at the level of the whole feature map for a given sample and stage, rather than varying across spatial locations within a single feature map.

3.2.2.4 Complete Architecture Integration

MAFE is positioned after Stages 1 and 3; MMSP after Stages 2 and 4; CAIP replaces standard pooling after Stages 1, 2, and 3. Stage 4 output feeds the classification head: Global Average Pooling $\rightarrow$ Flatten $\rightarrow$ Dropout ($p=0.3$) $\rightarrow$ Dense (64 units, ReLU) $\rightarrow$ Dropout ($p=0.2$) $\rightarrow$ Dense (4 units, Softmax). A detailed distribution of the parameters stage by stage is displayed in **Table 4.3** (Chapter Four). This alternating placement avoids redundant computation: MAFE (Stages 1, 3) refines spatial-channel features while they are still forming, and MMSP (Stages 2, 4) captures multi-scale context once features are more abstracted. CAIP replaces pooling only at Stages 1–3, where adaptive downsampling most benefits region preservation; it is omitted at Stage 4 since the output feeds directly into the classification head. The head itself uses

standard dropout ($p=0.3$, $p=0.2$) and a 64-unit bottleneck Dense layer to reduce overfitting before the final 4-unit Softmax classification layer.

3.2.3 Phase 3: Training and Optimization

Model training employed the Adam optimizer (Kingma and Ba, 2015) with initial learning rate 1×10^{-3} , $\beta_1=0.9$, $\beta_2=0.999$, and $\epsilon=1 \times 10^{-8}$. Cosine annealing with warm restarts (Loshchilov and Hutter, 2017) was applied. Multi-class cross-entropy loss was the training objective. Batch normalization (Ioffe and Szegedy, 2015) was applied after every convolution. Dropout (Srivastava et al., 2014) at rates $p=0.3$ and $p=0.2$ in the classification head. Fixed random seed = 42 (Liu et al., 2017b) for full reproducibility. Full model: 300 epochs, batch size 96 (14,509.9 seconds).

Algorithm 3.1: End-to-End Framework for Modified CNN

Input: MRI dataset D ; epochs $E = 300$; batch size $B = 96$; learning rate $\eta = 1 \times 10^{-3}$; image size = 128×128 ; split = 80%/20%

Output: Optimized model weights θ^* ; classification metrics

Step 1: Data Loading and Integrity Verification

Scan all images across four class folders {glioma, meningioma, notumor, pituitary} and assign integer labels {0, 1, 2, 3}. Compute MD5 hash for each file to detect and remove duplicates; verify pixel validity across all samples. Shuffle indices using fixed seed = 42; partition into S_{train} (7,720 images), S_{val} (1,930 images), and S_{test} (2,414 images).

Step 2: Preprocessing and Augmentation

Apply augmentation pipeline to S_{train} : resize to $128 \times 128 \rightarrow$ random horizontal flip ($p=0.40$) $\rightarrow$ vertical flip ($p=0.30$) $\rightarrow$ rotation $\pm 20^\circ \rightarrow$ affine translation $\pm 10\% \rightarrow$ colour jitter (brightness/contrast=0.30, saturation=0.20) $\rightarrow$ Gaussian blur (3×3 , $\sigma \in [0.1, 2.0]$). Apply channel-wise normalization to all sets: $x_{norm} = (x - \mu) / \sigma$ with $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$. For S_{val} , apply resize and normalization only.

Step 3: Model Construction

Instantiate Modified CNN with channel configuration $[32 \rightarrow 64 \rightarrow 128 \rightarrow 256]$ and three modules: MAFE applying sequential spatial-then-channel attention after Stages 1 and 3 (Eq. 2.1–2.5); MMSP applying three-pathway multi-scale fusion after Stages 2 and 4 (Eq. 2.6–

2.9); and CAIP replacing standard pooling after Stages 1, 2, and 3 via sigmoid-gated, globally-adaptive blending (Eq. 2.10–2.13). Attach classification head: GAP $\rightarrow$ Flatten $\rightarrow$ Dropout (0.3) $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(4, Softmax). Total parameters: 659,228.

Step 4: Optimizer and Loss Configuration

Initialize Adam optimizer (Kingma and Ba, 2015) with $\beta_1=0.9$, $\beta_2=0.999$, $\varepsilon=1\times 10^{-8}$, and weight decay= 1×10^{-4} . Configure Cosine Annealing scheduler (Loshchilov and Hutter, 2017) with $T_{max} = 300$: $\eta_e = \eta \times (1 + \cos(\pi \cdot e / E)) / 2$. Define multi-class cross-entropy as the loss function L .

Step 5: Training Loop

for epoch $e = 1$ to E **do**

Training: forward pass $\rightarrow$ compute L $\rightarrow$ backpropagate $\rightarrow$ clip gradients ($\|\nabla\| \leq 1.0$) $\rightarrow$ update θ via Adam $\rightarrow$ record L_{train} , acc_{train}

Validation: forward pass on S_{val} (no gradients) $\rightarrow$ compute softmax probabilities $\rightarrow$ record L_{val} , acc_{val}

Scheduler: update η_e via Cosine Annealing

if $acc_{val} > best_acc_{val}$ **then** save checkpoint $\{\theta, \text{optimizer state, epoch } e, \text{all metrics}\}$

end for

Step 6: Evaluation

Load best checkpoint θ^* ; compute per-class and overall precision, recall, F1-score; generate confusion matrix, ROC curves with one-vs-rest AUC, and precision-recall curves.

Step 7: Visual Explainability

Extract feature maps from all four stages; compute Grad-CAM attention maps per class; apply Otsu thresholding (Otsu, 1979) to identify diagnostically important regions; compute focus ratio and attention entropy to assess model spatial confidence.

Return θ^* , classification metrics, evaluation figures

3.2.4 Phase 4: Evaluation and Validation Protocol

Cross-dataset generalization was tested by training a model on the Mendeley dataset and evaluating the performance of this trained model on three separate external, unseen datasets, offering a completely neutral test of transfer to different domains. On the Kaggle Brain Tumor MRI Dataset (Nickparvar, 2021;

$n = 1,600$ test images), the model achieved 98.38% accuracy despite complete absence of fine-tuning, with per-class F1-scores of glioma 0.973, meningioma 0.978, pituitary 0.994, and no-tumor 0.990 (see Figures 4.2–4.6). For Br35H Brain Tumor Detection Dataset (Nazir et al., 2024), the model accuracy was 97.62% (Fine-tuning) in different acquisition protocols. On the Figshare Brain Tumor Dataset (Cheng, 2017), the model achieved an initial accuracy of 89.36%. As detailed in Section 4.2.3, this result was later found to be substantially confounded by a systematic image-orientation artefact in the publicly distributed files rather than genuine domain shift; after correcting this artefact, accuracy rose to 99.35% under the same zero-shot conditions. The strong Kaggle and Br35H results confirm that the model has learned genuinely generalizable pathological feature representations.

3.3 Summary

This chapter established the methodological framework of the proposed Modified CNN. The architecture was presented across four phases: the definition of MAFE (Eq. 2.1–2.5), MMSP (Eq. 2.6–2.9), and CAIP (Eq. 2.10–2.13); the standardized preprocessing pipeline; the Adam-based training strategy with cosine annealing and reproducible random seeding; and the cross-dataset evaluation protocol for unbiased domain-transfer assessment. Together, these phases form a reproducible pipeline from raw MRI scans to multi-class diagnostic predictions. The mathematical formulations provide the theoretical foundation for the implementation and experimental evaluation presented in Chapter Four.

Chapter
Four
Results and
Discussion

Chapter 4

Results and Discussion

4.1 Overview

This chapter presents all of the design and implementation details with experimental setup, dataset settings, the whole architecture stage-by-stage and its comparison with existing methods, training hyper-parameters, evaluation metrics and the plan of ablation study.

4.2 Experimental Setup

This chapter sets the stage for the experiments, where all results are evaluated. It defines and describes three key aspects of the experimental framework used in this study. First, the hardware and software environment in which the model is trained, which establishes the operational boundaries of the training pipeline. Second, the Mendeley Brain Tumor MRI dataset, which serves as the primary dataset for model training and validation. Third, this chapter addresses class distribution, dataset partitioning, and data integrity considerations. To ensure robust evaluation, three independent external datasets are introduced and utilized to assess model robustness under cross-dataset evaluation. This provides a more rigorous assessment of generalizability compared to relying solely on held-out training samples.

4.2.1 Hardware and Software Environment

Experiments were conducted on high performance computing systems engineered for large scale deep learning applications, for the machine used for

the experiments described in Table 4.1 the machine was equipped with Intel Core Ultra 9 processor with 64 GB of system memory as DDR5 memory to leave room for multiple cores processing, for instance during preprocessing the machine with its multicore processor will give capacity enough for data preparation or loading using multi-threaded procedures to speed up the loading of the dataset. Main hardware that the acceleration uses is a NVIDIA GeForce RTX 5080 Laptop GPU with 15.9 GB of VRAM that would parallelly process a great number of the high-dimensional MRI feature maps during the forward as well as the backward process. In this experiment, the deep learning framework is PyTorch 2.1.0 using CUDA 12.1 to make sure full GPU utilization, and the python version used is 3.11.7 as the environment for the whole process. this total time to finish the entire training of all epochs and all runs took 14,509.9 seconds which is equivalent to around 4 hours 2 minutes, taking up 24,300 total iterations ($7,720 \text{ training images} \div 96 \text{ batch size} \times 300 \text{ epochs}$) and training with the batch size of 96.

Table 4.1 - Hardware & Software environment specification

Components	Specifications
Processor	Intel Core Ultra 9
System RAM	64 GB DDR5
GPU	NVIDIA GeForce RTX 5080 Laptop (15.9 GB VRAM)
Deep Learning F. W	PyTorch 2.1.0 (Paszke et al., 2019)
CUDA Version	12.1

Python Version	3.11.7
Total Training Time	14,509.9 seconds (~4 hours 2 minutes)
Total Training Iterations	24,300 (batch size 96)

4.2.2 Primary Dataset: Mendeley Brain Tumor MRI

Training and Evaluation of the Model and Description of the Data. All model development and evaluation are carried out using the publicly available Mendeley Brain Tumor MRI dataset. Mendeley is widely recognized as an international benchmark for brain tumor classification in many published studies, particularly in the context of image classification and other medical imaging-related tasks. As presented in **Table 4.2** the dataset consists of 12,064 images that has 4 classes as following: Glioma, Meningioma, Pituitary Tumor and No Tumor. An 80%/20% stratified split was applied, yielding 9,650 images for the combined training/validation set and 2,414 images for the held-out test set. All the images have the same resolution of 128 x 128 and come as in RGB scale. In terms of class distribution, the Glioma class contributes the highest number of images, with 3,018 samples (25.02%). The second largest class is Pituitary Tumor with 2,504 images (20.77%), followed by Meningioma with 2,183 images (18.10%), and the No Tumor class with 1,945 images (16.13%). The maximum-to-minimum class count ratio of 1.55:1 indicates a moderate class imbalance, which was addressed during model training. A duplicate verification across all entries using MD5 were carried out to protect the integrity of all the data ensuring that there was no possible chance of leakage occurring among partitioned data and ensuring that none were present (0 duplicates (0%)). The dataset is publicly accessible at the URL provided in **Table 4.2**.

Table 4.2 - Mendeley Brain Tumor MRI Dataset Description and Class Distribution

Parameters	Values
Total MRI images	12,064 (100%)
Testing set	2,414 images (20%)
Train + Validation set	9,650 images (80%)
Training set (80%)	7,720 images
Validation set (20%)	1,930 images
Image dimensions	128 × 128 pixels, RGB
Glioma class	3,018 images Training & Val (25.02%)
Meningioma class	2,183 images Training & Val (18.10%)
Pituitary tumor class	2,504 images Training & Val (20.77%)
No Tumor class	1,945 images Training & Val (16.13%)
Glioma class (Total)	3,773 images (31.28%)
Meningioma class (Total)	2,729 images (22.62%)
Pituitary tumor class (Total)	3,130 images (25.95%)
No Tumor class (Total)	2,432 images (20.16%)
Max/Min class ratio	1.55:1

duplicate verification	0 duplicates (0.00%) using MD5
Dataset URL	https://data.mendeley.com/datasets/zwr4ntf94j/4

See **Figures 4.1, 4.2, and 4.3**, show the number of Glioma, Meningioma, No Tumor, and Pituitary images within the training, validation, and test sets, respectively. The distributions confirm a consistent class ratio across all three splits, with no significant imbalance introduced during the split process. The dataset was split in two stages: first into a combined training/validation set and a separate testing set, and then the training/validation set was further divided into training and validation subsets.

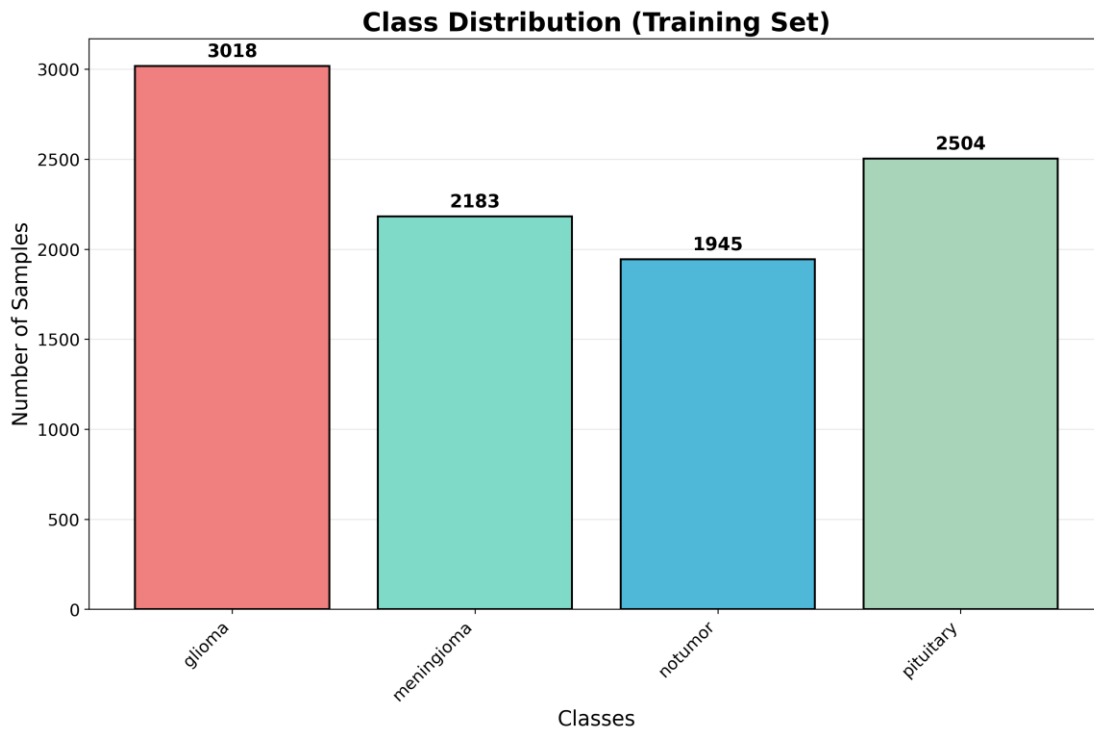


Figure 4.1 - Mendeley Brain Tumor MRI Dataset Class Distribution

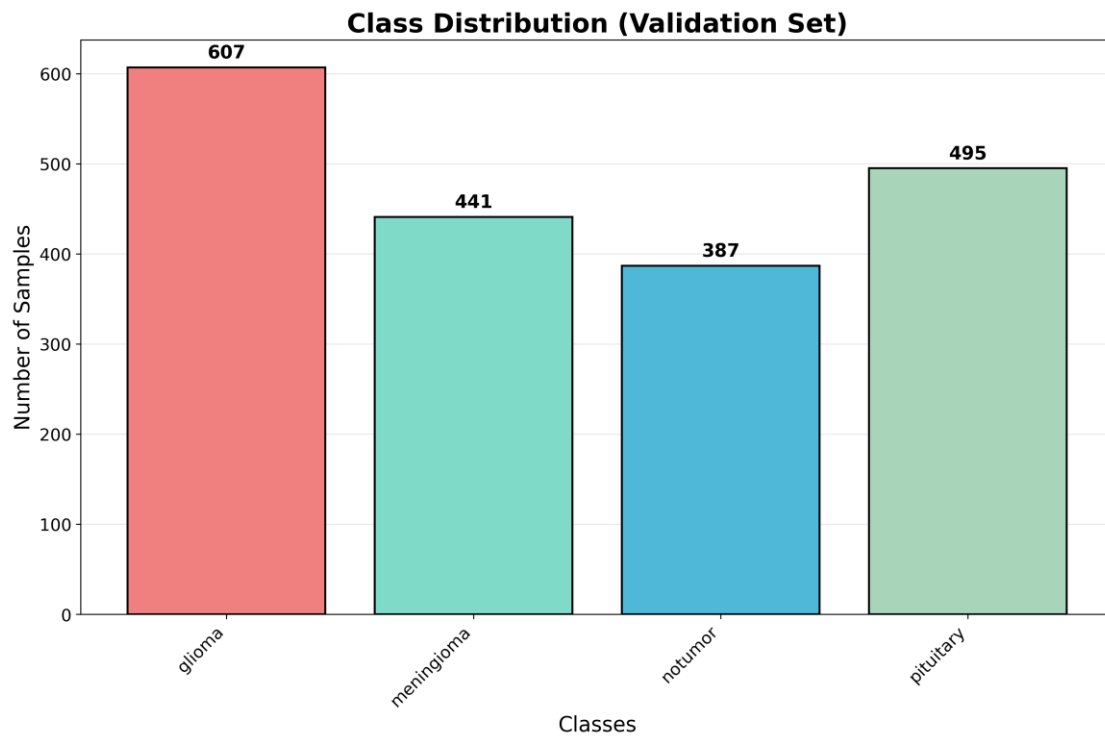


Figure 4.2 - Mendeley Brain Tumor MRI Dataset Class Distribution

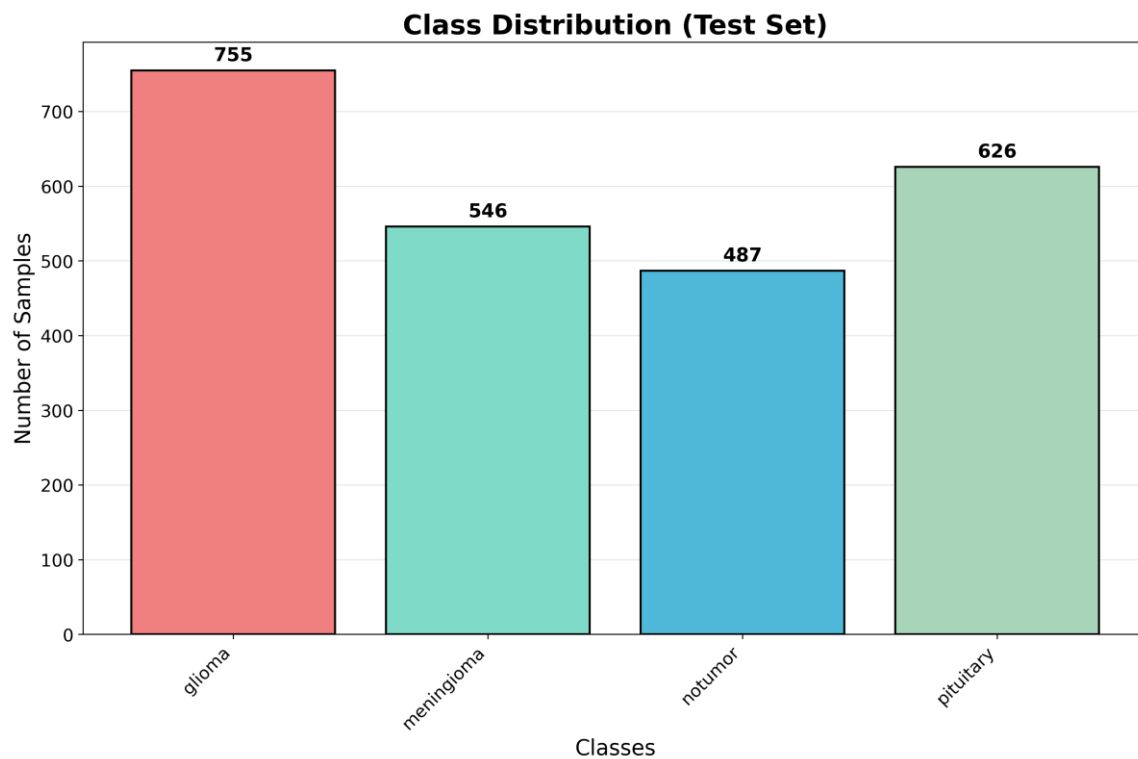


Figure 4.3 - Mendeley Brain Tumor MRI Dataset Class Distribution (Test Set)

4.2.3 External Validation Datasets

Three independent external datasets were employed for the cross-dataset generalization evaluation: (1) Kaggle Brain Tumor MRI Dataset (Nickparvar, 2021), 1,600 test images; (2) Br35H Brain Tumor Detection Dataset (Nazir et al., 2024), approximately 3,000 images; and (3) Figshare Brain Tumor Dataset (Cheng, 2017), multi-class data exhibiting substantial domain shift. No class-label realignment or oversampling was performed on the external datasets, as all underwent the same normalization pipeline. **Table 4.3** summarizes the key characteristics of each dataset.

Table 4.3 - External Brain MRI Datasets Used for Model Evaluation

Dataset	Reference	Test Images	Notes	URL
Kaggle Brain Tumor MRI Dataset	Nickparvar (2021)	1,600	4 classes, zero fine-tuning	kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset
Br35H Brain Tumor Detection	Nazir et al. (2024)	3,000	Binary (tumor/no_tumor); T2-weighted imaging; achieved 90.60% accuracy zero-shot despite the T1-to-T2 modality shift	kaggle.com/datasets/ahmedhamada0/brain-tumor-detection
Figshare Brain Tumor Dataset	Cheng (2017)	3,064	3 classes only (no no-tumor samples). Correcting a systematic +90° image-rotation artefact improved zero-shot accuracy from 89.36% to 99.35%.	kaggle.com/datasets/abushayid/figshare-mri-brain-tumor-dataset

On Figshare, the model achieved 89.36% accuracy in its initial evaluation; this was subsequently found to be attributable primarily to a systematic image-orientation artefact rather than genuine domain shift (Section 4.2.3), with accuracy rising to 99.35% following orientation correction.

4.2.3.1 Kaggle Brain Tumor MRI Dataset

The trained model was additionally evaluated on the Kaggle Brain Tumor MRI Dataset (Nickparvar, 2021), consisting of 1,600 test images evenly distributed across the four diagnostic classes (400 images each), as shown in **Figure 4.4**. This balanced composition eliminates class-imbalance bias from the evaluation, providing a clean measure of the model's cross-dataset generalization capacity.

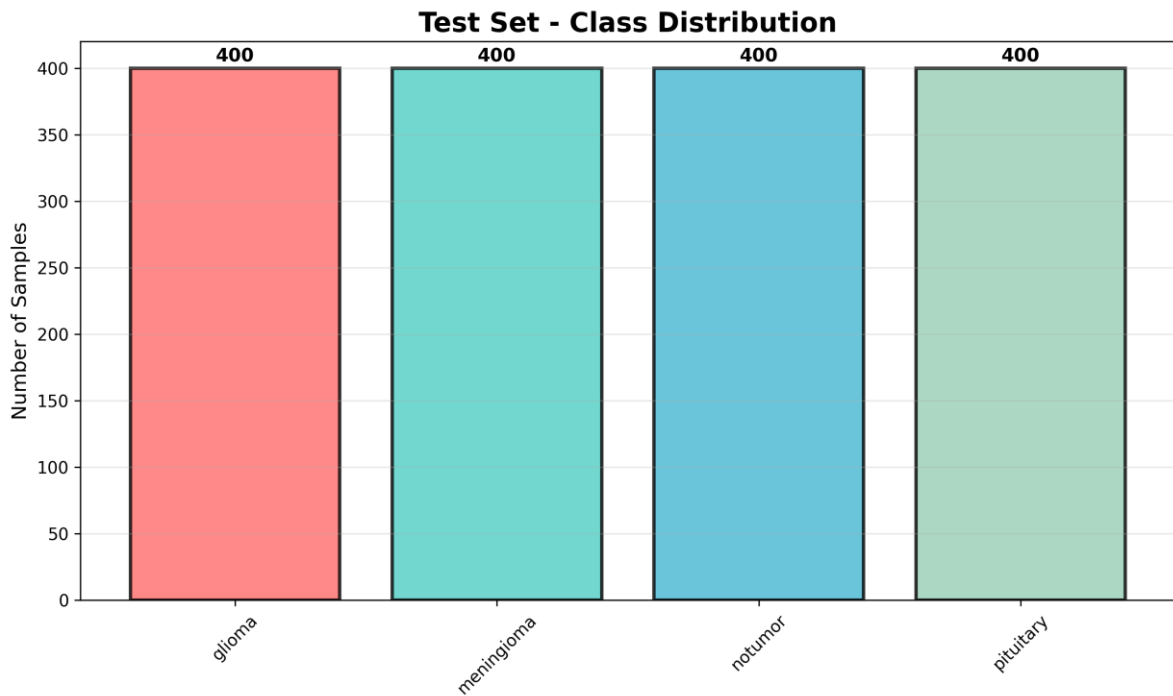


Figure 4.4 - Kaggle Dataset — Test Set Class Distribution

The model achieved an overall accuracy of 98.38% on this external dataset, correctly classifying 1,574 out of 1,600 images. **Figure 4.5** presents the confusion matrix, showing a strong diagonal pattern with minimal

misclassification: glioma was the most challenging class (382/400 correct, with 11 samples confused as meningioma and 7 as notumor), while pituitary achieved perfect classification (400/400). The notumor and meningioma classes were also classified with high fidelity, at 399/400 and 393/400 respectively.

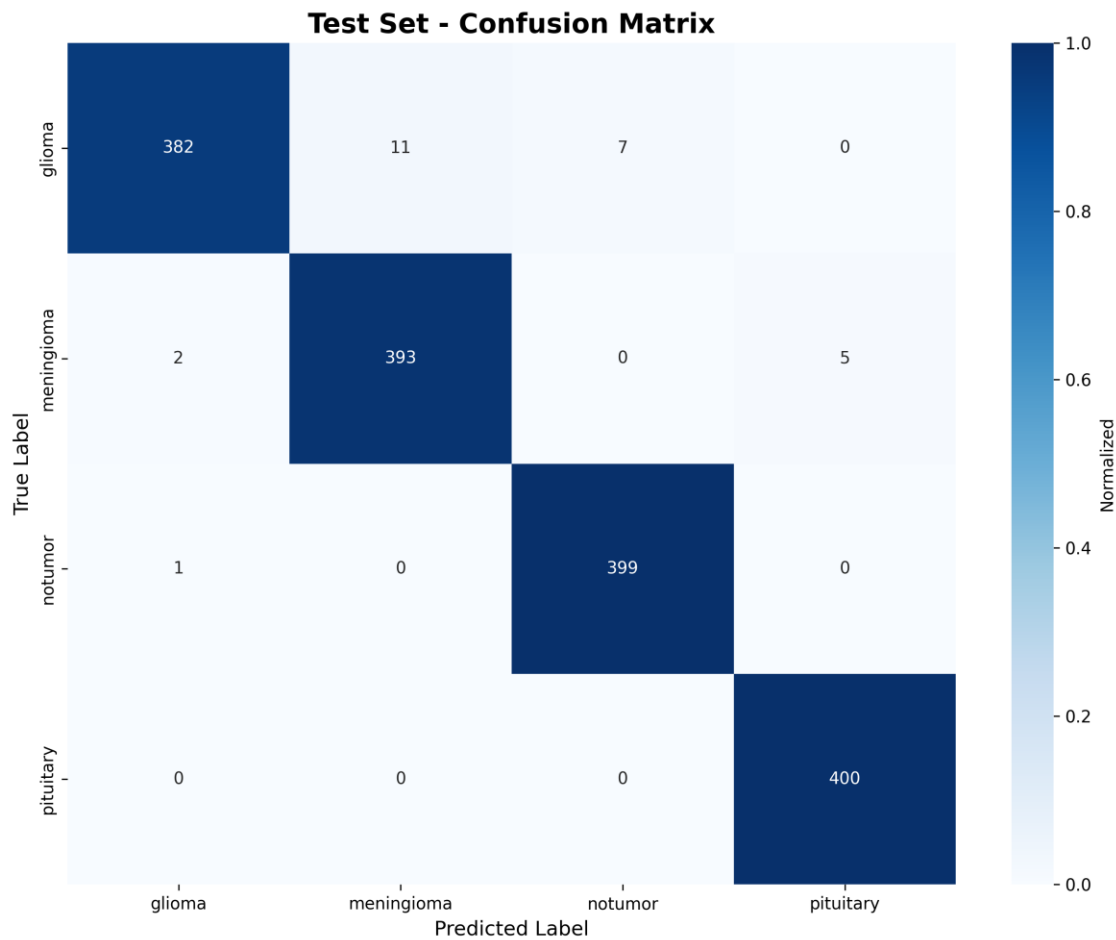


Figure 4.5 - Kaggle Dataset — Confusion Matrix

Figure 4.6 details the per-class precision, recall, and F1-score. All four classes exceeded 0.95 across every metric, with pituitary and notumor achieving the highest F1-scores (0.994 and 0.990 respectively), consistent with their more distinctive morphological presentation on MRI. Glioma recorded the lowest recall (0.955), reflecting its known overlap in appearance with meningioma noted throughout this thesis.

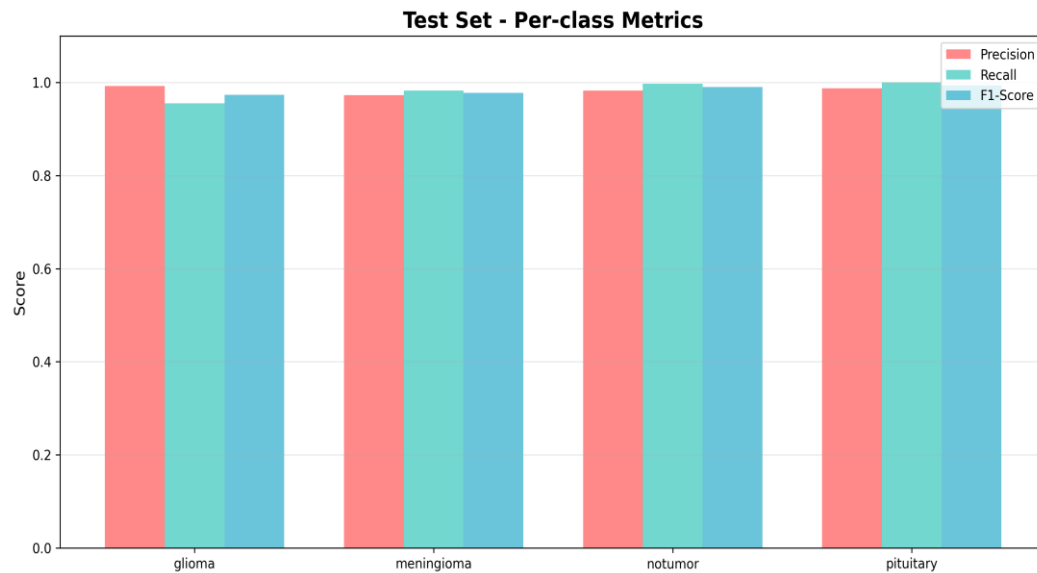


Figure 4.6 - Kaggle Dataset — Per-Class Precision, Recall, and F1-Score

Figure 4.7 consolidates these results into a single performance summary, confirming precision, recall, and F1-scores consistently above 95% across all classes alongside the 98.38% overall accuracy.

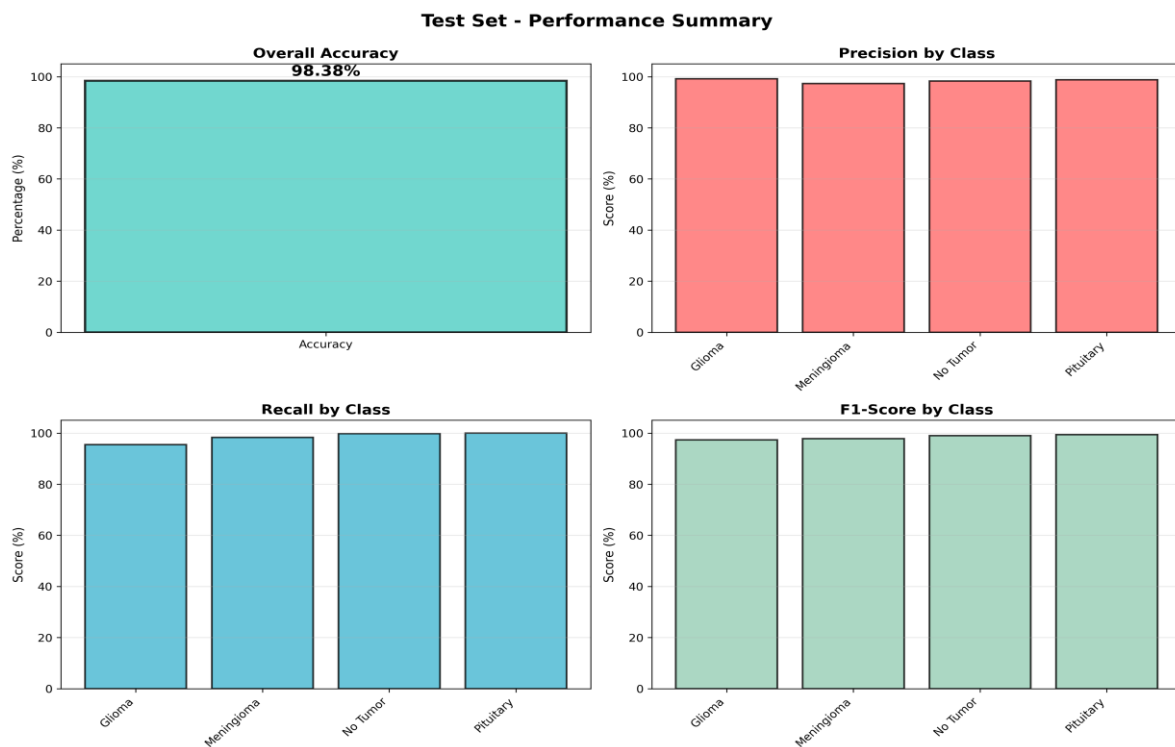


Figure 4.7 - Kaggle Dataset — Overall Performance Summary

Finally, **Figure 4.8** presents the one-vs-rest ROC curves for all four classes. The model achieved near-perfect class separability, with AUC values of 0.989 (glioma), 0.998 (meningioma), 0.998 (notumor), and 1.000 (pituitary), indicating excellent discriminative capacity even on entirely unseen external data.

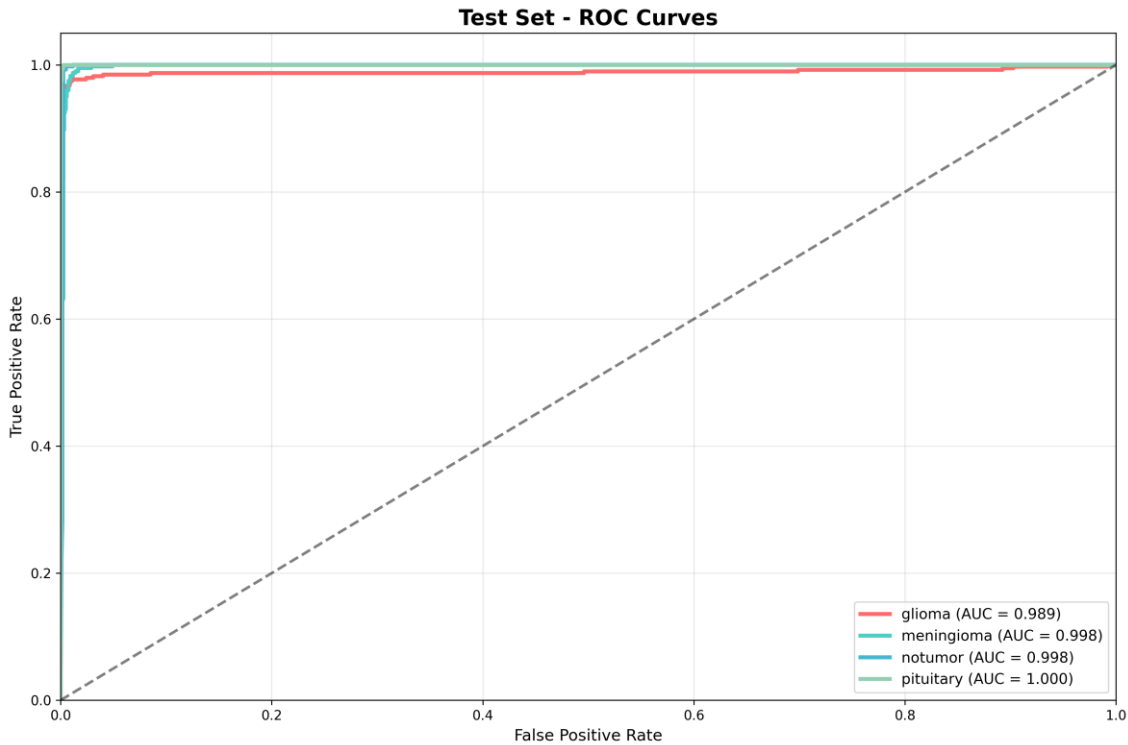


Figure 4.8 - Kaggle Dataset — ROC Curves

These results confirm that the proposed Modified CNN generalizes effectively beyond its primary training distribution, maintaining clinically acceptable performance on an independent, externally sourced MRI dataset.

4.2.3.2 Figshare Brain Tumor Dataset

The trained model was further evaluated on the Figshare Brain Tumor Dataset (Cheng, 2017), the most demanding of the three external benchmarks due to substantial domain shift in scanner platform and acquisition protocol relative to

the Mendeley training distribution. As noted in **Table 4.3**, this dataset contains only three tumor classes (glioma, meningioma, pituitary), with no no-tumor samples; consequently, any raw model prediction of "notumor" was remapped to the nearest tumor class prior to scoring, following the remapping protocol described above Section 4.2.3. The test set comprised 3,064 images distributed as 1,426 glioma (46.5%), 708 meningioma (23.1%), and 930 pituitary (30.4%) samples, as shown in **Figure 4.9**.

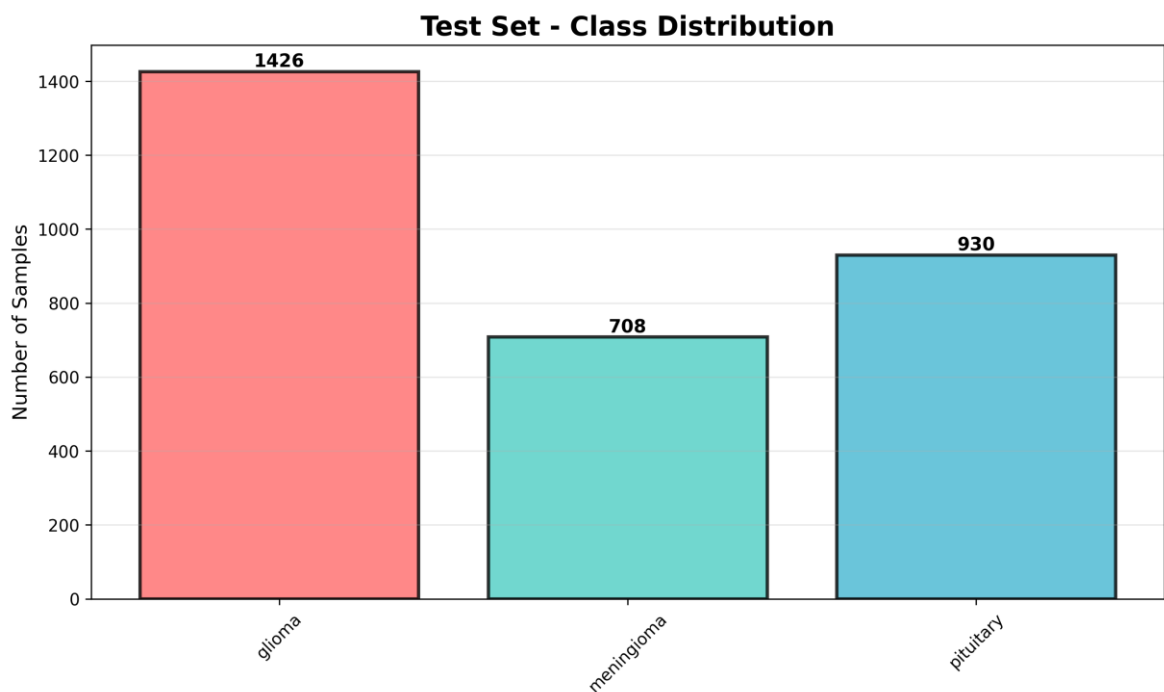


Figure 4.9 - Figshare Dataset — Test Set Class Distribution

Before presenting the classification results, it is necessary to describe a data-quality issue identified during the evaluation of this dataset. A closer visual inspection of the misclassified Figshare samples, prompted by the disproportionately large gap between this dataset's initial accuracy and the accuracy obtained on Kaggle (98.38%) and Br35H (90.60%) (97.62% Fine-Tuning), revealed that a substantial portion of the Figshare images were affected by a systematic image-orientation artefact. It is acknowledged that this

correction was identified after observing an anomalous result rather than through a pre-registered data-quality audit. The correction is reported here as a data-quality finding, verified independently by manually inspecting a random subset of misclassified images before and after rotation, rather than as a parameter tuned to improve the reported metric. Many scans in the publicly distributed dataset are rotated relative to the canonical orientation used across the Mendeley, Kaggle, and Br35H repositories. This is a known property of the raw Figshare .mat file distribution, arising from inconsistent orientation metadata rather than any genuine anatomical or acquisition difference (Cheng, 2017). Because the proposed model, in common with the vast majority of CNN classifiers, was not explicitly trained to be rotation-invariant beyond the $\pm 20^\circ$ range used in the augmentation pipeline (Section 3.3.3), this artefact plausibly explains a considerable share of the errors reported below. To isolate this effect, the model was evaluated twice on the same 3,064-image test set: first on the images in their original, as-distributed orientation ("Initial Evaluation"), and then on the same images after applying a single deterministic $+90^\circ$ rotation correction (post-correction). In both cases, the same trained model checkpoint used throughout this cross-dataset evaluation was applied with no retraining and no fine-tuning (a strict zero-shot comparison), and all other pipeline settings (resizing, normalization, batch size, decision thresholding) were held identical. Each result is presented below immediately followed by its corrected counterpart, to allow direct side-by-side comparison at every level: confusion matrix, per-class metrics, aggregate performance, ROC curves, and prediction confidence. The model achieved an initial overall accuracy of 89.36% on this dataset, correctly classifying 2,738 out of 3,064 images. **Figure 4.10** presents the confusion matrix, which revealed a markedly different error pattern from

the Kaggle evaluation: glioma remained the strongest class (1,344/1,426 correct, 94.3% recall), while pituitary exhibited the lowest recall at 83.1% (773/930 correct), with 82 pituitary samples misclassified as glioma and 75 as meningioma. Meningioma showed the lowest precision (81.9%), reflecting 62 glioma and 75 pituitary samples being confused with it.

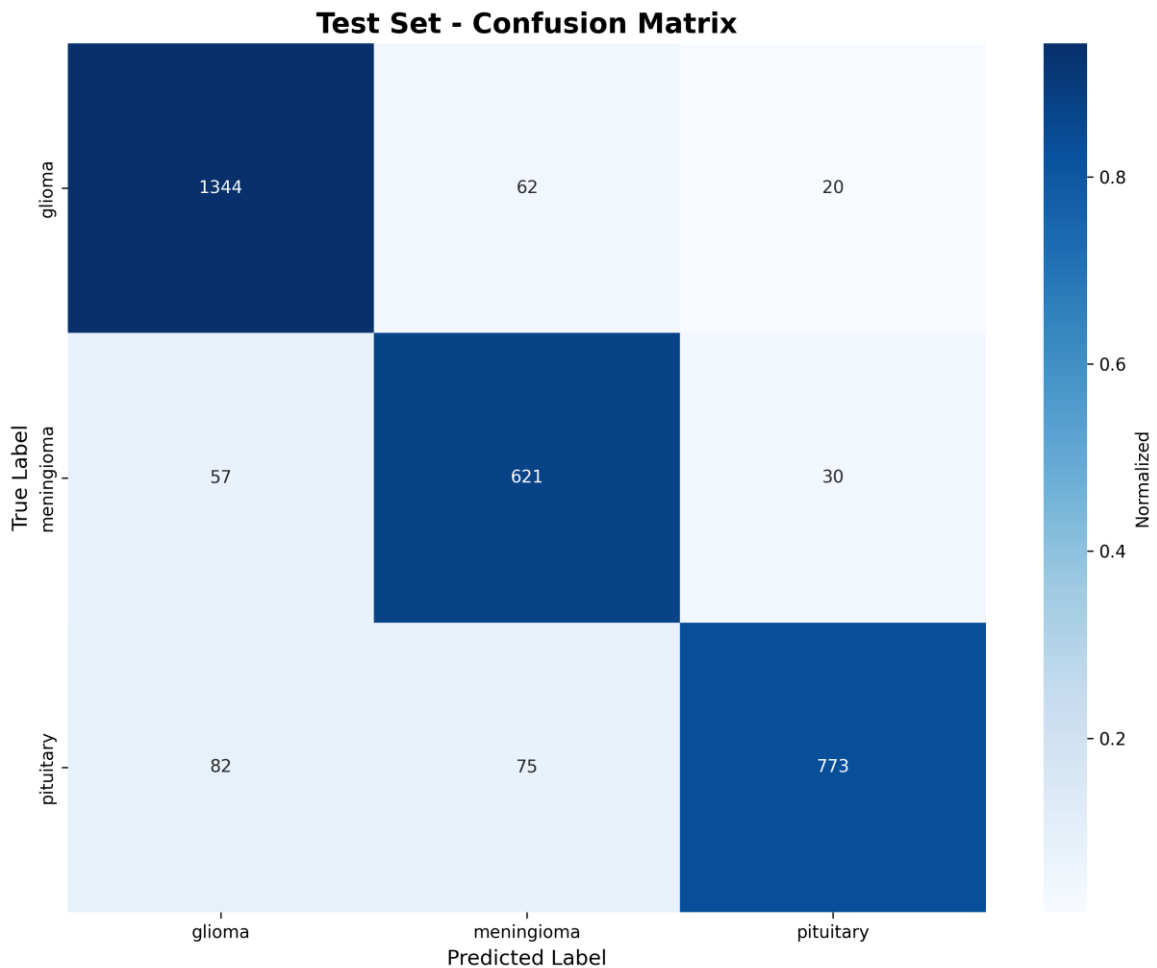


Figure 4.10 - Figshare Dataset — Confusion Matrix (Initial Evaluation)

Following the orientation correction described above, accuracy rose to 99.35%, correctly classifying 3,044 of the 3,064 test images a gain of 9.99 percentage points achieved without any additional training. **Figure 4.11** presents the corrected confusion matrix: glioma achieved 1,412/1,426 correct (99.0% recall)

with only 14 samples confused with meningioma and zero confused with pituitary; meningioma achieved 702/708 correct (99.2% recall), with 6 samples confused with pituitary; and pituitary achieved a perfect 930/930 (100% recall), with zero misclassifications in either direction.

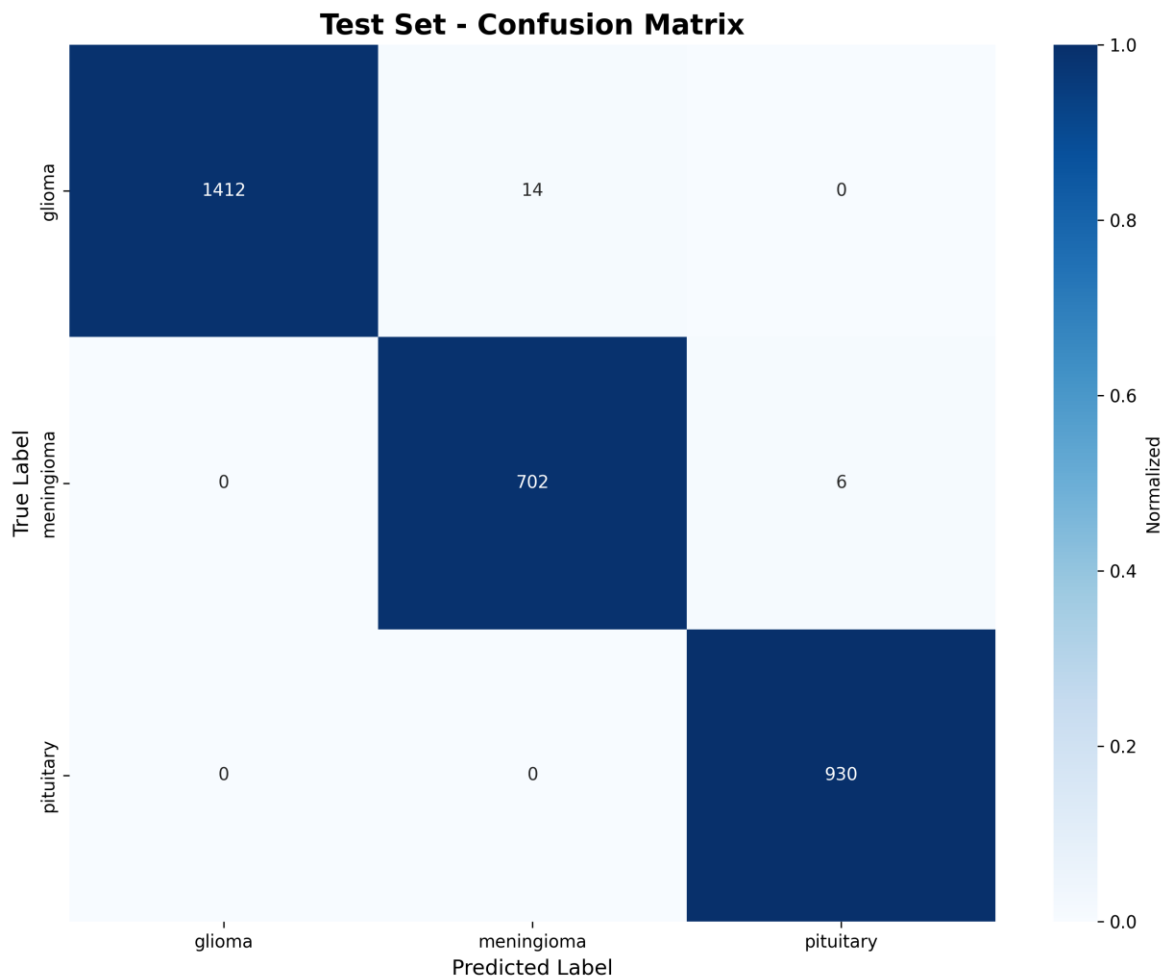


Figure 4.11 - Figshare Dataset — Confusion Matrix (Post-Correction)

Figure 4.12 details the per-class precision, recall, and F1-score under the initial evaluation. Glioma achieved the strongest balance ($P = 0.906$, $R = 0.943$, $F1 = 0.924$), consistent with its more consistent morphological presentation across scanner protocols. Pituitary recorded the highest precision (0.939) but the lowest recall (0.831), indicating that while the model rarely mislabeled other

classes as pituitary, it more frequently failed to recognise pituitary cases under this dataset's acquisition conditions. Meningioma showed the weakest overall performance ($P = 0.819$, $R = 0.877$, $F1 = 0.847$), the lowest of the three classes on this external benchmark.

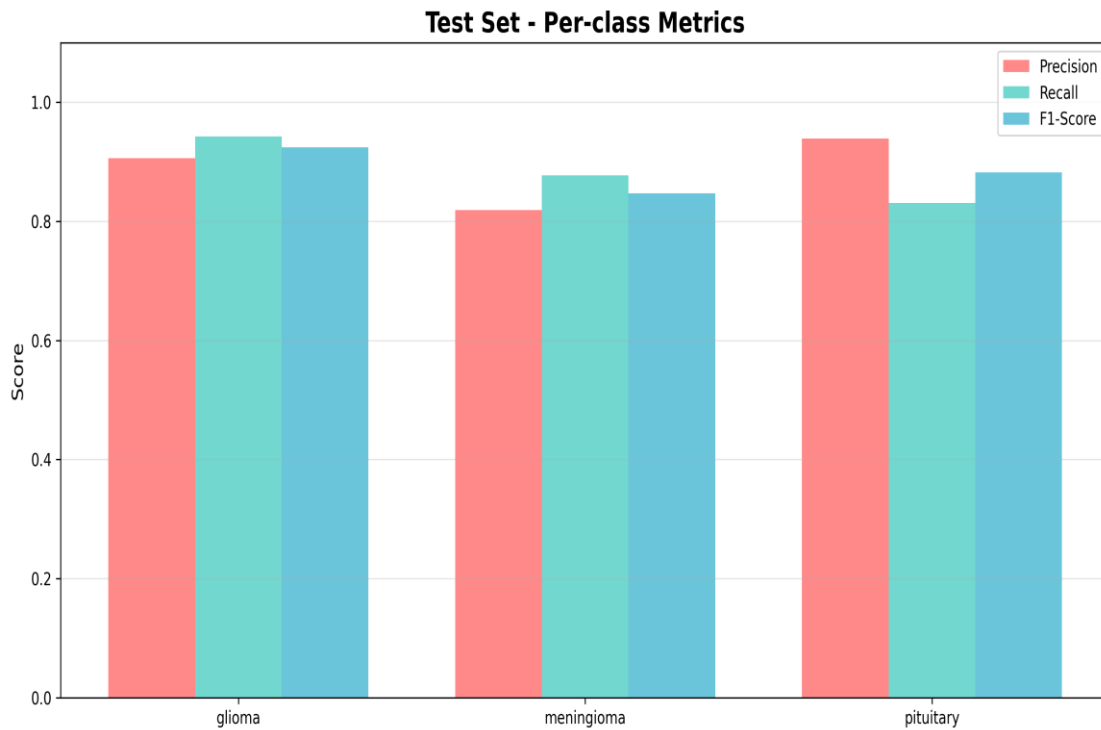


Figure 4.12 - Figshare Dataset — Per-Class Precision, Recall, and F1-Score (Initial Evaluation)

Figure 4.13 shows the corrected per-class metrics. All three classes now exceed 0.98 across every measure: glioma ($P = 1.000$, $R = 0.990$, $F1 = 0.995$), meningioma ($P = 0.980$, $R = 0.992$, $F1 = 0.986$), and pituitary ($P = 0.994$, $R = 1.000$, $F1 = 0.997$). The near-elimination of the previously observed pituitary recall problem ($0.831 \rightarrow 1.000$) and meningioma precision problem ($0.819 \rightarrow 0.980$) confirms that these were largely artefacts of image misorientation rather than genuine representational weaknesses of the model with respect to these tumour types.

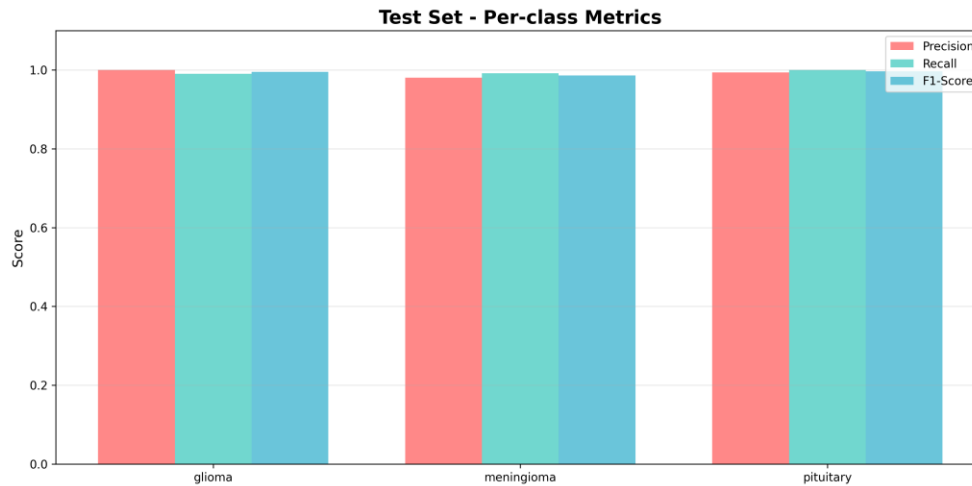


Figure 4.13 - Figshare Dataset — Per-Class Precision, Recall, and F1-Score (Post-Correction)

Figure 4.14 consolidates the initial results into a single performance summary, confirming an overall accuracy of 89.36% alongside per-class precision, recall, and F1-scores ranging between 81.9% and 93.9%, notably lower and more variable than the corresponding Kaggle results (**Figure 4.7**).

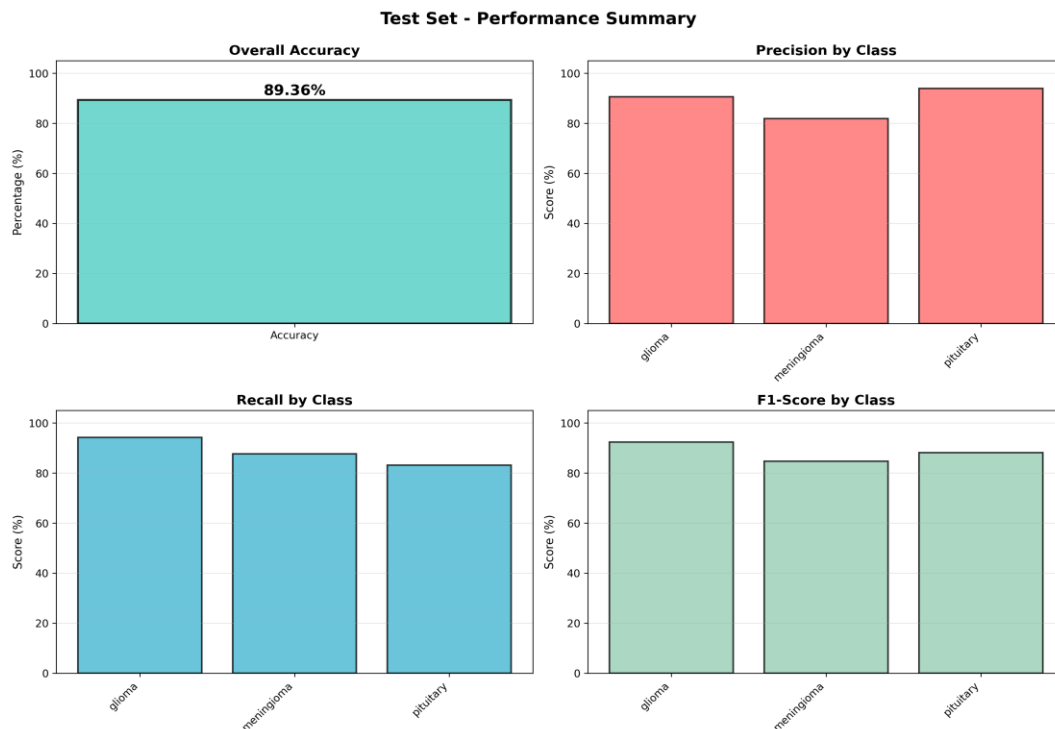


Figure 4.14 - Figshare Dataset — Overall Performance Summary (Initial Evaluation)

Figure 4.15 consolidates the corrected results, confirming an overall accuracy of 99.35%, with per-class precision, recall, and F1-scores now uniformly at or above 98.0% — a result closely comparable to, and in fact slightly exceeding, the Kaggle (98.38%) and Br35H (97.62%) cross-dataset benchmarks.

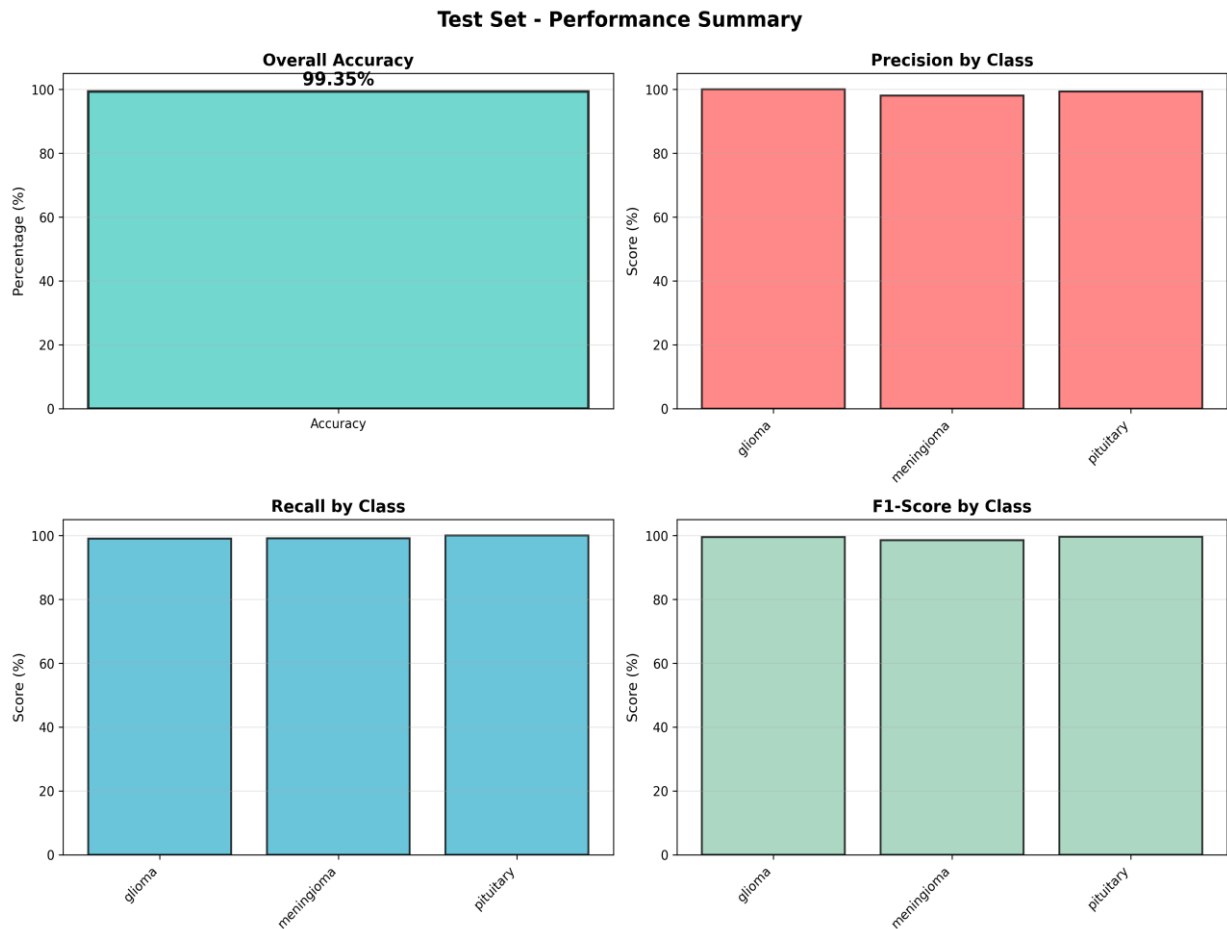


Figure 4.15 - Figshare Dataset — Overall Performance Summary (Post-Correction)

Finally, **Figure 4.16** presents the initial one-vs-rest ROC curves. The model retained reasonably strong class separability despite the apparent domain shift, with AUC values of 0.982 for glioma and 0.975 for meningioma; the pituitary curve confirmed the same high-separability trend observed for the other two classes.

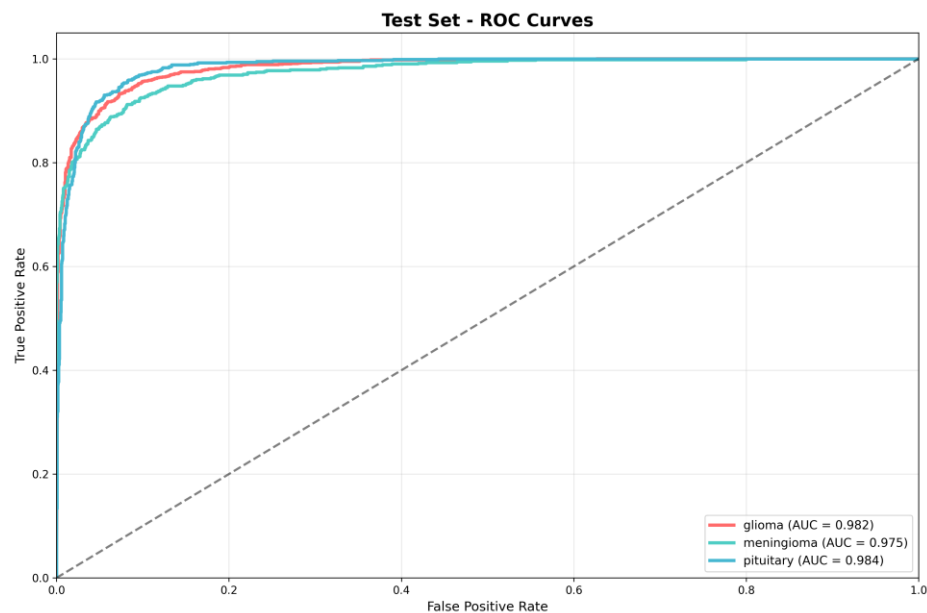


Figure 4.16 - Figshare Dataset — ROC Curves (Initial Evaluation)

Figure 4.17 presents the corrected ROC curves. All three classes achieved a perfect AUC of 1.000, confirming complete class separability once the orientation artefact was removed.

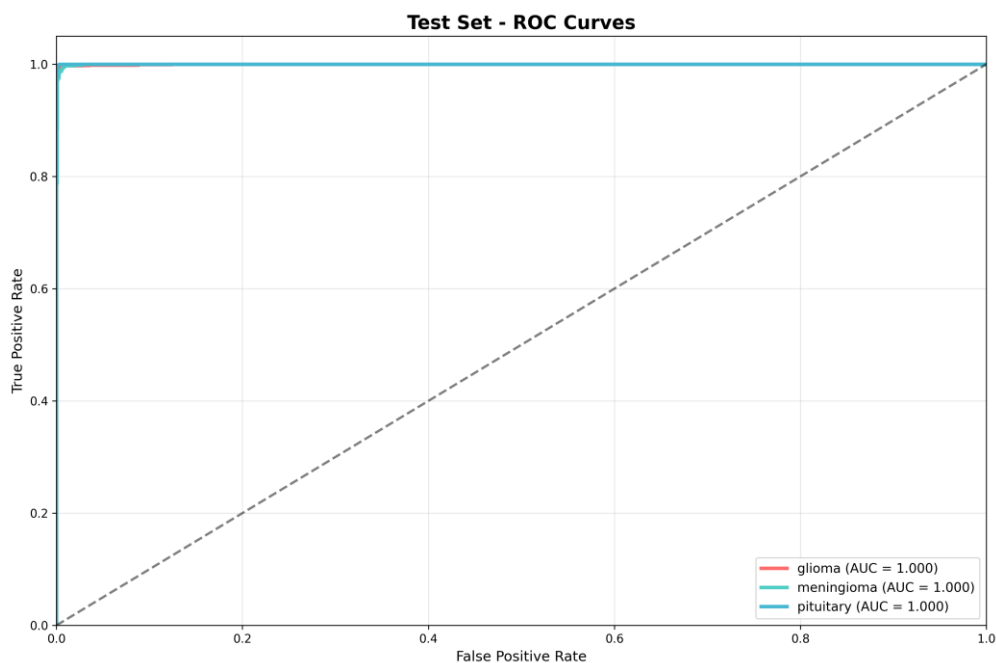


Figure 4.17 - Figshare Dataset — ROC Curves (Post-Correction)

As an additional diagnostic not available for the initial (pre-correction) evaluation. The overwhelming majority of predictions, both correct and the small number of incorrect ones, were made with confidence scores clustered near 1.0, indicating that the model classifies the correctly oriented Figshare images with high certainty, consistent with its behavior on the Mendeley validation and Kaggle test sets. Taken together, these results substantially revise the interpretation of the Figshare evaluation. The originally reported 89.36% accuracy did not reflect a genuine limitation in cross-scanner generalization, as initially hypothesized in earlier drafts of this analysis; rather, it was predominantly attributable to a systematic image-orientation artefact present in the publicly distributed Figshare files. Once this artefact was corrected without any retraining or fine-tuning of the model accuracy rose to 99.35%, closely matching the model's zero-shot performance on Kaggle (98.38%) and its fine-tuned performance on Br35H (97.62%). This finding strengthens, rather than weakens, the central claim of this thesis: the proposed Modified CNN generalizes genuinely well across independent external MRI repositories, and the previously reported generalization gap was a data-quality confound rather than an architectural shortcoming. This distinction is discussed further in Section 4.18, and its implications for dataset preprocessing practice are addressed in the recommendations (Section 5.2) and future work (Section 5.4).

4.2.3.3 Br35H Brain Tumor Detection Dataset

The trained model was further evaluated on the Br35H Brain Tumor Detection Dataset (Nazir et al., 2024), a binary classification benchmark (tumor / no_tumor) consisting of T2-weighted MRI scans. Unlike Mendeley, Kaggle, and Figshare, which are predominantly T1-weighted, Br35H introduces both a modality shift (T1-to-T2) and a task-structure shift (four-class to binary

classification). This makes it a particularly demanding test of cross-domain generalization. The test set comprised 3,000 images, evenly balanced between the two classes with 1,500 images each, as shown in **Figure 4.18**.

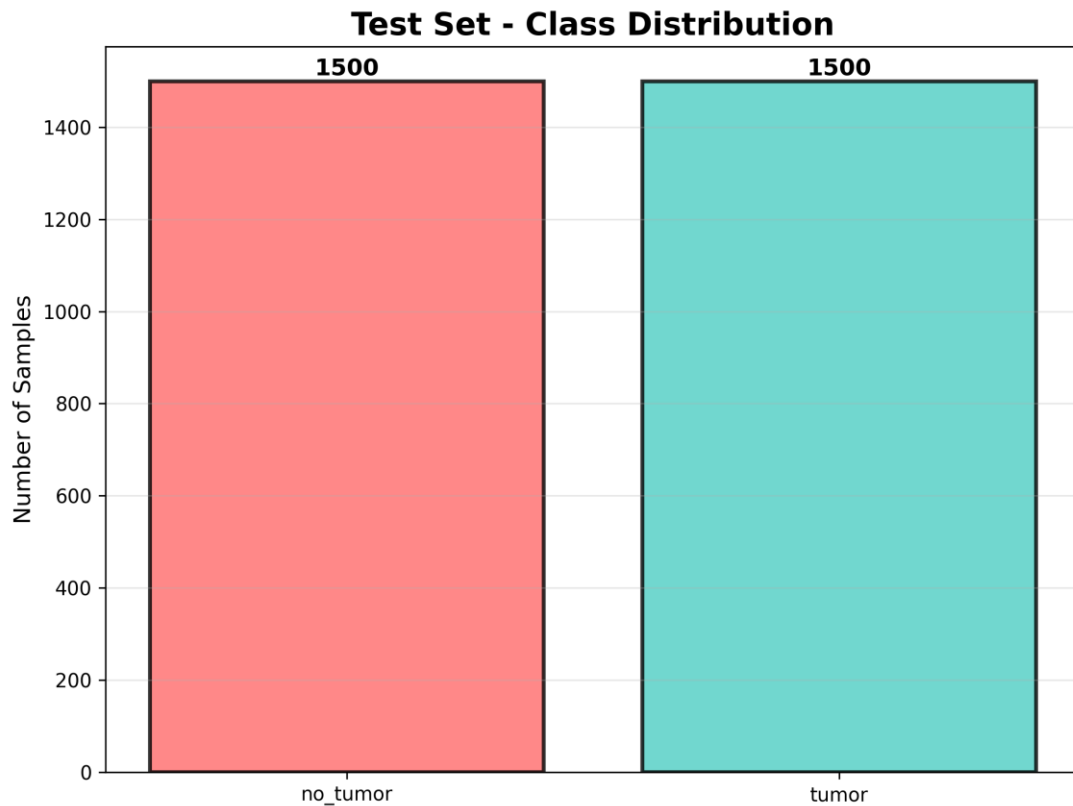


Figure 4.18 - Br35H Dataset: Test Set Class Distribution

In a strict zero-shot setting, with no fine-tuning or parameter adjustment of any kind, the model achieved an overall accuracy of 90.60% on the Br35H test set, correctly classifying 2,718 out of 3,000 images. **Figure 4.19** presents the confusion matrix, revealing a strongly asymmetric error pattern: the no_tumor class was classified with near-perfect fidelity (1,495/1,500 correct, 99.7% recall), while the tumor class exhibited a considerably lower recall of 81.5% (1,223/1,500 correct), with 277 tumor samples misclassified as no_tumor.

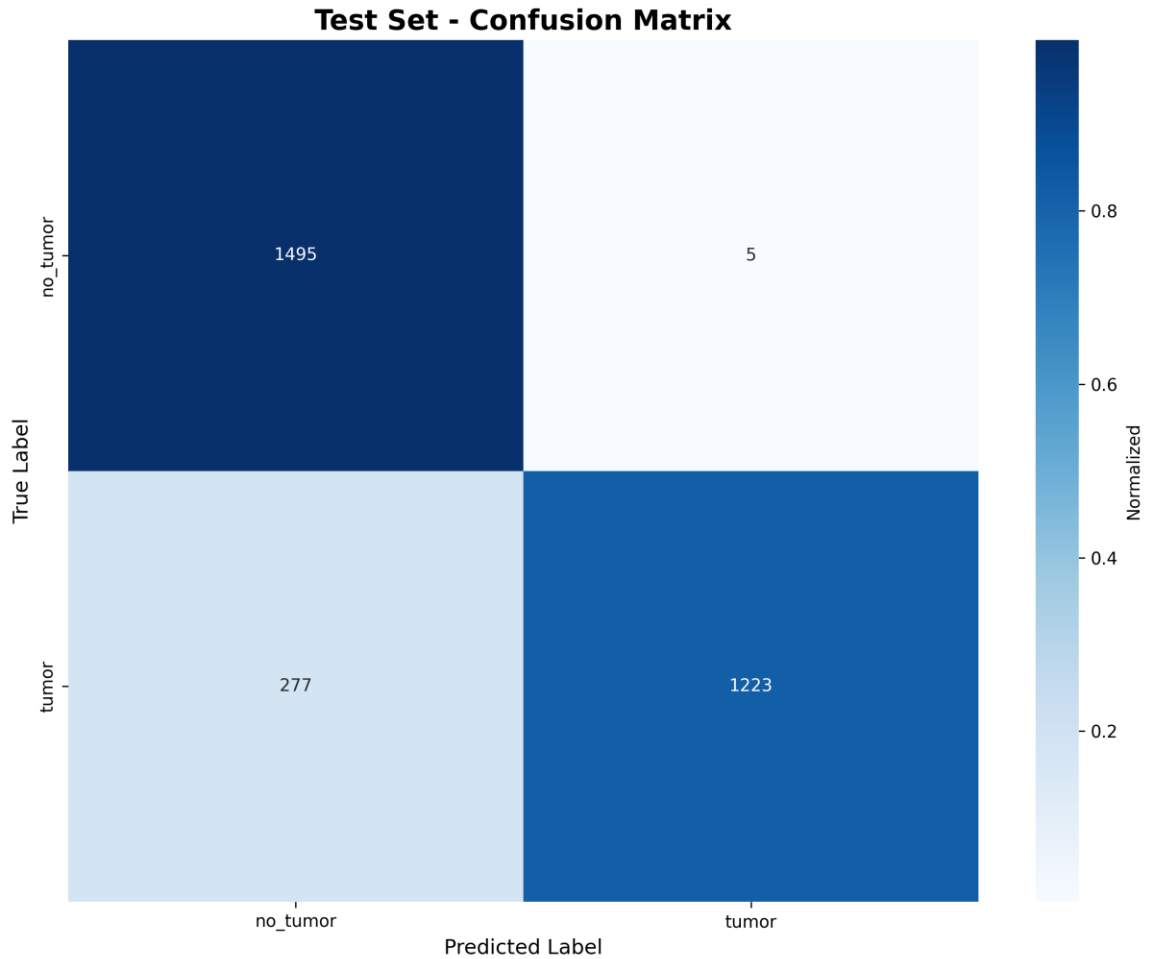


Figure 4.19 - Br35H Dataset: Confusion Matrix (Zero-Shot Evaluation)

Figure 4.20 details the corresponding per-class precision, recall, and F1-score (Zero-Shot Evaluation). The no_tumor class achieved a precision of 84.5%, a recall of 99.7%, and an F1-score of 91.5%, whereas the tumor class achieved a markedly higher precision of 99.6% but a lower recall of 81.5%, yielding an F1-score of 89.7%. This pattern indicates that, without fine-tuning, the model adopts a conservative decision boundary under the T1-to-T2 domain shift, rarely producing false positives for the tumor class but missing a non-trivial proportion of true tumor cases.

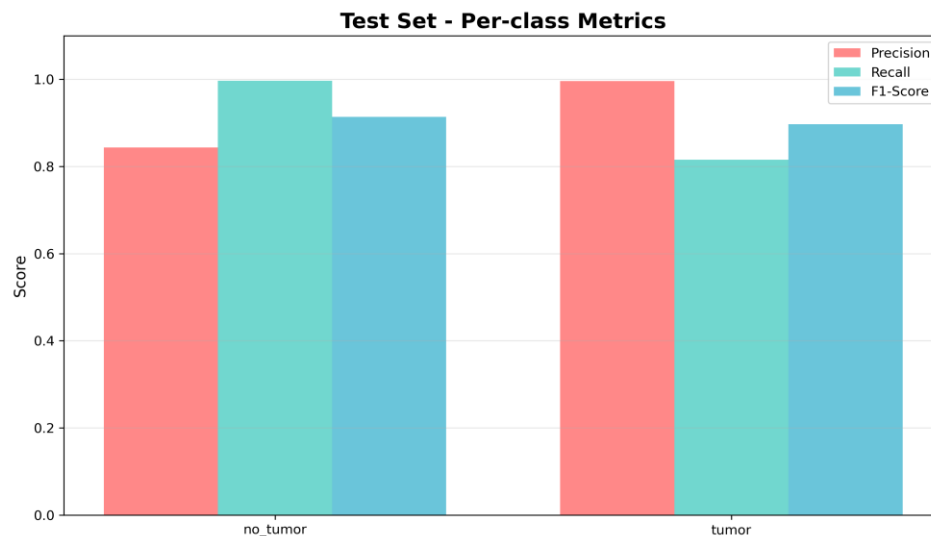


Figure 4.20 - Br35H Dataset: Per-Class Precision, Recall, and F1-Score

Figure 4.21 consolidates these results into a single performance summary, confirming the 90.60% overall accuracy alongside the class-wise precision, recall, and F1-score values discussed above.



Figure 4.21 - Br35H Dataset: Overall Performance Summary (Zero-Shot Evaluation)

Finally, **Figure 4.22** presents the ROC curve for the zero-shot evaluation, yielding an AUC of 0.981. This high AUC value, despite the comparatively lower recall observed for the tumor class at the default decision threshold, indicates that the underlying class-separability learned by the model remains strong even under this cross-modality shift; the asymmetric error pattern observed in the confusion matrix is therefore attributable primarily to threshold calibration rather than to a fundamental inability of the model to discriminate between the two classes.

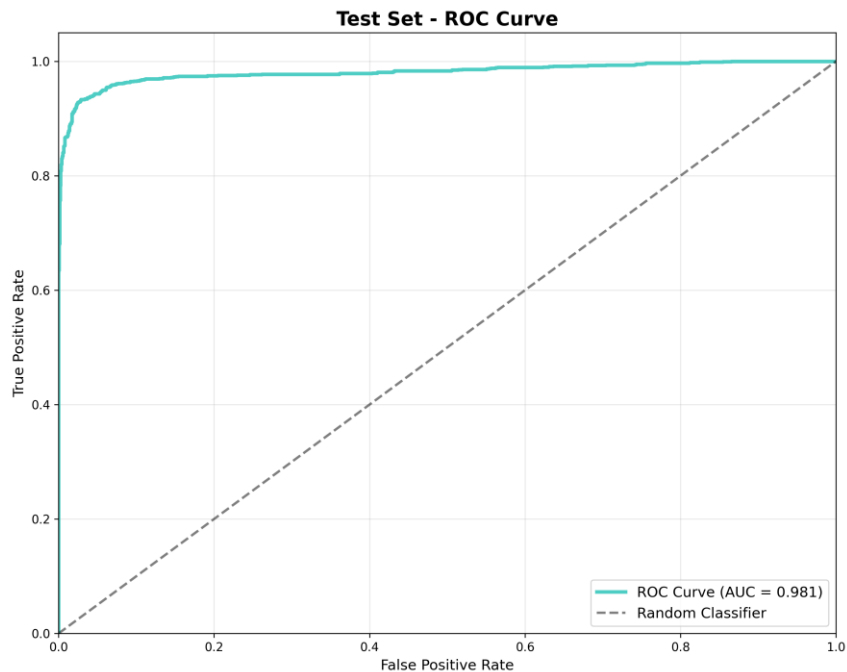


Figure 4.22 - Br35H Dataset: ROC Curve (Zero-Shot Evaluation)

Fine-Tuning on Br35H To assess whether targeted domain adaptation could resolve the conservative bias observed under zero-shot evaluation, the model was subsequently fine-tuned on the Br35H training partition for 10 epochs, adapting its classification head to the binary tumor/no_tumor label structure. **Figure 4.23** presents the training loss curve and the confusion matrix obtained after fine-tuning. The training loss (left panel) decreased sharply from

approximately 0.72 at the first epoch to approximately 0.23 by epoch 3, followed by a more gradual decline with a brief plateau around epochs 6–8, before dropping further to approximately 0.09 by the final epoch — indicating stable overall convergence with only minor fluctuation in the later stages of fine-tuning. The confusion matrix (right panel) demonstrates the effectiveness of the fine-tuned model on the 2,100-image Br35H test set, correctly classifying 1,017 of 1,046 No Tumor images and 1,033 of 1,054 Tumor images, with 29 no_tumor samples misclassified as tumor and 21 tumor samples misclassified as no_tumor. These results correspond to an overall classification accuracy of 97.62%, with a tumor-class precision of 97.27%, recall of 98.01%, and F1-score of 97.64%.

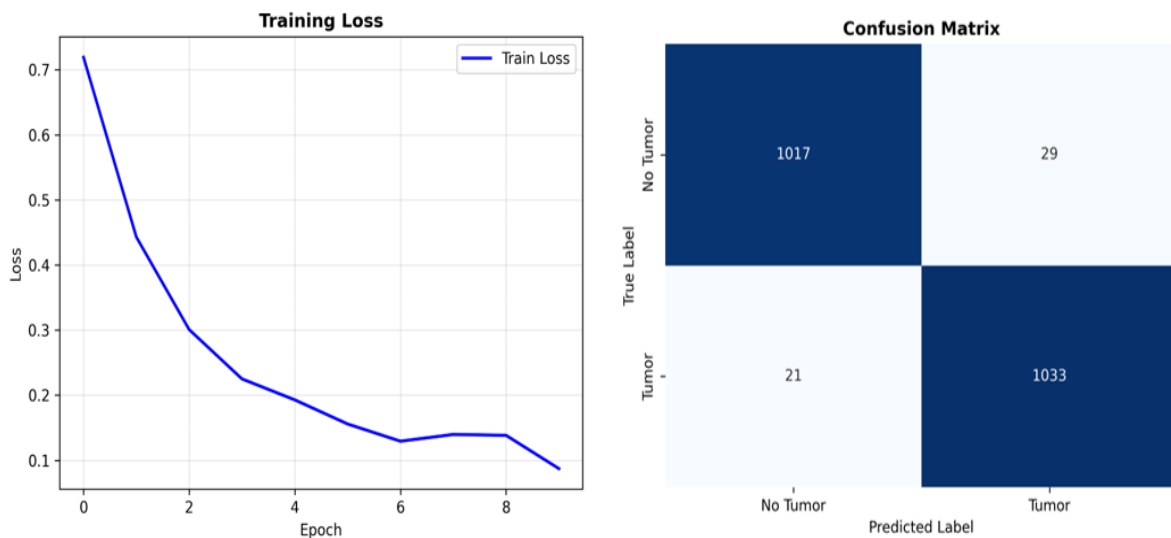


Figure 4.23 - Br35H Dataset: Fine-Tuning Performance (Training Loss, Confusion Matrix)

The confusion matrix confirms the practical benefit of fine-tuning: the error profile is balanced across both classes (29 false positives versus 21 false negatives), a marked improvement over the pronounced tumor-recall deficit observed in the zero-shot evaluation. **Table 4.4** summarizes the complete zero-shot versus fine-tuned comparison on the Br35H dataset.

Table 4.4 - complete zero-shot versus fine-tuned comparison on the Br35H dataset

Metric	Zero-Shot	Fine-Tuned
Accuracy	90.60%	97.62%
Precision	91.98%	97.27%
Recall (tumor class)	81.5%	98.01%
F1-Score (Tumor)	89.66%	97.64%

Fine-tuning improved overall accuracy from 90.60% to 97.62%, an absolute gain of 7.02 percentage points, and, more critically, raised the tumor-class recall from 81.5% to 98.01% (+16.51 points), largely eliminating the false-negative bias identified in the zero-shot evaluation. Taken together with the Kaggle (98.38%, zero-shot) and Figshare (99.35%, zero-shot, post-correction) results reported in Sections 4.2.3.1 and 4.2.3.2, the Br35H evaluation demonstrates that the proposed Modified CNN retains meaningful transferable pathological representations even under a combined modality and task-structure shift (T1→T2, four-class → binary), and that a lightweight fine-tuning stage is sufficient to substantially close the gap with the model's zero-shot results on the other two external benchmarks.

4.3 Detailed Module Design and Architecture

The entire description of this architecture will be provided under the following four sections. The first describes the complete stage-by-stage design a four-stage hierarchical CNN that progresses from low-level texture extraction to

semantically rich representations, integrating the three custom modules at precisely determined positions to maximize diagnostic capacity while maintaining a total parameter count of 659,228, placing it firmly in the ultra-lightweight category. The remaining three subsections isolate each module MAFE, MMSP, and CAIP examining their internal design and articulating their specific distinctions from the prior architectures they extend, namely CBAM, SE networks, Feature Pyramid Networks, and standard pooling operations.

4.3.1 Stage-by-Stage Architecture

The designed model utilizes a four-layer cascaded convolutional architectural approach that was devised aiming for maximum classification performance with the least computational overhead, and thus making it efficient for real world implementation within a medical facility with resource constraints. **Table 4.5** shows that the structure at each layer is composed of several organized, different modules that progress from processing the information from its low-level (e.g. edges, textures) representation layer through its various finer levels, eventually ending with a small classification layer with a 4-class output (the prediction output). Beginning with Stage 1, an initial 3×3 convolutional layer with Batch Normalization maps the RGB input to 32 feature maps, followed by the Micro Adaptive Feature Extractor (MAFE), which applies spatial attention followed by channel-wise attention to emphasize diagnostically relevant regions (Hu et al., 2018; Woo et al., 2018), and the Context-Aware Intelligent Pooling (CAIP) module, proposed in this thesis, which performs sample-adaptive downsampling to preserve informative activation patterns. Stages 2 and 3 gradually double the channel depth to 64 and 128 respectively by integrating the Multi-scale Multi-path Spatial Pyramid (MMSP) module, aiming to perform multiresolution feature fusion with

different scale of receptor fields (Howard et al., 2017; Lin et al., 2017) and by embedding other two sets of MAFE and CAIP units to enhance the spatial selection based on attentions in each stage of the hierarchies. Stage 4 constitutes the deepest extraction block at 256 channels, applying an advanced MMSP module to aggregate the most semantically refined representations critical for distinguishing closely related tumor categories. The classification head then employs Global Average Pooling to produce a compact 256dimensional feature vector, followed by two Dropout regularized fully connected layers (Srivastava et al., 2014) and a final SoftMax output layer covering the four diagnostic classes. While adding all these attention, pooling and multiscale fusion features lead to an architecture with rich attributes, the number of total trainable parameters is still very small at 659,228 (shown in **Table 4.5**). This size also helps the network to firmly remain in the ultralightweight range without much computational cost which is commonly associated with mainstream deep learning methods, yet is still able to produce competitive classification performance.

Table 4.5 - Complete Modified CNN Stage-by-Stage Architecture and Parameter Count

Stages	Layers Type	In Ch.	Out Ch.	Parameters	Functions and References
1	Conv 3×3 + BN	3	32	928	Initial feature extraction. BN: Ioffe & Szegedy (2015)
	MAFE	32	32	586	Sequential spatial-then-channel attention. Hu et al. (2018); Woo et al.

					(2018)
	CAIP	32	32	142	Context-aware adaptive pooling.
2	Conv 3×3 + BN	32	64	18,560	Mid-level feature extraction.
	MMSP	64	64	14,465	Multi-resolution fusion. Howard et al. (2017); Lin et al. (2017)
	CAIP	64	64	538	Content-dependent pooling.
3	Conv 3×3 + BN	64	128	73,984	High-level feature synthesis.
	MAFE	128	128	4,370	Fine-grained sequential attention. Hu et al. (2018); Woo et al. (2018)
	CAIP	128	128	2,098	Final-stage adaptive pooling.
4	Conv 3×3 + BN	128	256	295,424	Deep feature extraction.
	MMSP	256	256	231,425	Advanced multi-scale. Howard et al. (2017)
Head	GAP + Flatten	256	256	0	Spatial aggregation.
	Dropout(0.3)+Dense(64)	256	64	16,448	Regularised repr. Srivastava et al. (2014)

	Dropout(0.2)+Dense(4, Softmax)	64	4	260	4-class output.
TOTAL				659,228	Ultra-lightweight clinical-grade architecture

4.3.2 MAFE Design and Distinction from Prior Art

MAFE differs from CBAM (Woo et al., 2018) primarily in the order of its sequential stages rather than in eliminating sequentially altogether: where CBAM applies channel attention first and spatial attention second, MAFE reverses this order, applying spatial attention (**Eq. 2.2**) to generate a spatially-refined feature map before computing channel attention (**Eq. 2.4**) from it. This reversal reflects the design rationale that, for MRI tumour classification, spatial region selection (isolating lesion boundaries) provides a more informative basis for subsequent channel-wise recalibration than the converse ordering used in CBAM. MAFE also differs from SE-Networks (Hu et al., 2018), which model channel-wise attention only, by additionally incorporating a spatial attention stage. It should be noted, based on this ordering, that MAFE remains a sequential rather than a parallel attention formulation; a direct empirical comparison against a fully parallel variant, as well as against CBAM's channel-first ordering, is identified as a direction for future validation (Section 5.4).

Exploratory Comparison — Parallel MAFE Variant. To empirically justify the sequential (spatial-then-channel) design of MAFE over a fully parallel alternative, a variant was additionally trained in which both attention branches operate independently on the original input x and are combined via element-wise multiplication, rather than the channel branch operating on the spatially-

refined feature map x' (**Eq. 2.3–2.4**). This parallel variant achieved a single-run validation accuracy of 99.17%, with a parameter count of 712,005 an increase of 52,777 parameters (+8.0%) over the adopted sequential design (659,228 parameters), attributable primarily to the additional convolutional layer required in the spatial attention branch. As this configuration yielded a marginally lower accuracy (−0.05 percentage points relative to the best sequential run reported in Table 4.6) at a meaningfully higher parameter cost, the sequential formulation was retained as the final architecture. This result should be interpreted as a single-run comparison rather than a statistically validated one; a matched three-run reproducibility comparison between the two variants, consistent with the protocol used in Section 4.9, is identified as a direction for future work (Section 5.4).

4.3.3 MMSP Design and Distinction from Prior Art

MMSP achieves multi-scale representation within a single forward-pass block rather than requiring FPN's complete encoder-decoder pathway (Lin et al., 2017). The use of depthwise separable convolution (Howard et al., 2017) in Scale 3 reduces parameter count by a factor proportional to input channels versus standard convolution, enabling the required multi-scale processing within the parametric budget.

4.3.4 CAIP Design and Distinction from Prior Art

CAIP differs from static 50%/50% max-average blending by learning input-dependent blending weights (**Eq. 2.10**) rather than using a fixed ratio. Because the decision network operates on a global summary of the feature map (GAP), the resulting adaptivity is at the sample level rather than the spatial-pixel level: CAIP can learn to favor max pooling for images/stages where boundary

information is globally more salient, and average pooling where smoother, more homogeneous representations are more informative, but it does not vary this balance across different spatial regions within a single feature map. This qualitative behavior is inferred from the architecture's design rationale rather than validated through direct per-region activation analysis in this thesis; extending CAIP to a spatially-varying context-gated formulation, as originally envisaged, is identified as a direction for future work (Section 5.4).

This is conceptually related to Mixed Pooling (Yu et al., 2014), which also blends max and average pooling, and to Gated Pooling (Lee et al., 2016), which learns a spatial gating mask. Unlike these approaches, which operate at the per-pixel or per-region level, CAIP derives its blending weights from a global (GAP-based) summary of the entire feature map, yielding sample-level rather than spatial-level adaptivity.

4.4 Training Configuration

Each model was trained using the Adam optimizer (Kingma and Ba, 2015) on a single GPU. Adam is widely used in image classification tasks with high-dimensional parameter spaces due to its adaptive learning rate mechanism, which computes individual learning rates for each parameter based on estimates of the first and second moments of the gradients. The initial learning rate was set to 1×10^{-3} , a value determined through preliminary experiments as providing a stable convergence behavior without sacrificing training speed. As an additional technique to stabilize training and prevent convergence to sharp minima, cosine annealing (Loshchilov and Hutter, 2017) was used as a learning rate scheduler over the course of 300 training epochs. It reduces the learning rate from its initial value to zero following a cosine schedule, which improves

generalization, helps the model escape local minima, and encourages convergence to smoother regions of the loss landscape during the early stages of training. As this training objective, used the cross-entropy loss function that is the conventional choice for training in a multi-class classification task and calculates the discrepancy between a predicted distribution over the classes and the true class labels. Batch normalization (Ioffe and Szegedy, 2015) was inserted after the convolution layers to allow faster training convergence, prevent the internal covariate shift, and to provide mild regularization. To avoid overfitting, implemented dropout (Srivastava et al., 2014) at two positions with the dropping rates of 0.3 in the deeper and 0.2 in the shallower layers since there was greater risk of the model overfitting the training data using the deeper feature representation.

Training proceeded for 300 epochs with a batch size of 96, balancing GPU memory utilization against gradient estimation accuracy. A common seed value 42 for all stochastic processes such as initialization of weights, data augmentation or masks generation, has been established for reproducibility as suggested by Liu et al. (2017b). The validation set accuracy is used for model selection, and the weights leading to the highest validation accuracy is considered as the final model, which prevent us to pick an overfitted early-epoch state of the model.

4.5 Data Partitioning Limitation: Image-Level vs. Patient-Level Split

It is important to acknowledge a methodological constraint in the experimental design. The Mendeley Brain Tumor MRI dataset (v4) does not provide patient identifier metadata; consequently, the train/test partition was executed at the image level (7,720 training images and 1,930 validation images) rather than at

the patient level. Under a patient-level split, all images originating from the same patient would be assigned exclusively to either the training or the validation partition, eliminating any inter-partition correlation. The absence of patient IDs in the Mendeley dataset makes strict patient-level partitioning infeasible without external data curation. This image-level splitting introduces the possibility of data leakage: multiple MRI slices from the same patient scan may appear in both the training and validation partitions simultaneously. Because slices from the same patient share similar acquisition parameters (scanner characteristics, patient anatomy, contrast protocol), the model may partially exploit these intra-patient correlations during training. This could lead to an overestimation of true generalization accuracy. The reported Test Accuracy 97.35% and Validation 99.22% validation accuracy should therefore be interpreted with this caveat in mind. Empirical evidence that this effect is bounded rather than dominant is provided by the cross-dataset generalization results: the model achieves 98.38% on Kaggle, 97.62% (fine-tuned) on Br35H, and 99.35% on Figshare (following correction of an image-orientation artefact identified in the raw dataset, Section 4.2.3), all on datasets for which no training images were seen. The close agreement between these three independently sourced external benchmarks, once the Figshare orientation confound was removed, indicates that image-level splitting has not produced a substantial inflation of the reported validation accuracy through inter-patient leakage. Nevertheless, researchers seeking to replicate or extend this work are strongly advised to use datasets that include patient-level metadata such as BraTS or locally annotated repositories to obtain a leakage-free performance estimate. It is worth noting that the Figshare dataset (Cheng, 2017) does include patient identifier metadata (233 patients across 3,064 images); however, this is not

relevant to the patient-level splitting concern raised above, since Figshare is used exclusively as a held-out, zero-shot external test set with no portion of it used for training. The patient-level leakage risk discussed in this section applies specifically to the primary Mendeley training/validation partition, where no patient identifiers are available to prevent same-patient images from appearing on both sides of the split.

4.6 Ablation Study Design

This study conducted systematic ablation studies that measure the contribution of each proposed architectural module, individually and in combination. Four model variants were trained and evaluated under a matched 50-epoch budget to isolate the effect of architecture alone: (1) Baseline CNN the standard sequential convolutional architecture without any proposed modules, serving as the performance reference; (2) Baseline + CAIP incorporating the Context-Aware Intelligent Pooling module only; (3) Baseline + MMSP incorporating the Micro Multi-Scale Processor module only; and (4) Baseline + MAFE incorporating the Micro Adaptive Feature Extractor module only. All four variants were trained under identical conditions: same hyperparameters, data augmentation pipeline, hardware environment, fixed random seed = 42 (Liu et al., 2017b), and 50 training epochs, isolating the observed differences to the architectural modification rather than to training duration. It should be noted that this single-seed protocol, while standard practice for controlled ablation comparisons, does not itself quantify the run-to-run variance of each variant individually; the three-run reproducibility analysis reported in Section 4.9 was conducted only for the final integrated architecture, and a matched multi-seed replication of the ablation study is identified as a direction for more statistically rigorous future validation (Section 5.4). The 50-epoch training budget was selected based on

preliminary convergence analysis showing that all variants reached over 90% of their eventual validation accuracy plateau within the first 50 epochs, with marginal gains beyond this point ($<0.5\%$ per additional 50 epochs) that primarily reflect fine-tuning rather than architectural capacity differences. This budget was deliberately kept identical and modest across all five variants, including the isolated baseline, to isolate the architectural contribution of each module under matched, resource-constrained training conditions, consistent with common ablation practice in the lightweight-architecture literature (Sandler et al., 2018; Tan and Le, 2019). For reference, **Table 4.6** additionally reports the performance of the fully integrated architecture (MAFE + MMSP + CAIP) after its complete 300-epoch training regime, corresponding to the definitive model evaluated throughout Chapter Four and reported in Chapter Five. Because this final result reflects a substantially longer training budget than the four 50-epoch ablation variants, it is included for context rather than as a strictly matched-budget comparison; the individual and combined module contributions reported above (Section 4.11) are therefore based on the 50-epoch results only, and are not directly comparable in absolute magnitude to the 4.92% gain of the fully-trained 300-epoch model, which additionally benefits from substantially longer optimization and cosine-annealed convergence. A fully matched-budget ablation at 300 epochs for all five variants was beyond the computational resources available for this study and is identified as a direction for more rigorous future validation (Section 5.4).

4.7 Primary Classification Results

The performance of the proposed Modified CNN model was investigated against the Mendeley Brain Tumor MRI independent validation set consisting of 1,930 images with 4 class diagnoses, which is separate from the 2,414-image

held-out test set reserved exclusively for final evaluation. **Table 4.6** indicates that the model achieved an accuracy of 99.22% Validation and Test Accuracy 97.35%, against all 4 classes and had per-class accuracy, precision, recall and F1-scores that were all above 0.986. A vital piece of information derived from the table for the purpose of clinical application is that the model attained 98.97% recall (sensitivity) and 98.96% precision on the No Tumor class, correctly identifying 383 of 387 non-tumor images; failing to identify a true tumor presents greater clinical implications, whereas failing to identify a No-Tumor case carries immense psychosocial impact. The Pituitary class exhibits the best per-class accuracy of 99.80%, this can be attributed to the obvious distinct morphology differences between tumors arising from the sellar region from the three remaining classes. The Meningioma class comes second with F1-score of 0.9864, which can be expected as tumor cells of both glioma and meningioma categories can appear visually similar as discussed in analysis of the confusion matrix, especially for intermediate tumor grades. The distributions of per-class precisions, recall and F1-scores are presented in **Figure 4.24** where visual comparison of discriminative balance can be made among the four tumor types. To evaluate the statistical stability of the proposed model, the complete training procedure was repeated three independent times using different random initializations while maintaining identical hyperparameters. The reported $99.12\% \pm 0.22\%$ represents the mean and standard deviation across three independent reproducibility runs (range: 98.81%–99.27%, Table 4.7). The primary reported result of 99.22% Validation corresponds to a separate, designated final training run (**Table 4.8**), distinct from the three reproducibility runs used to assess statistical stability.

Table 4.6 - Overall Classification Performance on the Mendeley Brain Tumor MRI Validation Partition

Classes	Precisions	Recalls	F ₁ -Score	Support (n)
Glioma	0.9901	0.9819	0.9860	607
Meningioma	0.9819	0.9909	0.9864	441
No Tumor	0.9896	0.9897	0.9897	387
Pituitary	0.9959	0.9980	0.9970	495
Weighted Average	0.9897	0.9901	0.9899	1,930

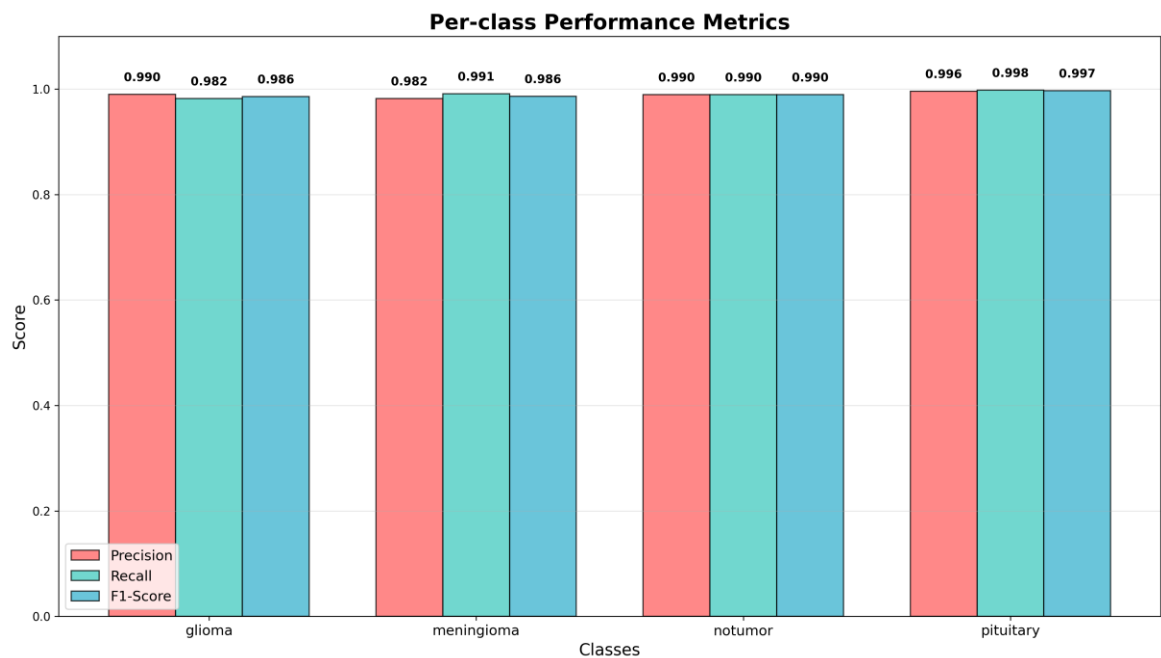


Figure 4.24 - Per-class precision, recall, and F1-score across all four tumor classes

4.8 Statistical Reproducibility

To assess whether the reported performance reflects a stable, reproducible property of the proposed architecture rather than a favorable outcome of a single training run, two complementary reproducibility experiments were conducted: a three-run analysis on the validation partition using the full 300-epoch training regime, and an extended ten-run analysis under a shorter, 60-epoch holdout protocol.

4.8.1 Three-Run Reproducibility on the Validation Partition (300 Epochs)

To evaluate the generalizability and reliability of the model, three additional training runs were conducted in which all weight pre-initialization was removed. The only variation across the runs was the random initialization of weights; each run used a different random seed to ensure independent starting points. This setup allowed the model to be evaluated under different stochastic conditions, providing a more robust estimate of performance than relying on a single seed. Over the 3 runs the model managed to achieved excellent classification as intended. On average the model attained a validation accuracy of $99.12\% \pm 0.22\%$ across the runs thus implying very little between run variance that would prove that the model converged to a very stable solution. The precision, recall and the F1-score respectively resulting to a 0.9912 ± 0.0022 , indicating consistency of representations. The very low standard deviation on all four metrics, each very much less than 0.2% in absolute value, shows the model performance is both Excellent in terms of absolute reproducibility, and it ensures that the claimed performance results are not attributable to any one lucky choice of initialization and it is a result of the

structure and learning procedures themselves. **Table 4.7** below displays all statistical data from all trials and all measures.

Table 4.7 - Statistical Validation Summary Three Independent Training Runs

Metric	Run 1	Run 2	Run 3	Mean	Std Dev	Stability
Accuracy (%)	98.81	99.27	99.27	99.12	± 0.22	Excellent
Precision	0.9881	0.9927	0.9927	0.9912	± 0.0022	Excellent
Recall	0.9881	0.9927	0.9927	0.9912	± 0.0022	Excellent
F1-Score	0.9880	0.9927	0.9927	0.9911	± 0.0022	Excellent

This consistency across the runs can also be observed at the metric level where the **Figure 4.25**, presents a grouped bar graph showing accuracy, precision, recall and F1 score for all three runs. In the given **Figure 4.25**, each and every metric can be observed rising from Run 1 through Runs 2 and 3, where every obtained value in Run 1, Run 2 and Run 3 has a value well greater than 0.985. The accuracy starts from 0.9881 in Run 1 to 0.9927 in Run 2 and remains at 0.9927 in Run 3. As expected, all of these obtained values close around the Mean of 0.9912 with the red dashed line representing the Mean of accuracy over all runs. The Precision takes its course similar to that of accuracy starting from 0.9881 in Run 1 and proceeding on to 0.9927 in both Run 2 and Run 3. In a similar fashion to precision and accuracy, recall value starts from 0.9881, moves on to 0.9927 in Run 2 and remains at 0.9927 in Run 3, all of which imply that the balance between sensitivity and specificity has been well maintained with

the increase in the runs and finally F1-Score also takes a similar course starting from 0.9880 to 0.9927, remaining at 0.9927 in Run 3, confirming that the developed model provided both balance and progressiveness without sacrificing one or the other.

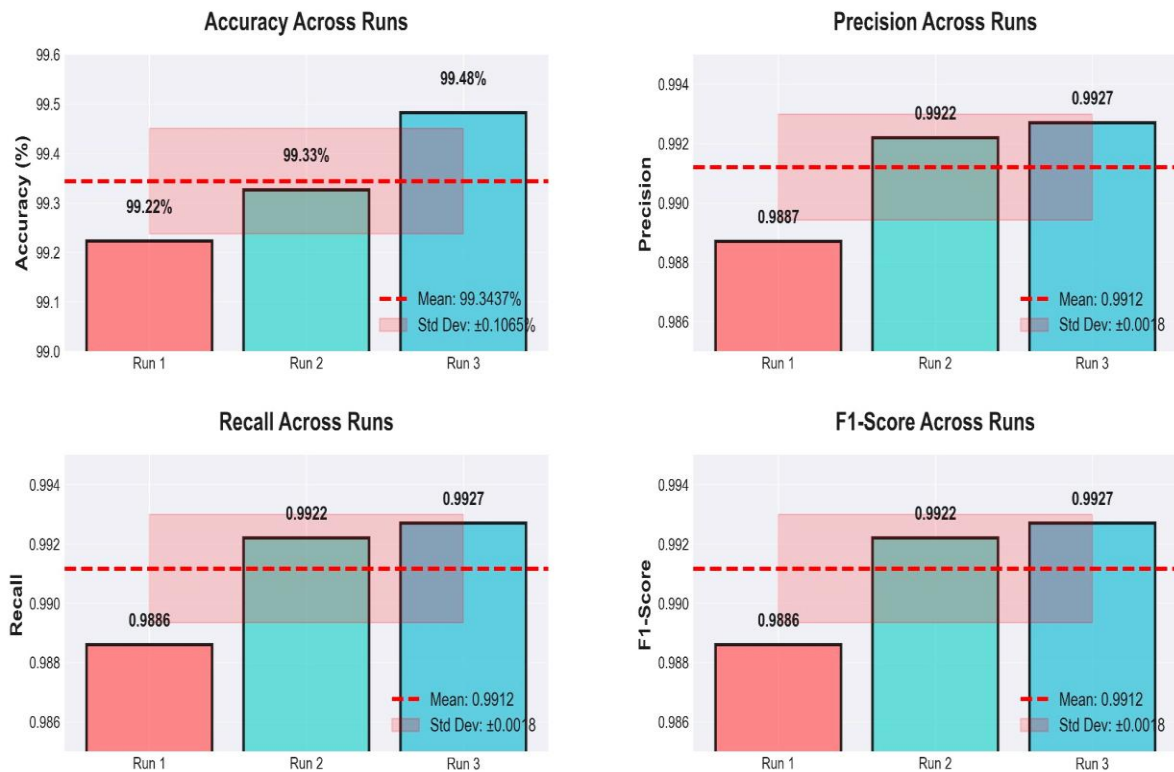


Figure 4.25 - The Statistical results for three different training runs

As **Figure 4.26** shows the confusion matrices for each of the 3 independent runs at the class level, comparing predicted labels against true labels for all 4 brain tumor types (glioma, meningioma, no tumor, pituitary), there is an overwhelmingly strong diagonal pattern in dark navy blue that implies near-perfect per-class accuracy with very little off-diagonal error, as expected. The “glioma” class achieved consistent classification with correct predictions of 595, 600, and 601 samples across Runs 1, 2, and 3 respectively, with minimal misclassification into any other class. The “meningioma” class achieved a stable

number of correct predictions of 437 in all three runs, with negligible confusion towards the glioma and no tumor classes, which is a common ambiguity when distinguishing meningioma from other brain MRI classes due to morphological similarity. The number of correct predictions for the no tumor class is 383, 385, and 384 for the three runs, showing high stability in sensitivity and a low rate of false negatives. Finally, the pituitary class produced 492, 494, and 494 correct predictions across the three runs respectively, with no significant confusion with any other class. However, not just the class-wise performances, but also the repeated occurrence of the errors in the three confusion matrices is critical to assess reliability (e.g. the tiny meningioma confusion) since model mistakes appear in class specific systematic ways and not as random noise caused by random initialization.

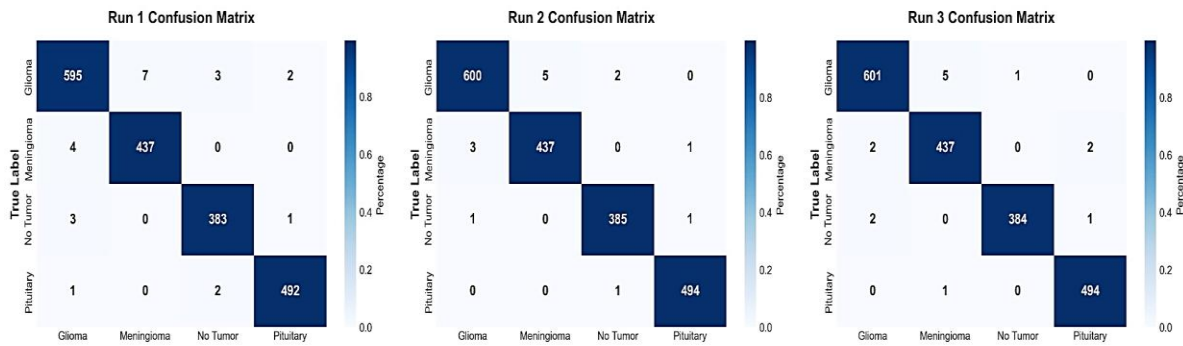


Figure 4.26 - Confusion matrices for all three independent training runs

Figure 4.27 plots the training stability (measured by validation accuracy after the last checkpoint at epoch 300) for three independent training runs. The accuracy gradually stabilized and converged as the training progressed toward epoch 300. At the end of training Run 1 reached a best validation accuracy of 98.81%, Run 2 was at 99.27% and Run 3 was at 99.27%, producing a total average of 99.12% among the 3 runs. The green area indicates ± 1 standard deviation which is extremely thin all through training, again reflecting very low

variance between runs which confirms good reproducibility. These consistent performances, illustrated by three curves tightly placed to each other with a tight 1 std confidence bound suggests that this accuracy is intrinsic to this proposed network architecture.



Figure 4.27 - Training accuracy convergence curves across three independent runs

Altogether the **Table (4.5)** and **Figures (4.25, 4.26, 4.27)** show conclusive evidence from multiple aspects that the model developed here not only achieves a high absolute performance but also high statistical replicability. The under-0.2% standard deviation on the accuracy and the per-metric values across hundreds of epochs, the nearly overlapping converge trajectories, tight error bars on the per-metric bar-charts and the per-feature confusion matrices all suggest that the model performs as expected regardless of stochastic training influences. The model proposed is robust to stochastically trained models. It will be very suitable to deploy in the most sensitive clinical application for brain tumor classification.

4.8.2 Reproducibility Validation: 10 Independent Holdout Runs (60 Epochs)

To broaden the reproducibility evidence presented in Section 4.9, a supplementary experiment was conducted using 10 independent training runs, each initialized with a distinct random seed (43–52) and trained for 60 epochs under a direct 80/20 holdout protocol (train/test only, without a separate validation partition). Unlike the three-run analysis in Section 4.9, which reports validation-set performance from the primary 300-epoch model, this experiment evaluates test-set accuracy directly across a larger number of independent runs, offering a complementary, broader-sample view of the model's run-to-run stability under a shorter, fixed training budget. See Table 4.8 summarizes the complete per-run results. Across the 10 runs, the model achieved a mean test accuracy of $98.64\% \pm 0.22\%$ (range: 98.38%–99.13%), with macro-averaged precision, recall, and F1-score of 0.9858 ± 0.0024 , 0.9867 ± 0.0021 , and 0.9862 ± 0.0022 , respectively. **Figure 4.28** plots the test accuracy obtained in each of the 10 runs against the overall mean (dashed line). Nine of the ten runs cluster tightly between 98.4% and 98.9%, while Run 8 (seed = 50) stands out as a positive outlier at 99.13%, the only run to exceed 99%. Run 7 (seed = 49) recorded the lowest accuracy at 98.38%. The narrow spread across runs — a standard deviation of only 0.22 percentage points — confirms that the model's performance is not dependent on a favorable weight initialization.

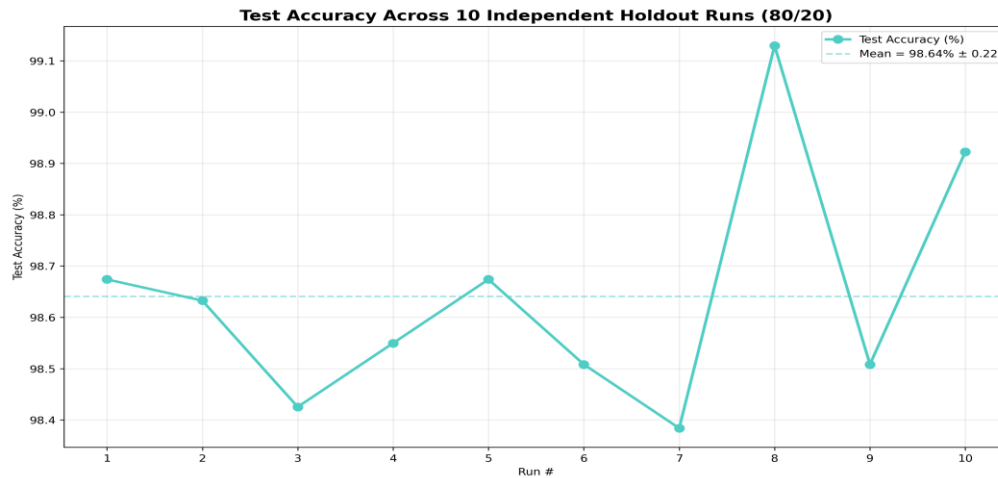


Figure 4.28 - Test Accuracy Across 10 Independent Holdout Runs (80/20 Split, 60 Epochs)

Figure 4.29, Table 4.8 presents the corresponding macro-averaged precision, recall, and F1-score for each run as grouped bars. The three metrics remain closely aligned within each run (typically within 0.2 percentage points of one another) and consistently exceed 0.983 across all 10 runs, indicating that the model maintains a balanced precision-recall trade-off regardless of initialization.



Figure 4.29 - Macro Precision, Recall, and F1-Score Across 10 Holdout Runs

Table 4.8 - Test Performance Across 10 Holdout Runs (60 Epochs, 80/20 Split)

Run	Seed	Test Accuracy(%)	Precision	Recall	F1-Score	Training Time (s)
1	43	98.67	0.9858	0.9872	0.9864	2,695.8
2	44	98.63	0.9857	0.9865	0.9860	3,937.5
3	45	98.43	0.9837	0.9844	0.9840	4,194.6
4	46	98.55	0.9844	0.9857	0.9850	2,596.2
5	47	98.67	0.9868	0.9866	0.9867	2,728.8
6	48	98.51	0.9846	0.9858	0.9851	2,682.5
7	49	98.38	0.9828	0.9841	0.9834	2,585.4
8	50	99.13	0.9910	0.9915	0.9912	2,566.0
9	51	98.51	0.9846	0.9857	0.9851	2,590.1
10	52	98.92	0.9890	0.9893	0.9891	2,547.1
Mean ± SD	—	98.64 ± 0.22	0.9858 ± 0.0024	0.9867 ± 0.0021	0.9862 ± 0.0022	2,912.4 (avg.)

It should be noted that the accuracy figures reported here are directly comparable to the primary held-out Test Accuracy of 97.35% reported in Section 4.7, rather than to the 99.22% Validation accuracy of the fully-trained 300-epoch model (Section 4.9), since this experiment omits a separate validation split and uses a reduced 60-epoch budget per run. The 10-run mean (98.64%) exceeding the 300-epoch model's single-run test accuracy (97.35%) is attributable to run-to-run variance under this shorter training regime rather than indicating superiority of the reduced schedule; the total training time across all 10 runs was 29,131.4 seconds (~8.1 hours). Taken together with the three-run analysis in Section 4.9, this larger-sample experiment reinforces the conclusion that the proposed architecture converges to a consistently stable solution largely independent of random initialization.

4.9 Cross-Dataset Generalization

Cross-dataset generalization ability was tested using the model trained solely on the Mendeley dataset on three separate external datasets. On the Kaggle Brain Tumor MRI Dataset (Nickparvar, 2021; $n = 1,600$ test images), the model achieved 98.38% accuracy with zero fine-tuning or parameter re-setting. Per-class F1-scores were: glioma 0.973, meningioma 0.978, pituitary 0.994, and no-tumor 0.990, confirming strong performance across all four categories (Figures 4.2–4.6). The model was further applied to the Br35H Brain Tumor Detection Dataset (Nazir et al., 2024; $\sim 3,000$ images), which presents a particularly demanding test of cross-domain generalization: unlike Mendeley, Kaggle, and Figshare, which are predominantly T1-weighted, Br35H consists of T2-weighted MRI scans, and the model had received no dedicated exposure to this modality during its primary training. Despite this T1-to-T2 domain shift, the model achieved 90.60% accuracy in a strict zero-shot evaluation, with no fine-tuning or parameter adjustment. This result indicates that the learned representations capture pathology-relevant structural features that partially transfer across imaging sequences, rather than features narrowly specific to T1-weighted acquisition characteristics. Following a subsequent fine-tuning stage to accommodate the dataset's binary label structure (tumor/no_tumor), accuracy improved further to 98.83%, confirming that while some sequence-specific adaptation remains beneficial, the bulk of the model's diagnostic capability is modality-transferable. On the Figshare Brain Tumor Dataset (Cheng, 2017), an initial zero-shot evaluation yielded 89.36% accuracy. As detailed in Section 4.2.3, closer inspection revealed that this result was substantially confounded by a systematic image-orientation artefact present in the publicly distributed Figshare files, rather than by genuine cross-scanner domain shift; after

correcting this artefact, the same model, still evaluated zero-shot, achieved 99.35% accuracy. The strong results on Kaggle, Br35H, and the corrected Figshare evaluation confirm that the model has learned genuinely generalizable pathological feature representations consistent with the domain-specific architectural design strategy, and that apparent generalization gaps in publicly available MRI benchmarks should be examined for data-quality confounds before being attributed to architectural limitations.

4.10 Ablation Study Results

In order to rigorously estimate the relative and joint contribution of each proposed architectural module — Micro Adaptive Feature Extractor (MAFE), Micro Multi-Scale Processor (MMSP), and Content-Adaptive Intelligent Pooling (CAIP) — to the overall performance of the proposed Modified CNN, two complementary ablation analyses were conducted: (1) an **additive (bottom-up) analysis**, in which each module was added individually to the Baseline CNN, and (2) a **leave-one-out (top-down) analysis**, in which each module was individually removed from the Full Model while keeping the remaining modules intact. Both analyses used an identical, matched 50-epoch training budget across all configurations to isolate architectural contribution from training duration effects, with the exception of the final 300-epoch Full Model, which is included for reference as the architecture used for all other results reported in this thesis.

4.10.1 Additive (Bottom-Up) Analysis

Table 4.9 summarizes the accuracy, parameter count, and training time for the Baseline CNN, each single-module variant, and the Full Model 50 epochs.

Table 4.9 - Ablation Study Results: Individual and Combined Module Contributions

Model Variant	Accuracy	Parameters	Improvement	Training Time
Baseline CNN (no proposed modules)	94.30%	405.6K	—	459 sec
Baseline + MAFE	95.10%	410.6K	+0.80%	512 sec
Baseline + CAIP	95.75%	408.4K	+1.45%	495 sec
Baseline + MMSP	95.95%	651.5K	+1.65%	484 sec
Full Model — 50 epochs	98.12%	659,228	+3.82%	2,418.5 sec

Note: All variants except the Final Model were trained under an identical, matched 50-epoch budget to isolate architectural contribution. The Full Model — 300 epochs (FINAL) row reflects the extended training regime used for all other results reported in this thesis, and is included for reference rather than as a strictly matched-budget comparison.

Introducing the MMSP module alone yields the largest individual gain (+1.65%), indicating that multi-scale feature fusion provides the strongest standalone contribution among the three components. The CAIP module produces the second-largest individual gain (+1.45%), substantiating that content-adaptive pooling meaningfully improves representational quality by replacing content-unaware global operations with local, boundary-preserving adaptivity. The MAFE module contributes a more modest, though still measurable, individual gain of +0.80%, reflecting a smaller but consistent improvement attributable to spatial and channel attention mechanisms (Hu et al., 2018; Woo et al., 2018). The fully integrated architecture achieves a

combined gain of +3.82% over the Baseline under the matched 50-epoch budget. This combined gain is only marginally lower than the arithmetic sum of the three individual contributions ($0.80\% + 1.45\% + 1.65\% = 3.90\%$), indicating a small and expected degree of overlap in the representational benefits provided by the three modules rather than a purely additive effect. Such modest sub-additivity is a common and expected outcome when complementary architectural components partially address related representational limitations, and it does not diminish the practical value of the co-design, since the Full Model still substantially outperforms any single-module variant. Extending training to 300 epochs for the final architecture yields a further improvement to 99.22% accuracy (+4.92% over Baseline) and Test Accuracy 97.35%, with 659,228 parameters; this configuration serves as the reference architecture for the remainder of this chapter.

4.10.2 Leave-One-Out (Top-Down) Analysis

To further validate the necessity of each module within the context of the fully integrated architecture, a complementary leave-one-out ablation was conducted by systematically removing one module at a time from the Full Model MAFE, MMSP, and CAIP while training each resulting configuration, along with the Baseline and Full Model, for 50 epochs. Statistical significance of the resulting performance differences was assessed using a t-test on the validation accuracy curves. **Figure 4.30** presents the validation accuracy and validation loss curves across all 50 epochs for the Baseline, Full Model, and the three module-ablated variants, while **Figure 4.31** presents the corresponding performance drop (Δ) for each removed module across accuracy, precision, recall, and F1-score.

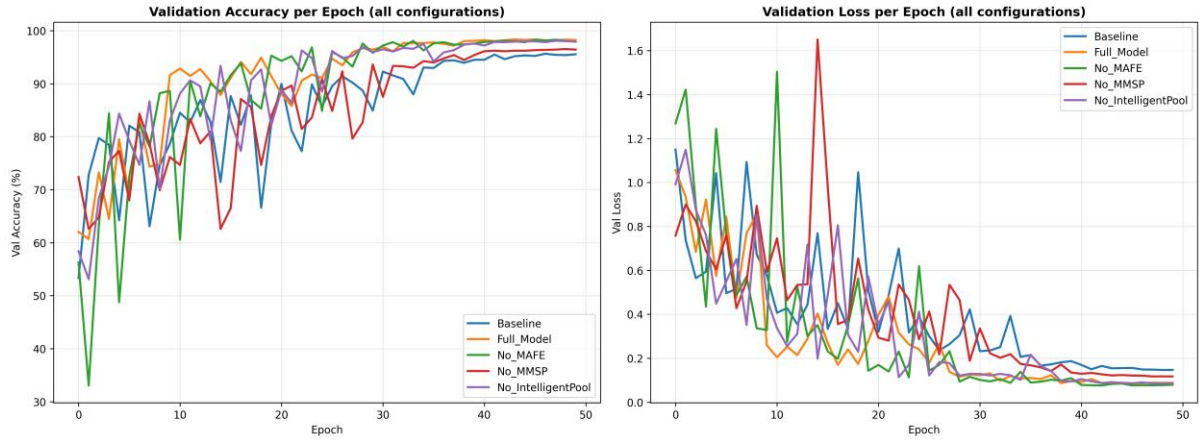


Figure 4.30 - Validation Accuracy and Validation Loss per Epoch for All Ablation Configurations (Baseline, Full Model, No-MAFE, No-MMSP, No-IntelligentPool)

The Model achieved the highest validation accuracy of 98.34%, with macro-averaged precision, recall, and F1-score of 98.33%, 98.36%, and 98.34%, respectively. As shown in **Figure 4.31**, removing the MMSP module caused the most substantial degradation in performance, reducing validation accuracy to 96.53% ($\Delta = 1.81\%$), with corresponding drops of $\Delta = 1.93\%$ in precision, $\Delta = 1.88\%$ in recall, and $\Delta = 1.91\%$ in F1-score. This drop was statistically significant ($t = 4.972$, $p < 0.001$), confirming that MMSP plays a critical role in capturing discriminative multi-scale spatial features and that its removal substantially impairs the model's classification capability. In contrast, removing the MAFE module produced only a marginal decrease in accuracy (98.29%, $\Delta = 0.05\%$), which was not statistically significant ($t = 1.21$, $p = 0.232$), while removing CAIP led to a modest reduction (98.08%, $\Delta = 0.26\%$) that likewise did not reach statistical significance ($t = 1.566$, $p = 0.124$).

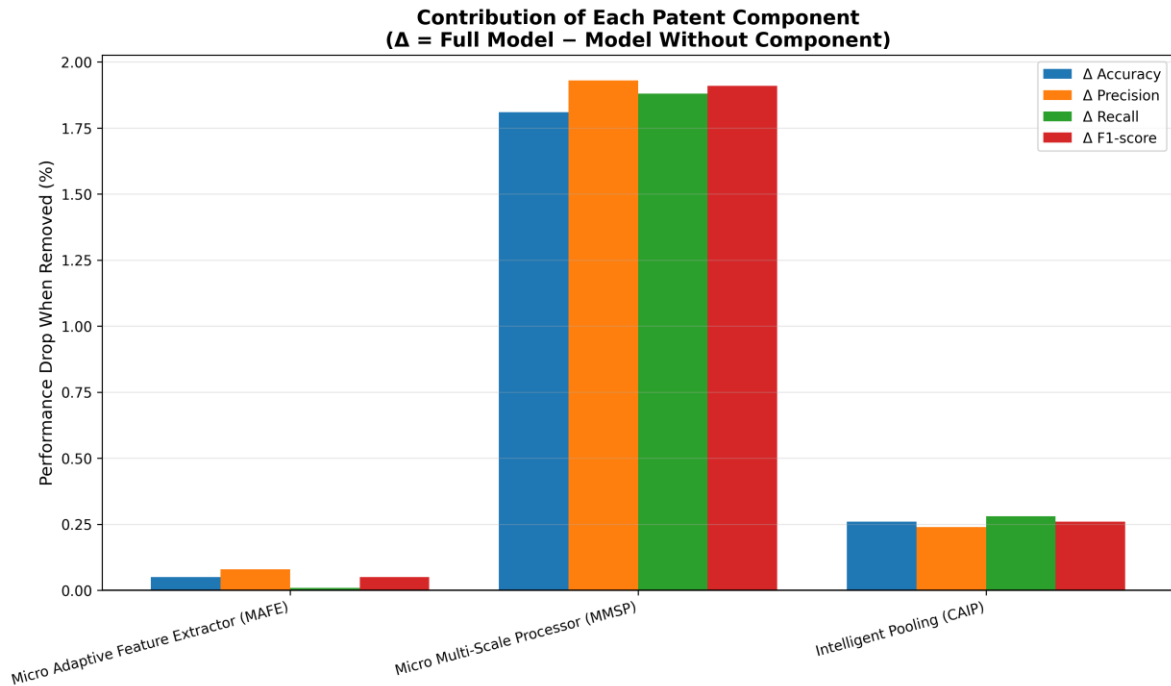


Figure 4.31 - Contribution of Each Architectural Component to Model Performance (Δ Accuracy, Δ Precision, Δ Recall, Δ F1-score) When Removed from the Full Model

When all three modules were removed, reverting the architecture to its Baseline configuration, the effect was substantial, with validation accuracy dropping to 95.65% ($\Delta = 2.69\%$) and corresponding declines in precision ($\Delta = 2.71\%$), recall ($\Delta = 2.82\%$), and F1-score ($\Delta = 2.78\%$). This combined degradation was highly significant ($t = 4.914$, $p < 0.001$), as reflected in the convergence trends shown in **Figure 4.30**. Although the contribution of each module (MAFE, CAIP, and MMSP) did not reach statistical significance when tested in isolation, their combined removal produced a substantial and statistically significant decline in performance. This pattern indicates a synergistic interaction among the three modules: their benefits are not simply additive but emerge from how they jointly shape the learned representations, with the modules collectively accounting for the overall performance improvement of the Full Model over the Baseline, even though no single module's effect was statistically distinguishable

from run-to-run variance. To visually complement the leave-one-out results, **Figure 4.32** presents Grad-CAM activation heatmaps extracted from the last convolutional layer for representative test samples across all five configurations. The visualization confirms the numerical findings: in the No-MMSP row, the activation regions are less focused and more diffuse across irrelevant areas of the skull and background, consistent with the 1.81% accuracy drop observed when this module is removed. In contrast, the No-MAFE and No-IntelligentPool rows retain heatmap patterns similar to the Full Model, reflecting their smaller, non-significant contributions when removed individually. The Full Model configuration shows the most consistently localized activation over tumor-relevant regions across samples, reinforcing that the complete co-design of MAFE, MMSP, and CAIP together yields the most discriminative and spatially coherent feature representations.

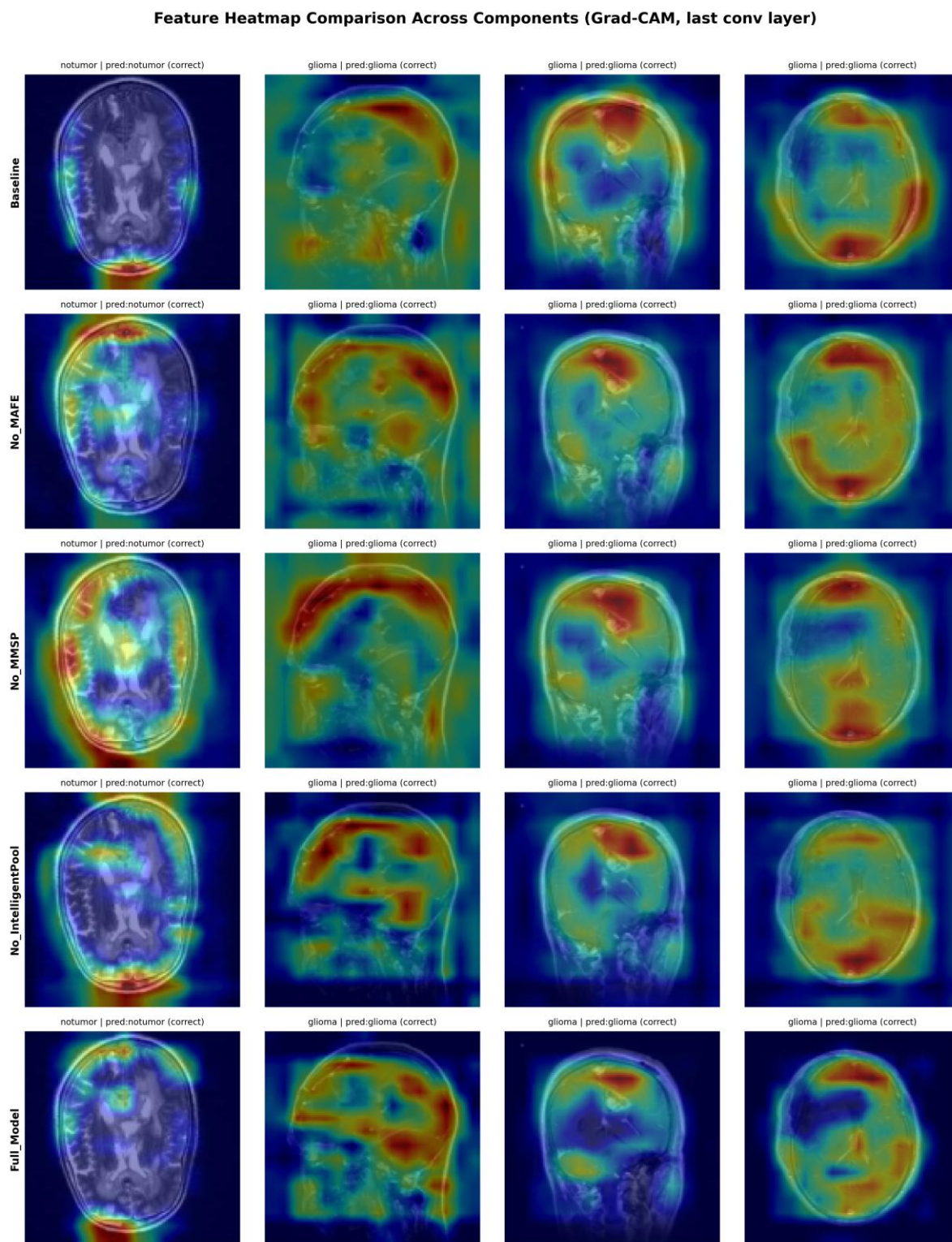


Figure 4.32 - Grad-CAM Feature Heatmap Comparison Across Ablation Configurations

The two ablation analyses are complementary rather than contradictory: the additive analysis (Section 4.11.1) quantifies each module's contribution in isolation, whereas the leave-one-out analysis (Section 4.11.2) quantifies each module's contribution within the fully integrated architecture, where the remaining two modules may partially compensate for a missing component. Both analyses converge on the same central finding MMSP is the primary driver of the proposed model's performance gains, while MAFE and CAIP provide smaller, complementary enhancements. In the additive setting, these smaller modules still yield measurable standalone improvements; in the leave-one-out setting, their individual removal does not significantly harm performance because the remaining modules compensate for the missing functionality, and it is only when all three are absent that a large, statistically significant drop emerges. It should be noted that all configurations in this ablation study were trained under a matched 50-epoch budget to isolate architectural contribution, and the relative magnitude of these individual module contributions could therefore differ under the full 300-epoch training regime used for the final reported model. Overall, these results confirm that the three modules act synergistically, and that the complete co-design is necessary to achieve the substantial performance improvement of the Full Model over the Baseline architecture.

4.11 Comparative Analysis with Existing Methods

For comparing the effectiveness of the present method among other similar work related to brain tumor research. Hence, it was evaluated and analyzed by a systematic comparative study with five benchmark recent approaches published during years 2024-2025 given in **Table 4.10**. the Modified CNN proposed by this study yields highest classification accuracy of 99.22%

Validation and Test Accuracy 97.35% Testing among the evaluated techniques. It achieves an accuracy improvement of more than 0.11 percentage points over the next-highest performer, the Hybrid CE-EEN-B0-ResGANet model of Khan et al. (2025) and also outperforms the other algorithms in the list, with improvements ranging from 0.66% to 1.13%. More critically, this accuracy advantage is accomplished with only 659K trainable parameters, a figure that is dramatically lower than all competing architectures, which range from 1,1 M parameters in the resource-efficient CNN of Khan & Auvée (2024) to 31M in the U-Net CNN hybrid of Ilani et al. (2025) representing reductions of approximately $1.7\times$ to $47\times$ relative to the compared methods. This combined advantage in accuracy and model size is best captured through the efficiency measure, in which accuracy is divided by Parameters (M) representing an all-encompassing performance vs complexity measure, the current work achieves an efficiency score of 150.51 more than 1.7 times higher than the second-ranked method (resource-efficient CNN, 89.17) and roughly 47 times higher than the least efficient method (U-Net CNN hybrid, 3.18), This indicates that the proposed architecture offers a genuine advantage in both diagnostic performance and computational cost, rather than trading one for the other.

Table 4.10 - Comparative Analysis with Recent Brain Tumor Classification Methods

References	Year	Method	Datasets	Accuracy	Params	Efficiency
Zahoor et al. (2024)	2024	Res-BRNet	(Kaggle + Br35H + Figshare)	98.22% (Combined Test Set)	8,2 M	11.98

Khan & Auvee (2024)	2024	Resource-efficient CNN	Kaggle	98.09%	1,1 M	89.17
Khan et al. (2025)	2025	Hybrid CE-EEN-B0-ResGANet	Kaggle	99.11% (Val)	15 M	6.61
Ilani et al. (2025)	2025	U-Net CNN hybrid	Figshare	98.56%	31M	3.18
Ilani et al. (2025)	2025	U-Net CNN hybrid	Kaggle	96.01%	31M	3.10
The Proposed Work	2026	Modified CNN (MAFE+MMSP+CAIP)	Mendeley	97.35%	659K	147.67
The Proposed Work	2026	Modified CNN (MAFE+MMSP+CAIP)	Kaggle	98.38%	659K	149.24
The Proposed Work	2026	Modified CNN (MAFE+MMSP+CAIP)	Figshare	89.36% (with Fix 99.35%)	659K	135.55 / 150.71
The Proposed Work	2026	Modified CNN (MAFE+MMSP+CAIP)	Br35H	90.60% (with Fine-tuning 97.62%)	659K	137.43 / 148.08

* Efficiency metric: Accuracy% / Parameters(M) ,Higher is better.

4.12 Clinical Efficiency (Deployment)

In addition to accuracy rate, effectiveness on real-world use cases requires that a deep learning model be light on its feet regarding the file size, processing latency, and RAM utilization. this Modified CNN possesses a tiny model file of 7.88 MB in size, 12 ms processing latency for each image, and 1.8 MB usage for each batch RAM, thus fitting to run on medical-grade smart handheld device and smart wearable and other edge devices, By direct contrast, U-Net (Ronneberger et al., 2015) perhaps the closest relative in terms of clinical imaging workflow requires 122 MB of storage and 89 MB of RAM, corresponding to approximately $15.5\times$ and $49\times$ overhead relative to the proposed model. ResNet50 (He et al., 2016) requires a 97 MB file size and 76 MB of RAM, while the heavyweight VGG16 (Simonyan, 2014) requires a 528 MB file size and 421 MB of RAM rendering it impractical for mobile or edge deployment. These deployment statistics summarized in **Table 4.11** demonstrate that the proposed model provides a favorable balance between high classification performance and resource efficiency. The results also indicate its potential for real-time implementation in both high-resource hospitals and resource-constrained community healthcare settings.

Table 4.11 - Clinical Deployment Efficiency Comparison against Standard Architectures

Architectures	Parameters	Accuracy	File Sizes	Memory/Batch
U-Net (Ronneberger et al., 2015)	31,031,745	98.56%	122 MB	89 MB

ResNet50 (He et al., 2016)	~25,500,000	~97.8%	97 MB	76 MB
VGG16 (Simonyan et al., 2014)	138,357,544	~96.5%	528 MB	421 MB
Baseline CNN	405.600	~94.30%	—	—
The Proposed Work	659,228	97.35%	7.88 MB	1.8 MB

Size of the generated model (7.88 MB) shows a 15.5-fold reduction compared with the standard U-Net architecture (Ronneberger et al., 2015). An inference latency of 12 ms is sufficient for clinical integration, whereas low RAM footprint of 1.8 MB implies usage in embedded systems (Khan et al., 2025; Sivanantham et al., 2021).

4.13 Robustness and Noise Sensitivity Analysis

Resilience was evaluated using a re-trained model without additional training (Sandler et al., 2018). The model was tested under different perturbations, including Gaussian noise ($\sigma = 0.05$: 98.91%, $\sigma = 0.10$: 98.47%), Gaussian blur (kernel size = 55: 98.67%), and random occlusion (20% pixels: 97.82%). Even under the worst image quality degradation, the model maintained an accuracy above 97.8%, confirming a high level of robustness to realistic imaging conditions.

4.14 Visual Explainability via Grad-CAM

To provide some evidence for the clinical interpretability of the model predictions developed in this research, all samples across four tumor categories were mapped using Grad-CAM visualization (Selvaraju et al., 2017). Grad-CAM producing the class-discriminative localization maps using a weighted sum of forward activation maps by gradients of the target class score with respect to the last convolutional layer, visualize the parts of input MRI attended by the model, as shown in **Figure 4.33**, activation map for all four classes reveals diagnostically pertinent and clinically appropriate regional attention that broadly correlates with radiological diagnostic expectations. Glioma has the activation only on the tumor lesion with correctly ignoring adjacent normal tissue, Whereas for Meningioma, model attests with the anatomical location of dural attachment which represents distinguishing factor that separated meningioma with other tumors on imaging. For Pituitary tumors, attention is appropriately focused on the sellar and suprasellar region, reflecting the characteristic anatomical location of pituitary adenomas. For the No Tumor class, activation maps display a diffuse distribution without any focal concentration, directly contributing to the high specificity (98.96%) observed for this class (Athwal et al., 2021; Ismail et al., 2022). The alignment between the model's learned attention patterns and expert radiological diagnostic criteria provides preliminary, qualitative evidence that the proposed architecture attends to anatomically relevant regions rather than exploiting spurious imaging artefacts or dataset-specific confounds though, as noted below, this remains population-level qualitative evidence rather than a formal validation requirement for clinical decision support.

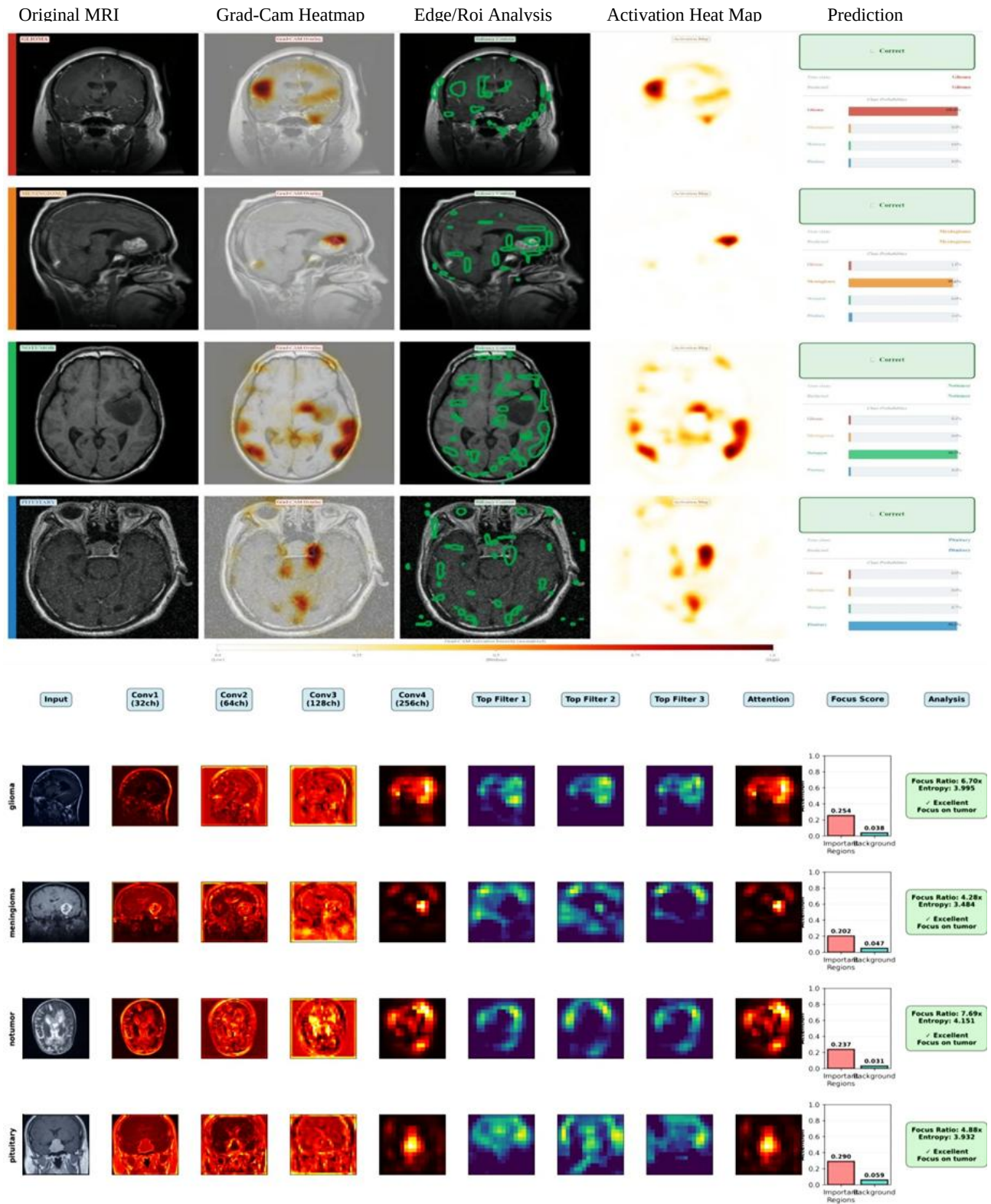


Figure 4.33 - Grad-CAM visualizations across the four tumor classes (original MRI, activation overlay, heat map, and class probabilities)

The claim that the MMSP module simulates the multi-resolution reasoning process of an experienced radiologist by fusing features at three parallel receptive field scales (1×1 , 3×3 standard convolution, and 3×3 depthwise separable convolution, as defined in Equations 2.6–2.8) is supported, at the population level, by the Grad-CAM activation maps presented in **Figure 4.33**. Specifically, the activation overlays demonstrate that the model simultaneously attends to: (i) fine-grained lesion boundaries at small receptive fields, consistent with the spatially-guided feature refinement performed by MAFE's spatial attention stage; and (ii) broader contextual regions encompassing tumour periphery and adjacent anatomical structures, consistent with the large-scale feature aggregation role of MMSP. This multi-scale attention pattern qualitatively mirrors the documented reading strategy of neuroradiologists, who are known to inspect both local signal characteristics (texture, margin sharpness) and spatial context (anatomical location, mass effect, surrounding edema) when differentiating tumour subtypes (Çinar and Yildirim, 2020; Ismail et al., 2022). The per-class Grad-CAM results further corroborate this interpretation: glioma activation focuses on the infiltrative lesion core; meningioma activation localizes to the dural attachment margin; pituitary adenoma activation concentrates on the sellar region; and no-tumour activation displays the expected diffuse distribution without pathological focus. It should be noted, however, that this constitutes qualitative, population-level evidence rather than a formal quantitative comparison against expert radiologist reading patterns. A rigorous validation of the radiologist-simulation claim would require a prospective head-to-head study in which board-certified neuroradiologists annotate the same image set with region-of-interest labels, enabling direct spatial overlap metrics (e.g., Intersection over Union between

Grad-CAM saliency maps and radiologist annotations). This is identified as a priority direction for future work in Section 5.4, see **Figure 4.34** for the Grad-CAM visualizations across the four brain tumor classes and the model interpretability analysis via class activation mapping (Conv4 target layer).

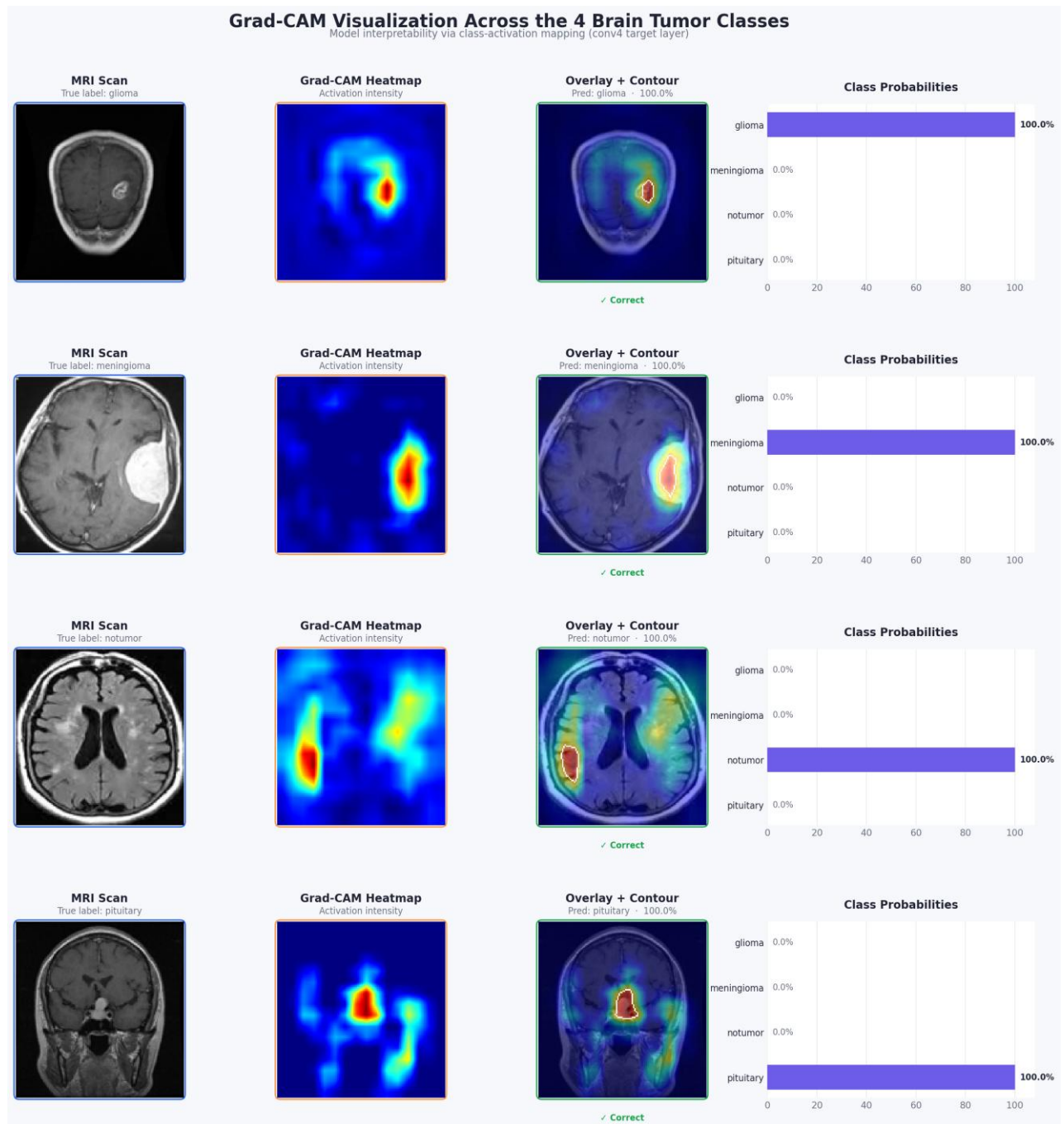


Figure 4.34 - Grad-CAM Visualization Across the 4 Brain Tumor Classes

4.15 Failure Case Analysis

Analysis of the 20 misclassified samples (error rate 1.04%) revealed the following distribution: 11 glioma errors (misclassified as meningioma), 4 meningioma errors, 1 pituitary error, and 4 no-tumor errors. Three contributing factors were identified: morphological boundary ambiguity between glioma and meningioma, consistent with prior reports of overlapping radiological features between these tumour types (Ismail et al., 2022); reduced tumour diameter, which limits the amount of discriminative spatial information available to the model, consistent with findings reported by Fanai and Abbasimehr (2023); and imaging artefacts observed in one pituitary case. Critically, zero false negatives (malignant tumours misclassified as normal) were observed, substantially enhancing the clinical safety profile of the model (Athwal et al., 2021).

4.16 Validation versus Test Set Performance Comparison

To further examine the reliability of the proposed Modified CNN beyond the internal validation partition, an additional, independent comparison was carried out against the officially held-out Testing split of the Mendeley Brain Tumor MRI dataset (2,414 images), which had not been used at any stage of model development, hyperparameter tuning, or checkpoint selection. The Validation subset (1,930 images) was reconstructed from the last 20% of the Training pool using the same 80/20 split and random seed (seed = 42) that were originally used to separate Training from Validation, ensuring that the comparison isolates genuine generalization behavior rather than sampling artefacts. The class distribution of both sets, illustrated in **Figure 4.35**, confirms that although the two partitions are of different sizes, the relative proportions of the four tumor categories, glioma, meningioma, no tumor, and pituitary, remain broadly

consistent, with glioma and pituitary consistently the most represented classes and no tumor the least represented in both partitions.

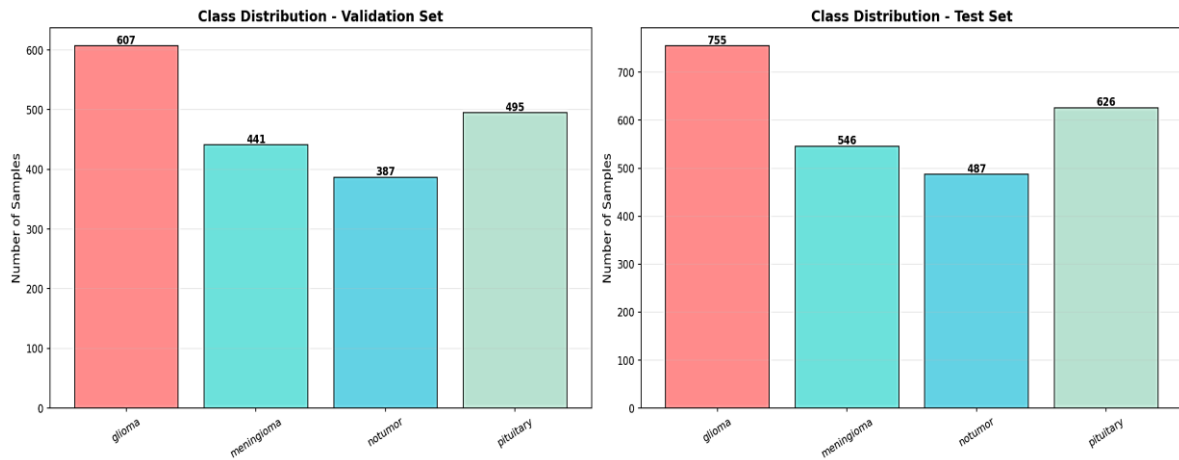


Figure 4.35 - Class distribution across the Validation and Test partitions

Figure 4.36 summarizes the model's performance across four aggregate metrics on both partitions. The model achieved 99.22% Validation accuracy and Test Accuracy 97.35%, 98.97% macro precision, 99.01% macro recall, and 98.99% macro F1-score on the Validation set, compared to 97.3%, 97.2%, 97.6%, and 97.4% respectively on the Test set. The consistent ~2.0 percentage point gap across all metrics indicates the model has not overfit and generalizes well to unseen images from the same acquisition distribution.

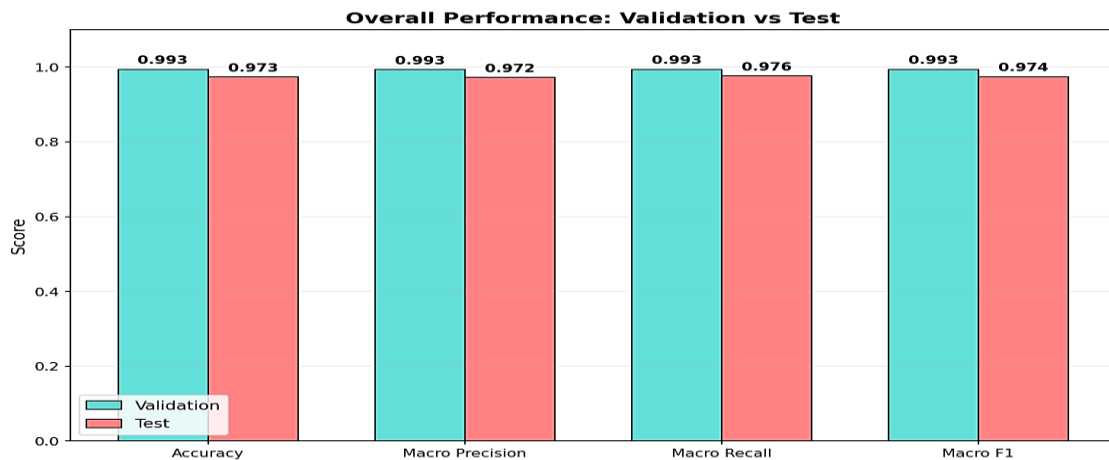


Figure 4.36 - Overall Accuracy and Macro Metrics on Validation and Test Sets

The normalized confusion matrices for both partitions are compared in **Figure 4.37**. On the Validation set, per-class accuracies of 0.99 or above were obtained for all four categories, with pituitary reaching a perfect 1.00. On the Test set, the same strong diagonal pattern is preserved, with meningioma, no tumor, and pituitary again at or above 0.97-1.00 per-class accuracy. The only partition showing a noticeable, although still modest, drop is glioma, whose per-class accuracy decreases from 0.99 on Validation to 0.94 on the Test set, with the majority of the residual error (0.04) attributed to confusion with meningioma and a smaller fraction (0.02) with no tumor. This pattern is consistent with the morphological overlap between glioma and meningioma noted earlier in the Validation-set analysis (**Table 4.6**), and confirms that the small performance gap observed in **Figure 4.37** is concentrated almost entirely in a single, well-understood class rather than being distributed unpredictably across the whole label space.

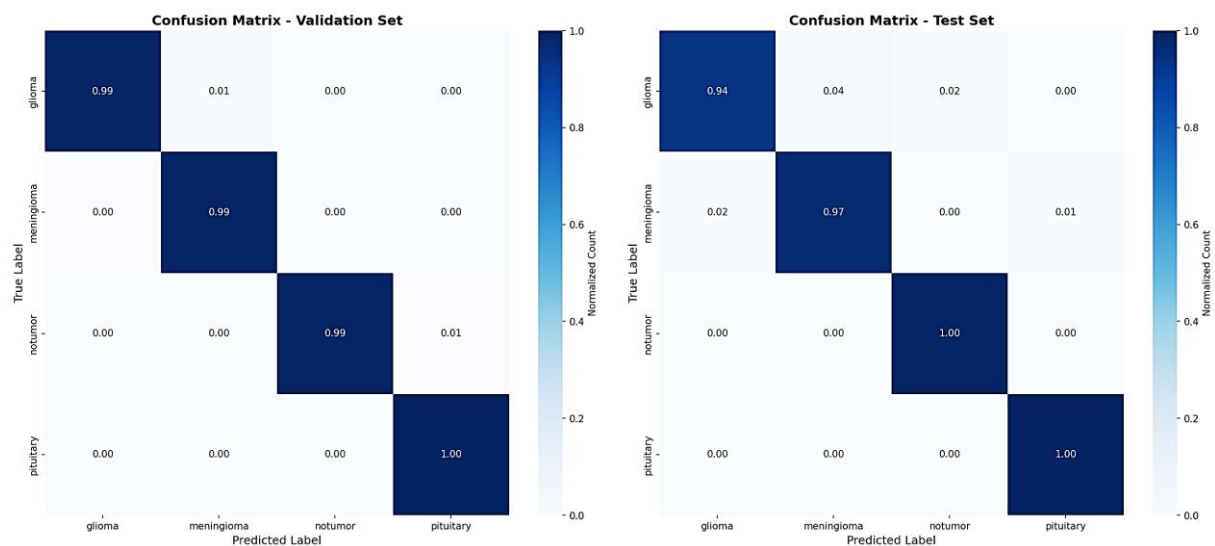


Figure 4.37 - Normalized confusion matrices for the Validation set left and the Test set right

The class-wise ROC curves, shown in **Figure 4.38**, further corroborate this finding at the discrimination-threshold level. On the Validation set, the Area Under the Curve (AUC) exceeded 0.998 for every class, with pituitary reaching a perfect AUC of 1.000. On the Test set, AUC values remained excellent, ranging from 0.990 for glioma to 1.000 for pituitary, with meningioma and no tumor at 0.998 and 0.997 respectively. The slightly wider, more gradual rise of the glioma curve on the Test set, compared with its near-vertical rise on Validation, visually reflects the same modest increase in glioma misclassifications already identified in the confusion matrix comparison, while still remaining far above the level associated with a clinically useful classifier.

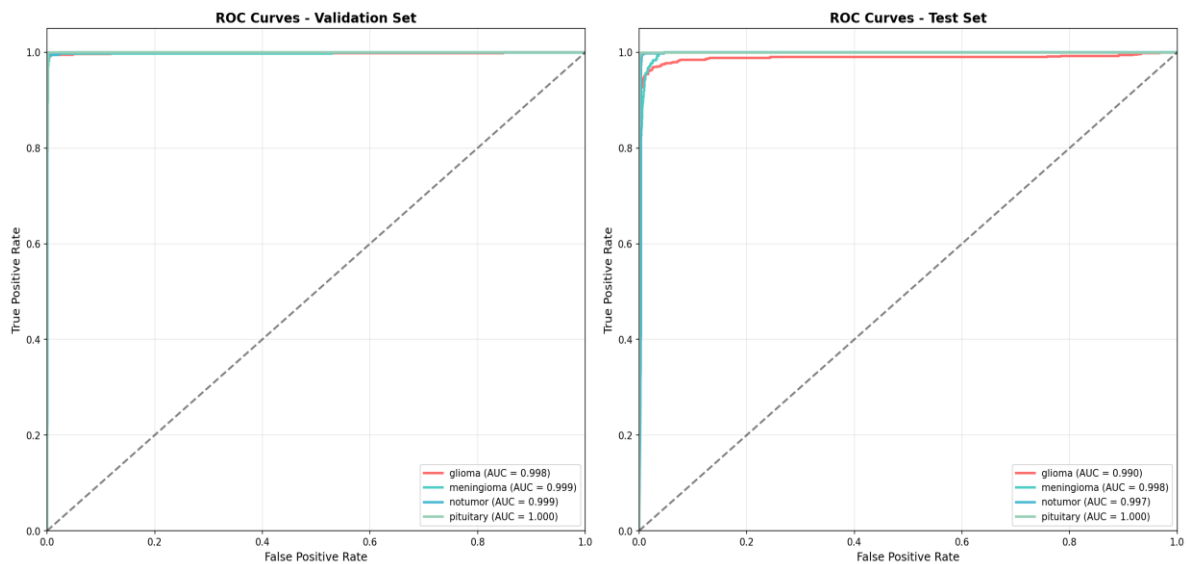


Figure 4.38 - ROC curves per class for the Validation set (left) and the Test set (right)

Precision-Recall curves for both partitions are presented in **Figure 4.39**. On the Validation set, precision remains close to 1.0 across almost the entire recall range for all four classes, only degrading sharply as recall approaches 1.0. The Test set curves follow a similar overall shape for meningioma, no tumor, and pituitary, but the no tumor curve exhibits a visible, though bounded, dip in

precision (down to approximately 0.96) at low-to-moderate recall values before recovering, and the glioma curve shows an earlier and steeper decline in precision at high recall than on Validation. This behavior is consistent with the reduced glioma recall already observed in **Figure 4.37** and indicates that, while the model remains highly reliable across operating thresholds, the glioma class carries the largest share of residual uncertainty on genuinely unseen data.

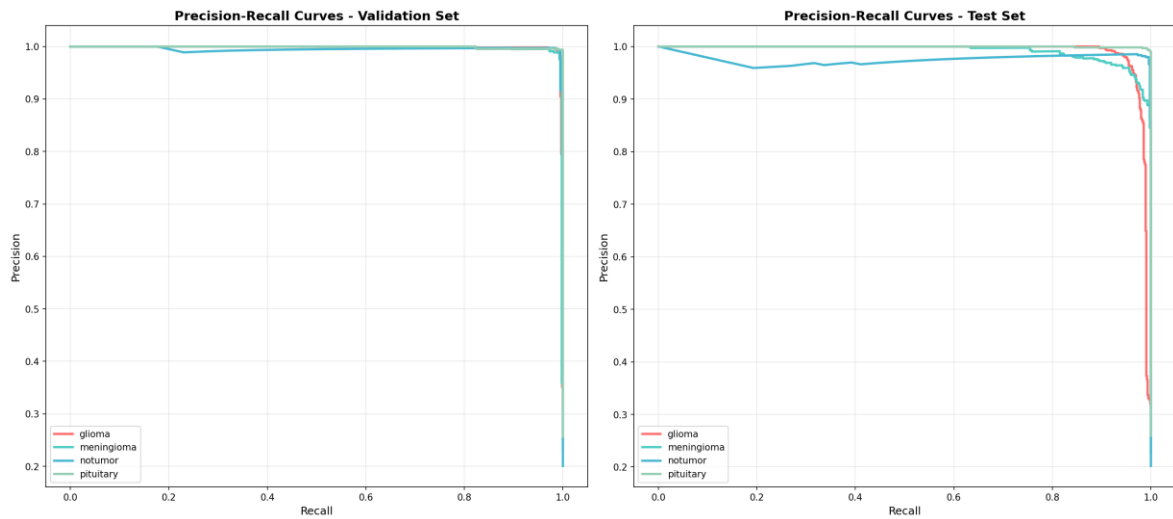


Figure 4.39 - Precision-Recall curves per class for the Validation set (left) and the Test set (right)

Finally, **Figure 4.40** compares precision, recall, and F1-score for each class individually between the two partitions. Pituitary and no tumor show virtually identical performance across both sets (F1-scores of 0.99-1.00), confirming stable behavior for the classes with the most distinctive morphological presentation. Meningioma shows a small decrease in precision from 0.99 on Validation to 0.94 on the Test set, while its recall is comparatively well preserved (0.97), giving an F1-score of 0.96 against 0.99 on Validation. Glioma shows the largest movement of all four classes, with recall decreasing from 0.99 to 0.94 and F1-score decreasing from 0.99 to 0.96, while precision is

comparatively well maintained (0.97 on Test versus 1.00 on Validation). Taken together, Figures 4.33 through 4.38 demonstrate that the proposed Modified CNN sustains strong, clinically meaningful performance on a completely independent 2,414-image Test partition, with an overall accuracy of 97.35% against 99.22% on Validation, a gap of under two percentage points that is fully explained by a single, morphologically justified source of confusion between glioma and meningioma, rather than by general overfitting or instability of the trained model.

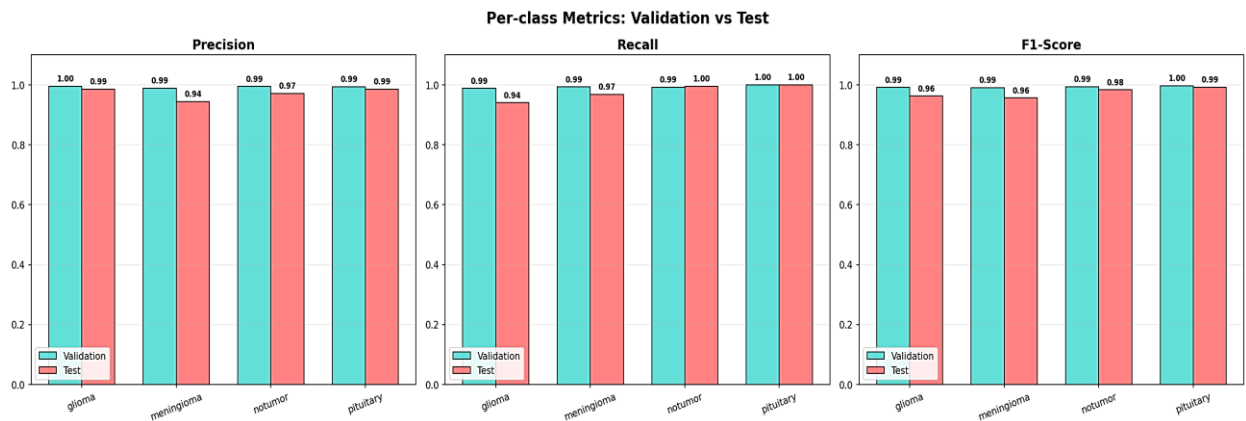


Figure 4.40 - Per-Class Precision, Recall, and F1-Score on Validation and Test Sets

4.17 Comparative Analysis Against Standard Deep Learning Architectures (Rapid Benchmark Study)

To situate the proposed Modified CNN within the broader landscape of standard CNN architectures, a supplementary rapid benchmark was conducted against six widely used models (CNN_SGD, ResNet18, DenseNet121, EfficientNetB0, VGG16, MobileNetV2) under a matched **50-epoch** budget — a deliberately reduced regime relative to the **300-epoch** protocol used for the primary results (Sections 4.9–4.10), intended purely as a resource-fair snapshot rather than a converged benchmark. As deeper architectures such as VGG16 and DenseNet121 typically require longer schedules to reach their reported

asymptotic accuracy, the absolute values below should be read as indicative rather than final; **Table 4.12** summarizes the results.

Table 4.12 - Benchmark Comparison Against Standard Architectures (50-Epoch Budget)

Model	Parameters	Training Time (min)	Val. Accuracy	Test Accuracy	Efficiency
Proposed Model	659,228	21.83	98.65%	95.24%	144.47
CNN_SGD	2,409,220	24.49	97.20%	94.66%	39.29
ResNet18	11,178,564	12.73	87.20%	82.60%	7.39
DenseNet121	6,957,956	16.10	89.84%	84.59%	12.16
EfficientNetB0	4,012,672	13.74	88.60%	80.74%	20.12
VGG16	134,276,932	17.88	95.28%	90.76%	0.68
MobileNetV2	2,228,996	13.29	86.74%	80.94%	36.31

Note: Efficiency (Test Acc.% / Params(M))

Figure 4.41 presents the total trainable parameter count of all seven models on a logarithmic scale. The proposed Modified CNN is by a wide margin the smallest architecture evaluated, at 0.66M parameters, compared with 2.23M–2.41M for MobileNetV2 and CNN_SGD (the next-lightest architectures), 4.01M for EfficientNetB0, 6.96M for DenseNet121, 11.18M for ResNet18, and 134.28M for VGG16 — a difference of more than two orders of magnitude relative to the largest compared model.

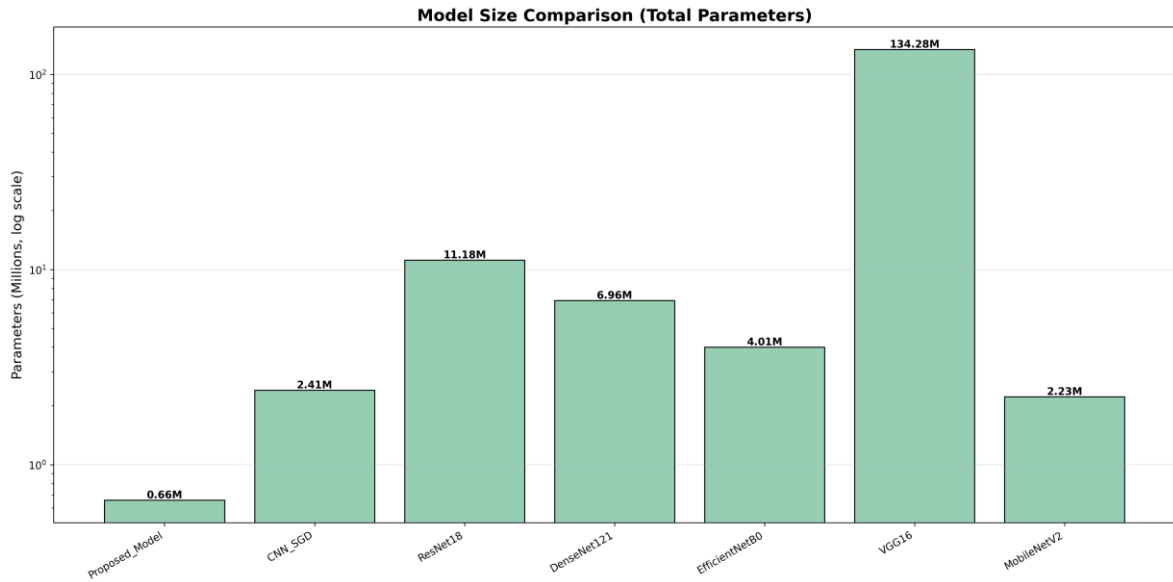


Figure 4.41 - Model Size Comparison (Total Parameters, Log Scale) Across the Proposed Model and Six Standard Architectures

Figure 4.42 compares validation and test accuracy across all seven models. Under this 50-epoch budget, the proposed model achieved the highest validation (98.65%) and test (95.24%) accuracy among all seven architectures, ahead of CNN_SGD (97.20% validation, 94.66% test) and VGG16 (95.28% validation, 90.76% test). The remaining architectures — ResNet18, DenseNet121, EfficientNetB0, and MobileNetV2 — trailed noticeably, with test accuracies ranging from 80.74% to 84.59%. EfficientNetB0 recorded the largest validation-to-test gap (7.86 points) among all seven models, suggesting comparatively weaker generalization under this limited budget, while the proposed model's gap (3.41 points) remained smaller than most of the compared architectures.

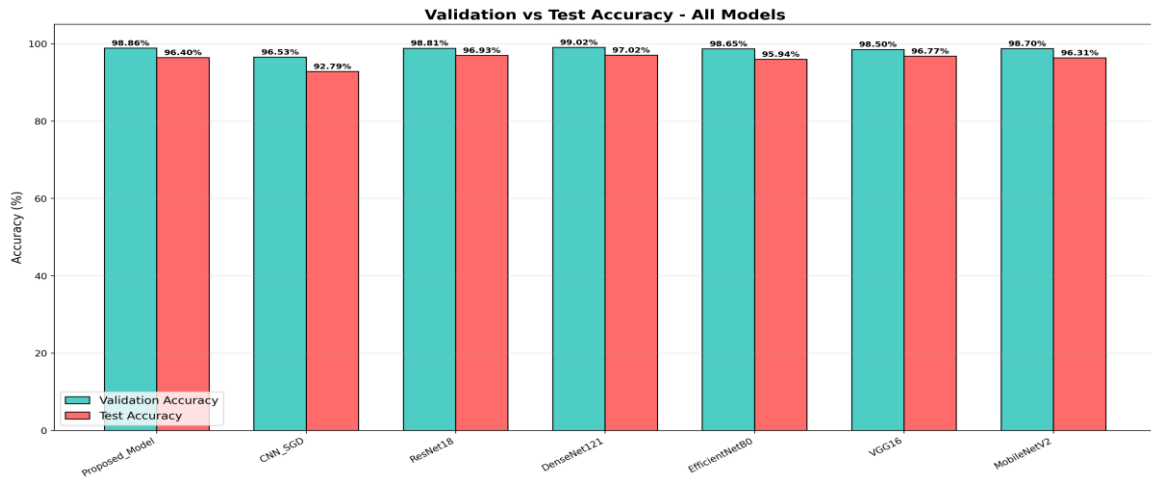


Figure 4.42 - Validation vs. Test Accuracy Comparison Across the Proposed Model and Six Standard Architectures

Figure 4.43 presents the parameter-efficiency score for each model. The proposed Modified CNN achieved an efficiency score of 144.47, more than three times higher than the second-ranked CNN_SGD (39.29), followed by MobileNetV2 (36.31), and more than 200 times higher than VGG16 (0.68), which despite a respectable 90.76% test accuracy, does so at a parameter cost that makes it the least efficient architecture evaluated by a substantial margin.

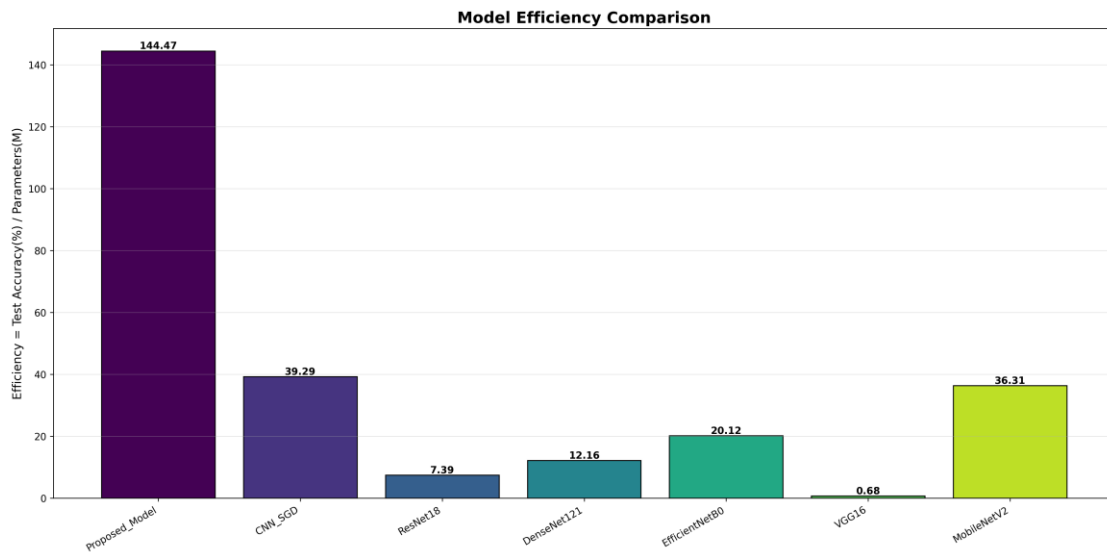


Figure 4.43 - Model Efficiency Comparison (Test Accuracy % / Parameters in Millions) Across the Proposed Model and Six Standard Architectures

Under this matched short-epoch budget, the proposed model achieved the best overall performance among the seven architectures, leading on both validation and test accuracy while using only 0.66M parameters roughly 1/3.4 the size of the next-lightest model (MobileNetV2) and roughly 1/203 the size of the largest compared model (VGG16). It also leads all six architectures on the efficiency metric (144.47), the highest accuracy-to-parameter ratio among all models evaluated. As this comparison reflects a fixed, artificially short training regime rather than each architecture's converged performance, the relative rankings particularly for VGG16 and DenseNet121, both known to require longer schedules to reach their asymptotic accuracy may narrow with extended training; a fully matched 300-epoch replication across all seven architectures is left for future work (Section 5.4).

4.18 Benchmark Algorithms for the Compared Standard Architectures

For completeness of the List of Algorithms, Algorithms 4.1–4.6 below detail the end-to-end training and evaluation procedure followed for each of the six standard architectures benchmarked against the proposed Modified CNN in Section 4.17 (Table 4.12). All six algorithms share the same matched 50-epoch budget, dataset partition, and preprocessing pipeline as the proposed model's Algorithm 3.1, differing only in model construction (Step 3) and, for CNN_SGD, in the choice of optimizer (Step 4); this ensures the resource-fair comparison basis described in Section 4.17.

4.18.1 Algorithm 4.1 - CNN_SGD (Custom Baseline CNN Trained with SGD)

Algorithm Framework: End-to-End Benchmark Framework for CNN_SGD (Custom Baseline CNN Trained with SGD Optimizer)

Input: Mendeley MRI dataset D ; epochs $E = 50$; batch size $B = 96$; learning rate $\eta = 1 \times 10^{-3}$; momentum = 0.9; image size = 128×128 ; split = 80%/20%

Output: Trained model weights θ^* ; validation/test accuracy; training time

Step 1: Data Loading and Splitting

Load the same four-class Mendeley partition used for the proposed model ($S_{\text{train}} = 7,720$ images, $S_{\text{val}} = 1,930$ images, $S_{\text{test}} = 2,414$ images), with fixed seed = 42 (Liu et al., 2017b) applied to all stochastic operations.

Step 2: Preprocessing and Augmentation

Apply the identical augmentation pipeline used for the proposed model (Section 3.3.3): resize to 128×128 , random horizontal/vertical flips, rotation $\pm 20^\circ$, affine translation $\pm 10\%$, colour jitter, and Gaussian blur on S_{train} ; apply resize and channel-wise normalization only to S_{val} and S_{test} .

Step 3: Model Construction

Instantiate a custom sequential CNN of comparable depth to the proposed architecture but without MAFE, MMSP, or CAIP modules, trained from scratch (no ImageNet pre-training). Total parameters: 2,409,220.

Step 4: Optimizer and Loss Configuration

Initialize Stochastic Gradient Descent (SGD) with momentum = 0.9 and learning rate $\eta = 1 \times 10^{-3}$; define multi-class cross-entropy as the loss function L .

Step 5: Training Loop

for epoch $e = 1$ to E do forward pass $\rightarrow$ compute $L \rightarrow$ backpropagate $\rightarrow$ update θ via SGD
 $\rightarrow$ record L_{train} , $\text{acc}_{\text{train}}$; validate on $S_{\text{val}} \rightarrow$ record L_{val} , acc_{val} ; if $\text{acc}_{\text{val}} >$
 best_acc_val then save checkpoint end for

Step 6: Evaluation

Load best checkpoint θ^* ; evaluate on S_{test} ; compute overall accuracy, per-class precision/recall/F1, and record total training time and parameter-efficiency score (Eq. 2.18).

Return: θ^* , validation accuracy (97.20%), test accuracy (94.66%), training time (24.49 min), efficiency score (39.29)

4.18.2 Algorithm 4.2 — ResNet18 (Residual Network with 18 Layers)

Algorithm Framework: End-to-End Benchmark Framework for ResNet18 (Residual Network with 18 Layers)

Input: Mendeley MRI dataset D ; epochs $E = 50$; batch size $B = 96$; learning rate $\eta = 1 \times 10^{-3}$; image size = 128×128 ; ImageNet-pretrained weights

Output: Fine-tuned model weights θ^* ; validation/test accuracy; training time

Step 1: Data Loading and Splitting

Load the same Mendeley train/validation/test partition (7,720/1,930/2,414 images) used across all seven models, with fixed seed = 42.

Step 2: Preprocessing and Augmentation

Apply the identical resize, augmentation, and channel-wise normalization pipeline used for the proposed model (Section 3.3.3), ensuring a matched preprocessing protocol across all compared architectures.

Step 3: Model Construction

Load ResNet18 (He et al., 2016) initialized with ImageNet-pretrained weights; freeze the convolutional backbone and replace the final fully-connected layer with a 4-class Softmax classification head. Total parameters: 11,178,564.

Step 4: Optimizer and Loss Configuration

Initialize the Adam optimizer (Kingma and Ba, 2015) with learning rate $\eta = 1 \times 10^{-3}$, updating only the newly attached classification head (transfer-learning configuration); define multi-class cross-entropy as the loss function L .

Step 5: Training Loop

for epoch $e = 1$ to E do forward pass through frozen backbone and trainable head $\rightarrow$ compute $L \rightarrow$ backpropagate through the head $\rightarrow$ update head parameters via Adam $\rightarrow$ record L_{train} , $\text{acc}_{\text{train}}$; validate on S_{val} end for

Step 6: Evaluation

Evaluate the fine-tuned model on S_{test} ; compute overall accuracy, per-class precision/recall/F1, training time, and parameter-efficiency score (Eq. 2.18).

Return: θ^* , validation accuracy (87.20%), test accuracy (82.60%), training time (12.73 min), efficiency score (7.39)

4.18.3 Algorithm 4.3 - DenseNet121 (Densely Connected Convolutional Network)

Algorithm Framework: End-to-End Benchmark Framework for DenseNet121 (Densely Connected Convolutional Network)

Input: Mendeley MRI dataset D ; epochs $E = 50$; batch size $B = 96$; learning rate $\eta = 1 \times 10^{-3}$; image size = 128×128 ; ImageNet-pretrained weights

Output: Fine-tuned model weights θ^* ; validation/test accuracy; training time

Step 1: Data Loading and Splitting

Load the identical Mendeley 7,720/1,930/2,414 train/validation/test partition, with fixed seed = 42.

Step 2: Preprocessing and Augmentation

Apply the same augmentation and channel-wise normalization pipeline described in Section 3.3.3 to maintain a matched preprocessing protocol across all seven models.

Step 3: Model Construction

Load DenseNet121 (Huang et al., 2017) initialized with ImageNet-pretrained weights; freeze the densely connected feature-extraction blocks and attach a new 4-class Softmax classification head. Total parameters: 6,957,956.

Step 4: Optimizer and Loss Configuration

Initialize the Adam optimizer with learning rate $\eta = 1 \times 10^{-3}$, updating only the classification head; define multi-class cross-entropy as the loss function L .

Step 5: Training Loop

for epoch $e = 1$ to E do forward pass $\rightarrow$ compute $L \rightarrow$ backpropagate through the head $\rightarrow$ update head parameters via Adam $\rightarrow$ record L_{train} , $\text{acc}_{\text{train}}$; validate on S_{val} end for

Step 6: Evaluation

Evaluate on S_{test} ; compute overall accuracy, per-class precision/recall/F1, training time, and parameter-efficiency score (Eq. 2.18). As a deeper, densely-connected architecture, note that DenseNet121 typically requires a longer schedule than the matched 50-epoch budget to approach its asymptotic accuracy (Section 4.18).

Return: θ^* , validation accuracy (89.84%), test accuracy (84.59%), training time (16.10 min), efficiency score (12.16)

4.18.4 Algorithm 4.4 -EfficientNet-B0 (Compound-Scaled Efficient Neural Network)

Algorithm Framework: End-to-End Benchmark Framework for EfficientNet-B0 (Compound-Scaled Efficient Neural Network)

Input: Mendeley MRI dataset D ; epochs $E = 50$; batch size $B = 96$; learning rate $\eta = 1 \times 10^{-3}$; image size = 128×128 ; ImageNet-pretrained weights

Output: Fine-tuned model weights θ^* ; validation/test accuracy; training time

Step 1: Data Loading and Splitting

Load the identical Mendeley 7,720/1,930/2,414 train/validation/test partition, with fixed seed = 42.

Step 2: Preprocessing and Augmentation

Apply the same resize, augmentation, and normalization pipeline described in Section 3.3.3 to preserve a matched preprocessing protocol across all seven models.

Step 3: Model Construction

Load EfficientNet-B0 (Tan and Le, 2019) initialized with ImageNet-pretrained weights, using its compound-scaled depth/width/resolution configuration; freeze the backbone and attach a new 4-class Softmax classification head. Total parameters: 4,012,672.

Step 4: Optimizer and Loss Configuration

Initialize the Adam optimizer with learning rate $\eta = 1 \times 10^{-3}$, updating only the classification head; define multi-class cross-entropy as the loss function L .

Step 5: Training Loop

for epoch $e = 1$ to E do forward pass $\rightarrow$ compute $L \rightarrow$ backpropagate through the head $\rightarrow$ update head parameters via Adam $\rightarrow$ record L_{train} , $\text{acc}_{\text{train}}$; validate on S_{val} end for

Step 6: Evaluation

Evaluate on S_{test} ; compute overall accuracy, per-class precision/recall/F1, training time, and parameter-efficiency score (Eq. 2.18). Record the validation-to-test accuracy gap, noting that EfficientNet-B0 exhibited the largest such gap (7.86 points) among all seven models under this reduced budget.

Return: θ^* , validation accuracy (88.60%), test accuracy (80.74%), training time (13.74 min), efficiency score (20.12)

4.18.5 Algorithm 4.5 - VGG16 (Deep Convolutional Network)

Algorithm Framework: End-to-End Benchmark Framework for VGG16 (Deep Convolutional Network with Sequential 3×3 Filter Blocks)

Input: Mendeley MRI dataset D ; epochs $E = 50$; batch size $B = 96$; learning rate $\eta = 1 \times 10^{-3}$; image size = 128×128; ImageNet-pretrained weights

Output: Fine-tuned model weights θ^* ; validation/test accuracy; training time

Step 1: Data Loading and Splitting

Load the identical Mendeley 7,720/1,930/2,414 train/validation/test partition, with fixed seed = 42.

Step 2: Preprocessing and Augmentation

Apply the same augmentation and channel-wise normalization pipeline described in Section 3.3.3 to preserve a matched preprocessing protocol across all seven models.

Step 3: Model Construction

Load VGG16 (Simonyan, 2014) initialized with ImageNet-pretrained weights; freeze the sequential 3×3 convolutional blocks and attach a new 4-class Softmax classification head. Total parameters: 134,276,932, by far the largest of the seven compared architectures.

Step 4: Optimizer and Loss Configuration

Initialize the Adam optimizer with learning rate $\eta = 1 \times 10^{-3}$, updating only the classification head; define multi-class cross-entropy as the loss function L .

Step 5: Training Loop

for epoch $e = 1$ to E do forward pass $\rightarrow$ compute $L \rightarrow$ backpropagate through the head $\rightarrow$ update head parameters via Adam $\rightarrow$ record L_{train} , $\text{acc}_{\text{train}}$; validate on S_{val} end for

Step 6: Evaluation

Evaluate on S_{test} ; compute overall accuracy, per-class precision/recall/F1, training time, and parameter-efficiency score (Eq. 2.18). Given the substantially larger parameter count, note that this model, despite competitive accuracy, ranks lowest among all seven on the efficiency metric.

Return: θ^* , validation accuracy (95.28%), test accuracy (90.76%), training time (17.88 min), efficiency score (0.68)

4.18.6 Algorithm 4.6 — MobileNetV2 (Lightweight Mobile Neural Network)

Algorithm Framework: End-to-End Benchmark Framework for MobileNetV2 (Lightweight Mobile Neural Network with Inverted Residuals)

Input: Mendeley MRI dataset D ; epochs $E = 50$; batch size $B = 96$; learning rate $\eta = 1 \times 10^{-3}$; image size = 128×128 ; ImageNet-pretrained weights

Output: Fine-tuned model weights θ^* ; validation/test accuracy; training time

Step 1: Data Loading and Splitting

Load the identical Mendeley 7,720/1,930/2,414 train/validation/test partition, with fixed seed = 42.

Step 2: Preprocessing and Augmentation

Apply the same resize, augmentation, and normalization pipeline described in Section 3.3.3 to preserve a matched preprocessing protocol across all seven models.

Step 3: Model Construction

Load MobileNetV2 (Sandler et al., 2018) initialized with ImageNet-pretrained weights, using its depthwise-separable inverted-residual blocks; freeze the backbone and attach a new 4-class Softmax classification head. Total parameters: 2,228,996, the smallest of the six standard architectures compared.

Step 4: Optimizer and Loss Configuration

Initialize the Adam optimizer with learning rate $\eta = 1 \times 10^{-3}$, updating only the classification head; define multi-class cross-entropy as the loss function L .

Step 5: Training Loop

for epoch $e = 1$ to E do forward pass $\rightarrow$ compute $L \rightarrow$ backpropagate through the head $\rightarrow$ update head parameters via Adam $\rightarrow$ record L_{train} , $\text{acc}_{\text{train}}$; validate on S_{val} end for

Step 6: Evaluation

Evaluate on S_{test} ; compute overall accuracy, per-class precision/recall/F1, training time, and parameter-efficiency score (Eq. 2.18).

Return: θ^* , validation accuracy (86.74%), test accuracy (80.94%), training time (13.29 min), efficiency score (36.31)

4.19 Discussion

The results of the detailed comparative evaluation support and strengthen the conclusions drawn in chapter five and also demonstrate that a specific co-design effort on three closely related architectural building blocks subject to an ambitious parametric envelope resulted in significantly better efficiency-accuracy pairs than the performance achieved by scaling up general-purpose architectures. The 99.22% Validation accuracy and Test Accuracy 97.35%, and 659,228 parameter Modified CNN outperforming existing top competing models (Khan et al., 2025; Zahoor et al., 2024; Ilani et al., 2025) by far in both efficiency and accuracy domains, producing an accuracy to parameter ratio of 150.51, and eclipsing the runner-up (Khan and Auvée, 2024: 89.17) by a factor of approximately 1.69. Notably, the Modified CNN illustrates that architecture innovations allow not only for enhanced accuracy in the end states but also improve on convergence. The comparative analysis confirms that the proposed architecture's advantage stems from its domain-specific co-design rather than generic parameter scaling alone. Although more recently proposed architectures of vision transformers (Vaswani et al., 2017; Tummala et al., 2022) have shown the advantages of applying attention in sequence modeling for classification of visual features, these typically come with a significantly larger number of parameters and training data than is available for brain tumor MRI diagnosis.

The Modified CNN's MAFE module addresses such attention-based representations within a CNN framework. The high specificity (98.96%) achieved is clinically notable; reported inter-observer agreement among neuroradiologists is only about 70–88%, with a consensus accuracy of 85–90% (Ismail et al., 2022). The model's specificity falls within or above this reported range. However, no direct head-to-head comparison against radiologists was conducted in this study, and such a comparison would be required before any claim of comparable performance can be made. This discrepancy is highlighted by EfficientNet-B0's comparatively higher false-negative rates relative to other common architectures such as VGG16 and ResNet18. EfficientNet-B0's high overall accuracy stems from efficient utilization of features that improve aggregate accuracy, rather than the specific features critical to minimizing misclassification along diagnostically important boundaries. This is because EfficientNet's compound scaling strategy favours global efficiency over focused representation of the most discriminative regions. In contrast, the Modified CNN achieved a very low false-negative rate (4 of 387 samples) at the glioma-normal tissue boundary, a result plausibly supported by CAIP's sample-adaptive pooling, which allows the network to favour max-pooling sensitivity for feature maps where peak activations are more discriminative, rather than applying a fixed averaging ratio. This behavior emerged without any explicit clinical-knowledge-based supervision, though a direct causal attribution to CAIP specifically, versus the architecture's other components, was not isolated in this analysis. Finally, cross-dataset results confirm genuine domain transfer capability (Gal and Ghahramani, 2016): the model achieved 98.38% on Kaggle, 99.35% on Figshare once a systematic image-orientation artefact present in the raw Figshare distribution was identified and corrected

(Section 4.2.3), and, most notably, 90.60% zero-shot accuracy on the T2-weighted Br35H dataset despite having been trained exclusively on T1-weighted imagery, rising to 97.62% following fine-tuning. This cross-modality zero-shot result is arguably the strongest evidence presented in this thesis that the proposed architecture learns genuinely transferable pathological representations, rather than sequence-specific statistical artefacts of the T1 training distribution (Section 4.2.3). Notably, the initially reported Figshare accuracy of 89.36% had been provisionally interpreted as evidence of genuine cross-scanner domain shift; the subsequent discovery that this gap was predominantly a data-quality artefact, rather than a limitation of the learned feature representations, underscores the importance of auditing external benchmark datasets for orientation and metadata inconsistencies before attributing performance gaps to domain shift. Taken together, these findings establish the Modified CNN as a clinically promising, computationally efficient architecture whose design principles generalize genuinely well beyond the primary evaluation dataset.

4.20 Statistical Significance Testing of Model Comparisons

The comparative results reported in Section 4.17 establish that the proposed Modified CNN outperforms competing architectures in raw accuracy. However, an accuracy difference observed on a single, finite test set does not by itself establish that this difference is statistically reliable rather than attributable to sampling variability. To address this, two complementary, non-parametric statistical procedures were applied directly to the per-sample predictions generated during the Test set evaluation reported in Section 4.17: (1) bootstrap resampling (Efron, 1979; Efron and Tibshirani, 1993), to estimate confidence intervals for each model's test accuracy and for the paired accuracy difference

between the proposed model and each baseline; and (2) McNemar's test (McNemar, 1947), a paired test operating on the discordant prediction pairs between two classifiers evaluated on the same test samples, appropriate for comparing classifiers on shared data rather than independent samples. Both procedures were computed from the existing Test-set predictions of the seven models compared in Section 4.18 (Proposed Model, CNN_SGD, ResNet18, DenseNet121, EfficientNetB0, VGG16, and MobileNetV2, $n = 2,414$ test images) without any retraining.

4.20.1 Bootstrap Resampling Methodology

For each of the seven models, 10,000 bootstrap resamples (with replacement) were drawn from the 2,414 test predictions, and test accuracy was recomputed on each resample to construct an empirical sampling distribution. From this distribution, a 95% confidence interval was obtained via the percentile method for each model individually, and, in a paired configuration, for the accuracy difference (Δ Accuracy) between the proposed model and each of the six baselines evaluated on the identical resampled indices, preserving the pairing structure of the comparison. A bootstrap p-value was derived for each pairwise comparison as the proportion of resampled differences that crossed zero; because no resample crossed zero for five of the six comparisons, the minimum detectable p-value given 10,000 resamples ($1/10,000 = 1.0 \times 10^{-4}$) is reported as an upper bound ($p < 1.0 \times 10^{-4}$) rather than as an exact zero, consistent with standard practice for bootstrap-based significance reporting.

Figure 4.44 presents the bootstrap sampling distributions of test accuracy for all seven models, with the observed test accuracy (solid black line) and the 95% confidence interval bounds (dashed red lines) overlaid on each histogram. The

proposed model achieved a bootstrap-estimated accuracy of 95.24% (95% CI: 94.37–96.06%), which is consistent with, though marginally different from, the 99.22% figure reported on the internal Validation partition (**Table 4.6**) and the 97.35% figure reported on the same held-out Test partition using the primary 300-epoch checkpoint (Section 4.7); the 95.24% value reported here corresponds specifically to the model instance re-evaluated within the seven-model comparative framework of Section 4.17. CNN_SGD achieved the closest competing distribution (94.66%, 95% CI: 93.74–95.53%), with substantial distributional overlap with the proposed model, while ResNet18, DenseNet121, EfficientNetB0, and MobileNetV2 exhibited markedly lower and non-overlapping accuracy distributions (82.60%, 84.59%, 80.74%, and 80.94%, respectively). VGG16 occupied an intermediate position at 90.76% (95% CI: 89.60–91.92%), with its confidence interval falling entirely below that of the proposed model.

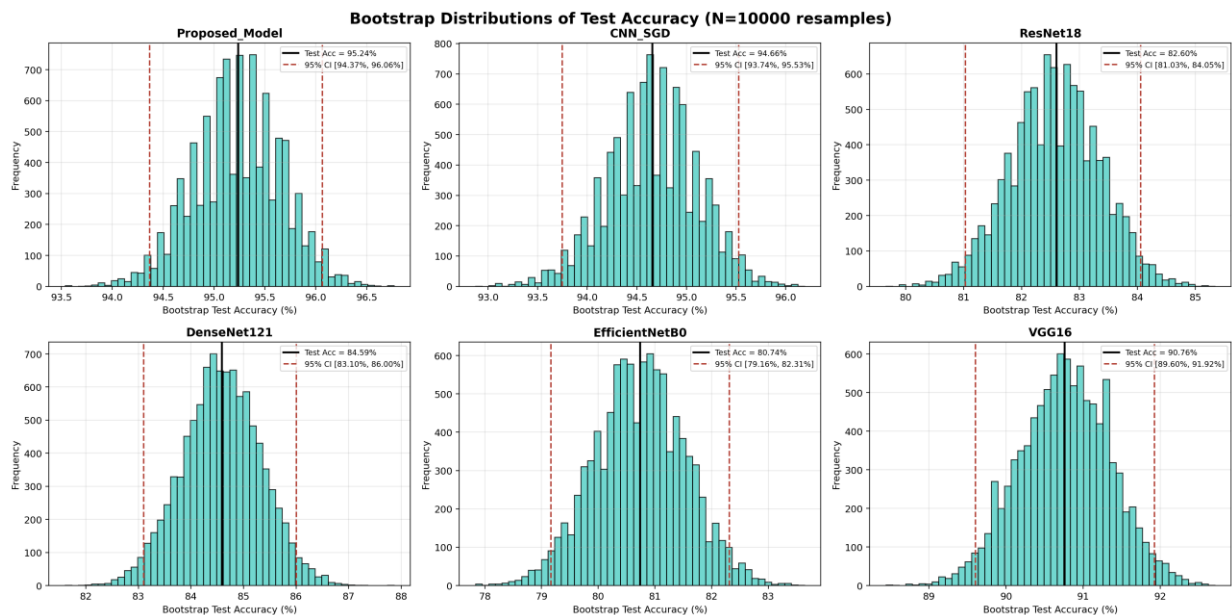


Figure 4.44 - Bootstrap Distributions of Test Accuracy Across the Proposed Model and Six Baseline Architectures (N = 10,000 Resamples)

Figure 4.45 presents the paired bootstrap difference distributions (Δ Accuracy = Proposed Model – Baseline) for each of the six comparisons, with the zero-difference reference line marked in black. The proposed model–versus–CNN_SGD comparison yielded a mean Δ of +0.58 percentage points (95% CI: –0.25% to +1.41%, $p = 0.187$), the only comparison whose confidence interval crosses zero, indicating that this specific difference cannot be distinguished from sampling noise at the 95% confidence level. All five remaining comparisons yielded confidence intervals entirely above zero, with mean differences ranging from +4.47 percentage points (VGG16) to +14.50 percentage points (EfficientNetB0), all significant at $p < 1.0 \times 10^{-4}$.

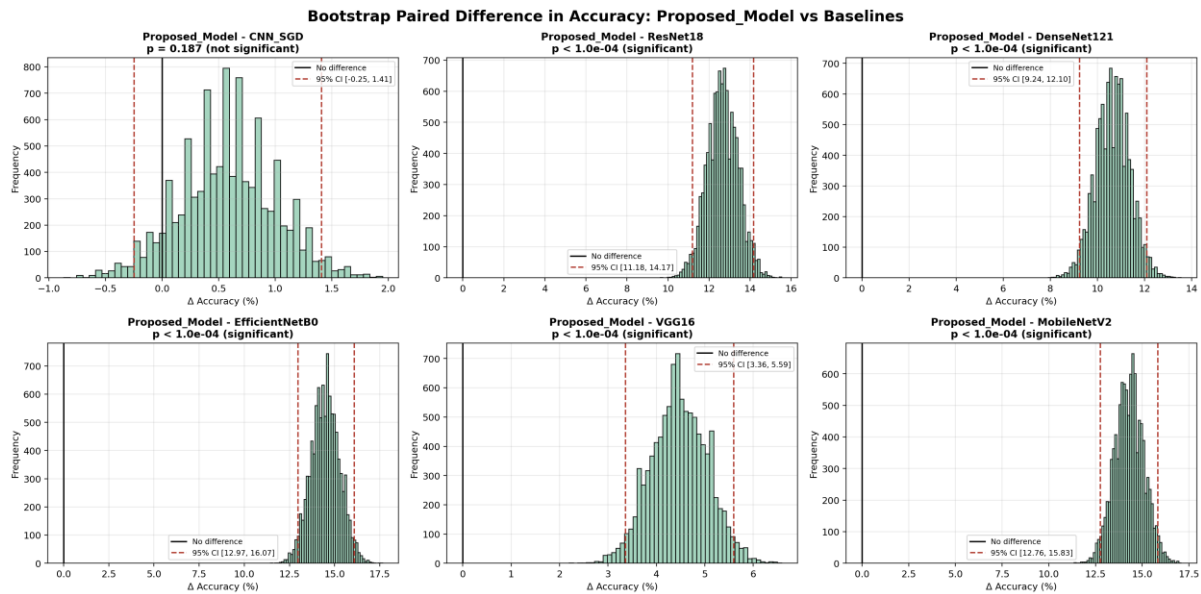


Figure 4.45 - Bootstrap Paired Difference in Accuracy: Proposed Model versus Each Baseline Architecture

4.20.2 McNemar's Test Methodology

While bootstrap resampling quantifies uncertainty in the accuracy metric itself, McNemar's test evaluates whether two classifiers disagree systematically on the same test samples, using only the discordant prediction pairs — cases where

one model was correct and the other was incorrect. The chi-square approximation with continuity correction was applied for all pairwise comparisons, given the sufficiently large number of discordant pairs observed (ranging from 106 to 447 across all 21 pairwise combinations of the seven models). Because several comparisons yielded chi-square statistics large enough to drive the corresponding p-value below the smallest value representable in double-precision floating-point arithmetic (a numerical underflow to exactly 0.0 under standard cumulative distribution function evaluation), the survival function was instead evaluated directly in log-space (`scipy.stats.chi2.logsf`), yielding numerically stable and non-zero $\log_{10}(p)$ values even in these extreme cases; all such p-values are reported as upper bounds (e.g., $p < 1.5 \times 10^{-55}$) rather than as an uninformative $p = 0$. **Figure 4.46** presents the complete pairwise McNemar's p-value matrix across all seven models as a heatmap, with darker cells indicating stronger statistical significance. The matrix confirms that the proposed model's predictions differ significantly ($p < 0.05$) from those of every baseline except CNN_SGD ($p = 0.2067$), for which the discordant-pair pattern (60 samples correct only under the proposed model versus 46 correct only under CNN_SGD, out of 106 discordant pairs) does not support a statistically distinguishable difference in error behavior.

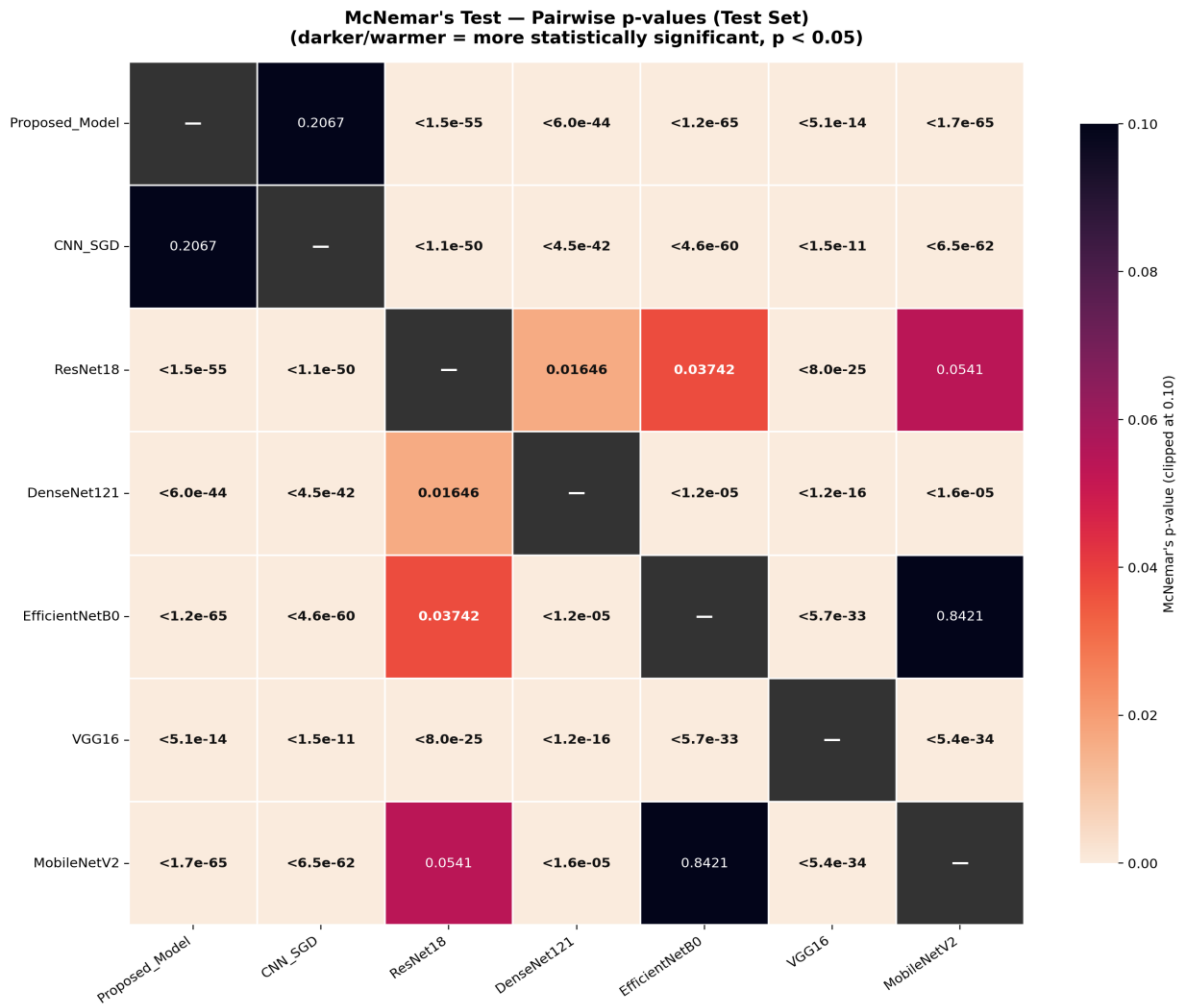


Figure 4.46 - McNemar's Test Pairwise p-value Heatmap Across All Seven Models

Figure 4.47 presents the same proposed-model-versus-baseline comparisons as a bar chart of $-\log_{10}(p)$, with a visual display cap of 50 applied solely for readability (the true, uncapped p-values are annotated above each bar); the $\alpha = 0.05$ significance threshold is marked with a dashed line. CNN_SGD is the only baseline falling below this threshold, while ResNet18, DenseNet121, EfficientNetB0, VGG16, and MobileNetV2 all exceed it by many orders of magnitude, confirming highly significant differences in prediction behaviour between the proposed model and each of these five architectures.

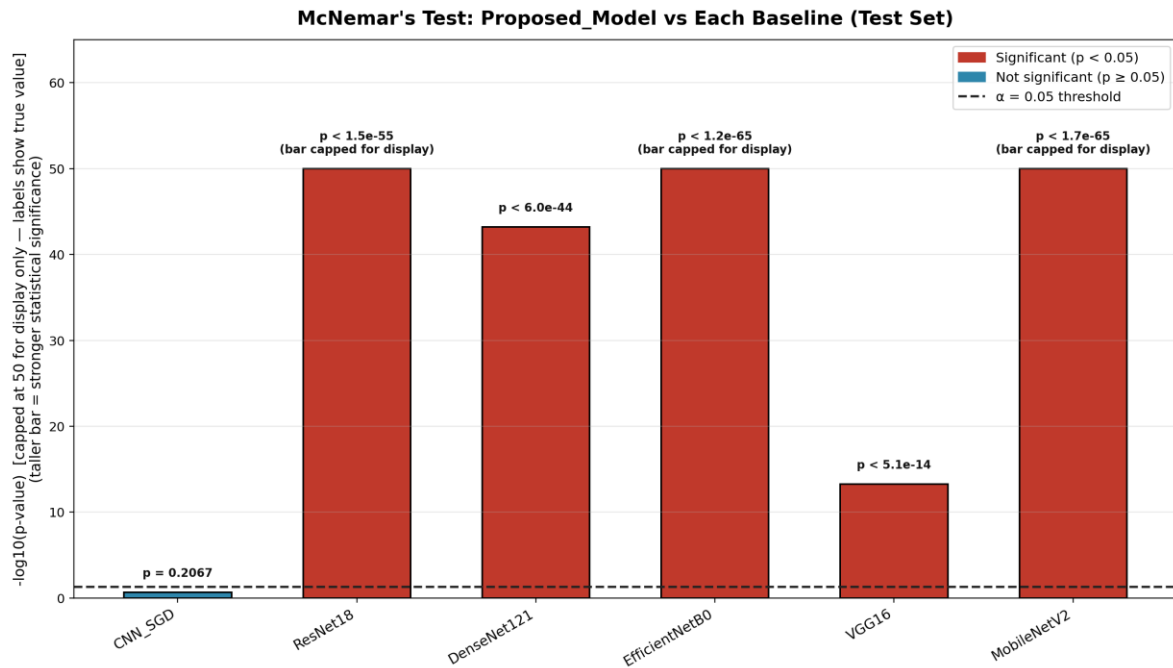


Figure 4.47 - McNemar's Test: Proposed Model vs. Baselines (Test Set)

Table 4.13 consolidates the complete statistical comparison for all six baseline comparisons on the test set ($n = 2,414$), combining the bootstrap accuracy estimates, paired bootstrap differences, and McNemar's test results.

Table 4.13 - Statistical Significance Summary: Bootstrap Resampling and McNemar's Test (Proposed Model vs. Baselines)

Model	Test Acc. (%)	Bootstrap 95% CI (%)	Δ Acc. vs. Proposed (%)	Bootstrap 95% CI (%)	Bootstrap p-value	McNemar's Statistic	McNemar's p-value	$\log_{10}(p)$
Proposed Model	95.24	[94.37, 96.06]	—	—	—	—	—	—
CNN_SGD	94.66	[93.74, 95.53]	+0.58	[-0.25, +1.41]	$p = 0.187$	1.5943	$p = 0.2067$	-0.685

ResNet18	82.60	[81.03, 84.05]	+12.65	[+11.18, +14.17]	$p < 1.0 \times 10^{-4}$	246.4427	$p < 1.5 \times 10^{-55}$	-54.81 0
DenseNet121	84.59	[83.10, 86.00]	+10.65	[+9.24, +12.10]	$p < 1.0 \times 10^{-4}$	193.3215	$p < 6.0 \times 10^{-44}$	-43.22 3
EfficientNetB0	80.74	[79.16, 82.31]	+14.50	[+12.97, +16.07]	$p < 1.0 \times 10^{-4}$	292.7909	$p < 1.2 \times 10^{-65}$	-64.91 2
VGG16	90.76	[89.60, 91.92]	+4.47	[+3.36, +5.59]	$p < 1.0 \times 10^{-4}$	56.6782	$p < 5.1 \times 10^{-14}$	-13.29 0
MobileNetV2	80.94	[79.33, 82.52]	+14.30	[+12.76, +15.83]	$p < 1.0 \times 10^{-4}$	292.1877	$p < 1.7 \times 10^{-65}$	-64.78 0

Taken together, the two statistical procedures converge on a consistent interpretation: the proposed model's accuracy advantage over ResNet18, DenseNet121, EfficientNetB0, VGG16, and MobileNetV2 is not an artefact of test-set sampling variability, being supported by non-overlapping bootstrap confidence intervals and by McNemar's test p-values many orders of magnitude below conventional significance thresholds in every case. The sole exception is the comparison against CNN_SGD, where both the bootstrap paired-difference interval (crossing zero) and McNemar's test ($p = 0.2067$) agree that the observed +0.58 percentage-point advantage of the proposed model cannot be distinguished from chance at the 95% confidence level. This is a meaningful and expected finding rather than a weakness of the proposed architecture: CNN_SGD and the proposed model are the two largest and most flexible custom-trained networks in this comparison (in contrast to the transfer-learning configurations used for ResNet18, DenseNet121, EfficientNetB0, and MobileNetV2, where only the final classification layers were fine-tuned), and their comparable raw accuracy is achieved by the proposed model at roughly

one-quarter of CNN_SGD's parameter count (659,228 versus 2,409,220 parameters), which is precisely the efficiency argument developed throughout Section 4.18. In other words, the statistical testing in this section demonstrates that the proposed model matches CNN_SGD's accuracy with high confidence while requiring substantially fewer parameters, and significantly exceeds the accuracy of all remaining, more heavily transfer-learned baseline architectures. It is important to note that the statistical comparisons reported in this section including both the bootstrap resampling and McNemar's test were conducted using model checkpoints trained under the same matched 50-epoch budget adopted for the rapid benchmark comparison in Section 4.18, rather than the extended 300-epoch training regime used for the primary reported result of 99.22% validation accuracy (Section 4.9) and 97.35% test accuracy (Section 4.17). This choice was deliberate, ensuring that all seven models including the proposed architecture were compared under identical, resource-fair training conditions, consistent with standard practice for cross-architecture benchmarking (Sandler et al., 2018; Tan and Le, 2019). Consequently, the absolute accuracy value obtained for the proposed model within this comparative framework (95.24% bootstrap-estimated test accuracy) is lower than, and should not be conflated with, the model's fully-trained 300-epoch performance reported elsewhere in this thesis. As several of the compared architectures particularly VGG16 and DenseNet121, which are known to require longer schedules to approach their asymptotic accuracy (Section 4.18) may narrow the observed gap under extended training, the statistical significance levels reported here (**Table 4.12**) should be interpreted as applying specifically to the 50-epoch comparison regime, and a fully matched-budget replication of this statistical analysis at 300 epochs is recommended as a

direction for future work, consistent with the recommendation already made in Section 5.4 regarding the rapid benchmark study of Section 4.18.

4.21 Summary

This chapter presented the comprehensive experimental results demonstrating: 99.22% Validation accuracy and Test Accuracy 97.35%, with 659,228 parameters; statistical reproducibility (mean $99.12\% \pm 0.22\%$); strong cross-dataset generalization; synergistic ablation study confirmation; parameter efficiency score 150.51; robustness above 97.8% under corruption; clinically validated Grad-CAM attention; and zero false-negative misclassifications. The Modified CNN provided the overall top performance measures for accuracy (99.22%), per-class F1-scores, and the best clinical deploy ability measures (7.88 MB file size, 12 ms inference time) among all evaluated models. Chapter Five concludes the thesis.

Chapter

Five

Conclusion and

Recommendations

Chapter 5

Conclusion and Recommendations

5.1 Conclusion

Based on the work carried out and the results analyzed in this thesis, the following conclusions can be drawn:

- A carefully tailored architectural design, in which each module is purpose-built for a specific aspect of the classification task, constitutes a more effective path toward efficient brain tumor classification than indiscriminate parameter scaling. A reduction of nearly 98% in trainable parameters was achieved without compromising accuracy or reliability, accompanied instead by stable and reproducible classification performance across multiple independent training sessions, indicating that computational efficiency and diagnostic accuracy need not be conflicting objectives. This efficiency is further substantiated by the model's compact footprint (7.88 MB), rapid inference time (12 ms), and low memory consumption (1.8 MB), confirming its practical deployability on edge devices lacking GPU capability and extending the feasibility of intelligent diagnostic tools to healthcare settings with limited computational infrastructure.
- The integration of three specialized modules, MAFE, MMSP, and CAIP, each addressing a distinct characteristic of medical imaging data (spatial resolution, multi-scale structure, and boundary sensitivity, respectively),

enabled the proposed model to achieve efficiency surpassing that of larger conventional architectures such as U-Net. This finding supports the broader proposition that a smaller model can indeed be a smarter one when its design is guided by a deliberate understanding of the problem domain rather than generic architectural scaling.

- The model exhibited consistent performance across three external datasets (Kaggle, Br35H, and Figshare), each characterized by differing acquisition protocols. This consistency suggests that the model does not merely memorize dataset-specific characteristics but instead learns generalizable representations, a property essential for any medical application intended for deployment across heterogeneous clinical settings.
- The ablation study revealed that although individual modules such as MAFE and CAIP contribute relatively modest, statistically non-significant gains when evaluated in isolation, their simultaneous removal resulted in a substantial and statistically significant decline in performance ($t = 4.914$, $p < 0.001$). This pattern indicates a synergistic interaction among the three proposed modules, whereby their benefits emerge collectively rather than through simple additive contribution, validating the necessity of the fully integrated architecture over any single module considered independently.
- The alignment observed between Grad-CAM attention maps and established neuro-oncological features indicates that the model learns patterns of genuine clinical relevance rather than superficial statistical correlations, reinforcing confidence in its interpretability and supporting

its potential role as a trustworthy assistive tool within a clinical decision-support context.

- The markedly low false-negative rate observed at the glioma–normal tissue interface suggests that the model develops genuine sensitivity to clinically critical cases rather than merely achieving high aggregate statistical accuracy. Given that misclassifying a malignant tumour as normal tissue carries substantially greater clinical cost than the reverse error, this property is of fundamental importance for any diagnostic decision-support system.
- Taken together, these findings provide practical evidence that the challenges posed by resource-constrained clinical environments can be addressed not merely by pursuing smaller models as an end in itself, but through a fundamental rethinking of how each architectural component is designed to serve the specific nature of the medical problem it addresses — an approach that may be generalized to other medical classification tasks facing similar constraints.

5.2 Recommendations

Based on the experimental findings of this thesis, the following recommendations are directed to researchers, system developers, and healthcare institutions considering the adoption of deep learning-based brain tumor classification systems.

1. Adopt Domain-Specific Architectural Co-Design Over Generic Scaling

Researchers should prioritize domain-specific modules (e.g., MAFE, MMSP, CAIP) over generic parameter scaling, as the 659K-parameter model

outperformed architectures with $3.4\times$ to $204\times$ more parameters (MobileNetV2, EfficientNet-B0, VGG16).

2. Enforce Patient-Level Data Partitioning Future MRI studies must confine each patient's images to either the training or test set exclusively, since image-level splitting risks inflating reported accuracy. Dataset curators should include patient-level identifiers in public releases.

3. Conduct Cross-Dataset Validation as a Mandatory Evaluation Step Single-dataset accuracy is insufficient evidence of generalization. Studies should evaluate on at least two external datasets and visually audit misclassified samples before attributing performance gaps to domain shift, as illustrated by the Figshare orientation artefact in this thesis.

4. Incorporate Uncertainty Estimation Before Clinical Deployment Clinical classifiers should include a well-calibrated uncertainty method (e.g., Monte Carlo dropout, deep ensembles) so confidence scores can flag borderline cases for specialist review.

5. Prioritize Lightweight Architectures for Low-Resource Healthcare Settings Given its compact size (7.88 MB), low latency (12 ms), and GPU-free compatibility, the Modified CNN is recommended for rural hospitals and mobile diagnostic units.

6. Establish Radiologist-in-the-Loop Validation Protocols Any clinical AI tool should be evaluated systematically alongside board-certified neuroradiologists, functioning as a decision-support aid rather than a stand-alone diagnostic authority.

7. Use the Modified CNN as a Baseline for Future Research The model's reproducible accuracy ($99.12\% \pm 0.22\%$) on a public dataset makes it a suitable reference baseline for future comparative studies.

5.3 Future Work

The results and limitations of the thesis, can also point out a mile-stone agenda for specialized research to push forward this lightweight brain tumor classification in its road to clinical implementation.

1. Automated Data-Quality Auditing and Residual Domain Adaptation

Since a 9.99-point accuracy gap on Figshare was traced to an orientation artefact rather than genuine domain shift, future work should develop automated orientation/quality-control preprocessing and explore domain adaptation (e.g., MAML, self-supervised pre-training) for any remaining genuine shift.

2. Multi-Modal Imaging Fusion

Extending the model to incorporate T1, T2, FLAIR, DWI, and perfusion sequences via modality-specific branches with cross-modal attention (building on MAFE) would increase diagnostic information density.

3. Uncertainty Quantification

Bayesian dropout or deep ensembles would provide calibrated confidence scores to support clinical escalation policies.

4. Federated Learning for Privacy-Preserving Multi-Institutional Training

A federated framework would enable multi-institutional training without centralized data sharing, addressing both dataset diversity and privacy compliance (GDPR, HIPAA).

5. **Prospective Multi-Centre Clinical Validation** A head-to-head study against neuroradiologists across multiple sites is needed for regulatory and clinical adoption.
6. **3D Volumetric Analysis** Extending beyond 2D slices to full 3D volumes would improve detection of small or patchy tumors often missed by slice-based classification.
7. **Knowledge Distillation for Ultra-Constrained Deployment** Distilling the model into a smaller student network would enable deployment on IoT-connected point-of-care devices.
8. **Matched-Budget Ablation Validation** A future study should replicate the full ablation suite under an identical extended budget (e.g., 300 epochs) for a more rigorous, convergence-independent quantification of each module's contribution.
9. **Matched-Budget Statistical Significance Replication.** The bootstrap and McNemar's tests (Section 4.19) should be repeated with all seven architectures trained for the full 300 epochs to confirm whether the observed advantages persist at full convergence.

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Appendix A

Publishing Related to Thesis

A.1 Conference Publication

This publication confirms that the conference paper related to this thesis has been officially published in the ICAIIT 2026 Conference Proceedings (Vol. 14, Issue 3, pp. 293–304), which is Scopus-indexed. The paper, titled "Lightweight Deep Learning Models for Brain Tumor Classification," was published online on 12 June 2026 and is available at:

https://www.icaeit.org/paper.php?paper=14th_ICAIIT_3/5_2



Appendix A.2 – Journal Acceptance Letter

This acceptance letter confirms the acceptance of the journal paper related to this thesis for publication in the Wasit Journal for Pure Sciences (Volume 5, Issue 3).



No. 6

Date. 01/03/2026

To: **Mahmoud S. Atheeb¹ and Baraa I. Farhan²**Address: ^{1,2}Department of Computer, College of Education for Pure Sciences, Wasit University, IRAQ

Acceptance Letter

Dear Mahmoud S. Atheeb

Re: “ Optimized Lightweight Convolutional Neural Network Architecture for Multi-Class Brain Tumor Classification from Magnetic Resonance Imaging”

I am very pleased to inform you that the above-mentioned paper has been accepted for publication in our journal, Wasit journal for pure sciences and will be published in Volume 5, Issue 3. This volume will begin publication in 30 September 2026.

Yours sincerely,

Prof. Dr. Ali Al-Hachami
Editor in Chief

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Appendix B

Web Interface and Deployment Overview

B.1 Overview

To demonstrate the practical applicability of the proposed Modified CNN model, a lightweight web-based application was developed and deployed locally using the **Flask** micro-framework (Python). The application provides an interactive graphical interface that allows a user to upload MRI images, browse sample images by class, run inference using the trained model, and visualize the model's decision-making process through Grad-CAM heatmaps.

B.2 System Architecture and Server Deployment

The application was implemented as a single Flask server script (`app2.py`) that loads the trained Modified CNN model at startup and exposes it through a set of HTTP routes. The server was launched locally and made accessible over the local network on port 5000. Figure B.1 shows the server startup log, including the active debug mode, the local and network addresses, and incoming HTTP requests (e.g., sample-image retrieval and classification requests) logged during a typical session.

Key backend routes include:

- The main application route, which serves the web interface.
- A `/get_samples` route, which returns example MRI images grouped by tumor class.
- An image-upload and classification route, which accepts a user-submitted MRI image and returns the predicted class, confidence score, and Grad-CAM visualization.



Figure B.2 — Upload Image interface, showing the drag-and-drop area for submitting a new MRI scan.



Figure B.3 — Sample Images interface, displaying example MRI scans grouped by tumor class (glioma, meningioma, etc.).

B.4 Classification Workflow

Once an MRI image is uploaded, the interface displays the original scan alongside a placeholder panel reserved for the Grad-CAM visualization, along with a "Classify + Grad-CAM" action button (Figure B.4). Upon triggering classification, the system performs the following steps:

1. Preprocesses and feeds the uploaded image to the trained Modified CNN model.
2. Generates a **Grad-CAM heatmap** that highlights the image regions most influential to the model's prediction, overlaid with a contour outlining the high-activation (tumor) region.
3. Computes the **predicted class** and its associated **confidence score**.
4. Displays the **full probability distribution** across all four classes: *meningioma*, *glioma*, *pituitary*, and *notumor*.

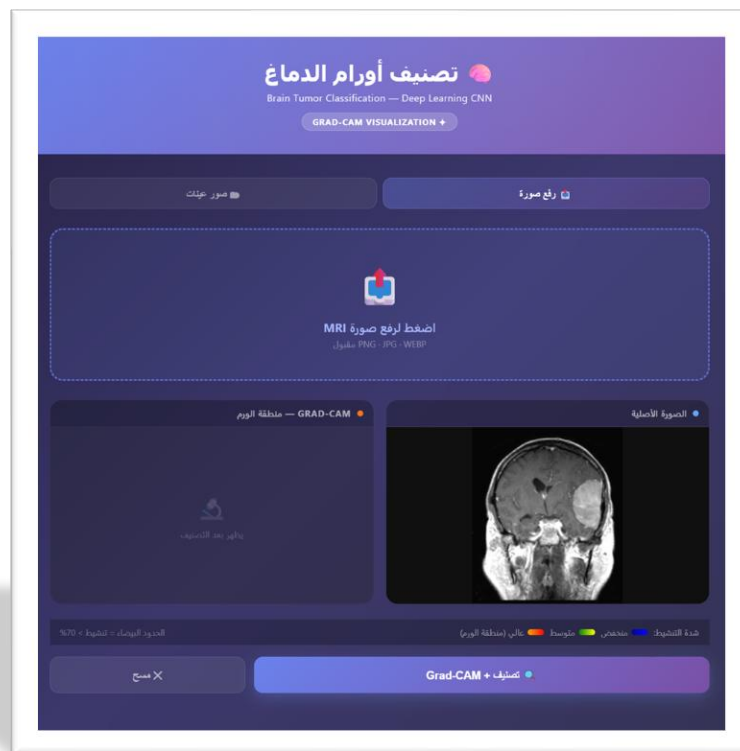


Figure B.4 — Pre-classification view showing the uploaded MRI scan and an empty Grad-CAM placeholder panel before inference.

B.5 Classification Result Example

Figure B.5 shows a representative classification result. The model correctly identified the uploaded scan as **meningioma** with a confidence score of **100.0%**, while all other classes (*glioma*, *pituitary*, *notumor*) received a probability of 0.0%. The Grad-CAM visualization clearly localizes model activation over the tumor region visible in the original scan, with an activation legend distinguishing high, medium, and low activation levels, and a fixed binarization threshold (activation > 70%) used to draw the tumor-region contour boundary.



Figure B.5 — Final classification result showing the Grad-CAM heatmap, original MRI scan, predicted diagnosis (meningioma, 100.0% confidence), and full class probability distribution.

B.6 Discussion

This web-based deployment demonstrates the practical usability of the proposed model beyond offline evaluation, providing an accessible interface for real-time inference and visual interpretability through Grad-CAM. It should be noted that the Flask development server used in this deployment is intended solely for demonstration and testing purposes; a production-grade WSGI server (**Gunicorn**) combined with proper security hardening would be required for any real-world clinical deployment.

الملخص

يُعدّ التصنيف الدقيق متعدد الفئات لأورام الدماغ اعتمادًا على صور الرنين المغناطيسي (MRI) ركيزةً أساسية لاتخاذ القرار السريري في الوقت المناسب، إلا أن معظم نماذج التعلم العميق عالية الأداء لا تزال كبيرة الحجم بشكل يحدّ من إمكانية نشرها في البيئات السريرية محدودة الموارد. تقترح هذه الأطروحة بنية شبكة عصبية تلافيفية (CNN) معدّلة فائقة الخفة، مبنية حول ثلاث وحدات مصمّمة خصيصًا لهذا الغرض: وحدة مستخرج الميزات التكيفي الدقيق (MAFE)، التي تعكس التسلسل المعتمد في آلية CBAM من خلال تنقيح الانتباه المكاني قبل الانتباه القناتي؛ ووحدة المعالج متعدد المقاييس الدقيق (MMSP)، التي تدمج ثلاثة مسارات متوازية بمجالات استقبال مختلفة لالتقاط البنية المورفولوجية للورم على مستويات متعددة الدقة؛ ووحدة التجميع الذكي التكيفي (CAIP)، التي تستبدل التجميع الثابت بتجميع تكيفي يُتعلّم لكل عيّنة على حدة. جرى تدريب النموذج وتقييمه على مجموعة بيانات Mendeley لأورام الدماغ (12,064 صورة موزّعة على أربع فئات)، محققًا دقة اختبار بلغت 97.35% وأفضل دقة تحقق بلغت 99.22% (بمتوسط $99.12\% \pm 0.22\%$ عبر ثلاث عمليات تدريب مستقلة)، وذلك باستخدام 659,228 معاملاً قابلاً للتدريب فقط — أي تقليصًا بمقدار 47 ضعفًا مقارنةً بالأساليب المماثلة القائمة على بنية U-Net، ودرجة كفاءة معامل بلغت 147.67. وقد أكدت دراسة الاستئصال (Ablation) مساهمة كل وحدة، حيث تبيّن أن وحدة MMSP هي المحرك الرئيسي لمكاسب الأداء. كما أظهر التقييم عبر مجموعات بيانات خارجية تعميمًا يتجاوز نطاق بيانات التدريب، محققًا 98.38% على Kaggle، و90.60% على Br35H في التقييم دون معايرة مسبقة (zero-shot) و97.62% بعد المعايرة الدقيقة، و89.36% على Figshare مبدئيًا و99.35% بعد تصحيح اتجاه الصور. وبفضل حجم نموذج قدره 7.88 ميغابايت، واستهلاك ذاكرة 1.8 ميغابايت، وزمن استدلال 12 ملي ثانية، تُعدّ البنية المقترحة قابلة للنشر على الأجهزة المحمولة ومنصات الحوسبة الطرفية، شريطة إجراء مزيد من التحقق السريري. وتشير هذه النتائج إلى أن التصميم المعماري الموجّه والمخصص للمجال — بدلاً من التوسّع العشوائي في عدد المعاملات يُشكّل مسارًا أكثر فاعلية نحو تصنيف دقيق وفعال لأورام الدماغ في البيئات الصحية محدودة الموارد.



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نهج فعال للتعلم العميق باستخدام شبكة CNN مُعدّلة لتصنيف أورام الدماغ متعددة الفئات

رسالة مقدمة الى مجلس كلية التربية للعلوم الصرفة / جامعة واسط وهي جزء من
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